



Nash bargaining for power purchase agreements: A structured approach to designing and pricing PPAs in modern energy markets



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DTU Wind & Energy Systems is a department of the Technical University of Denmark with a unique integration of research, education, innovation and public/private sector consulting in the field of wind energy. Our activities develop new opportunities and technology for the global and Danish exploitation of wind energy. Research focuses on key technical-scientific fields, which are central for the development, innovation and use of wind energy and provides the basis for advanced education at the education.

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Approval

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Abstract

Bilateral contracts are a central risk-hedging instrument in modern electricity markets, and long-term Power Purchase Agreements (PPAs) are increasingly critical for securing renewable energy investments. Yet their negotiation is complex, involving the allocation of price, volume, and risks. This thesis develops a cooperative game-theoretic framework to model bilateral PPA negotiations between a renewable generator and a consumer, using Nash bargaining theory to determine Pareto-efficient strike prices and contract volumes for both baseload and pay-as-produced (PAP) structures. The framework extends existing models by incorporating stochastic production, price volatility, capture rate effects, Conditional Value-at-Risk–based risk aversion, bargaining power asymmetries, and asymmetry of information.

Monte Carlo simulations of prices and production underpin a synthetic case study of a 30 MW wind farm negotiating with industrial consumers in Denmark's DK-2 market (2020–2025 data). Results show that Nash bargaining solutions deviate systematically from expected-price benchmarks, as risk aversion drives strike prices towards the least risk-averse party in negotiations, making the agreement proportionally fair. The bargaining outcomes highlight the difference in risk allocation between baseload and pay-as-produced contracts by showcasing different sizes in bargaining sets in different risk aversion combinations. Under symmetric expectations, price changes are the primary driver of baseload strike prices, while capture rate changes dominate PAP pricing. Under asymmetric expectations, the strike-price variation increases substantially, amplifying the impact of information divergence when negotiating. Additionally, the party with greater risk aversion experiences that the lesser risk-averse player's expectation dictates a negotiation.

The thesis makes four contributions: (1) extending barter-set theory to stochastic production and capture rates in baseload PPAs; (2) the first formal Nash-bargaining formulation for PAP contracts; (3) integrating risk aversion, bargaining power, and information asymmetry into a unified stochastic optimization model; and (4) conducting elasticity-based sensitivity analysis across key market parameters. These results provide both theoretical advances in cooperative bargaining under uncertainty and practical guidance for structuring equitable PPAs in renewable energy markets.

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Nomenclature

Sets and Indices

Symbol	Description	Unit
T	Set of time periods (years)	—
t	Time period index, $t \in T$	—
W	Set of scenarios	—
ω	Scenario index, $\omega \in W$	—
G	Generator (seller)	—
L	Load (buyer)	—
i	Player index, $i \in \{G, L\}$	—

Decision Variables

Symbol	Description	Unit
M	Contract volume (baseload)	MW
γ	Production share (pay-as-produced)	%
S	Strike price	EUR/MWh
ζ^i	Value-at-Risk for player i	EUR
η_ω^i	CVaR shortfall for player i in scenario ω	EUR
(S^*, M^*, γ^*)	Equilibrium contract terms	various

Parameters

Economic Parameters

Symbol	Description	Unit
$\lambda_{t,\omega}$	Electricity price in period t , scenario ω	EUR/MWh
$P_{t,\omega}^G$	Wind production in period t , scenario ω	MWh
$P_{t,\omega}^L$	Load consumption in period t , scenario ω	MWh
$CR_{t,\omega}^G$	Generator capture rate in period t , scenario ω	—
$CR_{t,\omega}^L$	Load capture rate in period t , scenario ω	—
π_ω	Probability of scenario ω	—

Bounds and Limits

Symbol	Description	Unit
M^R	Minimum contract volume	MW
M^U	Maximum contract volume	MW
γ^R	Minimum production share	%
γ^U	Maximum production share	%
S^R	Minimum strike price	EUR/MWh
S^U	Maximum strike price	EUR/MWh
S^{R*}	Viable minimum strike price	EUR/MWh
S^{U*}	Viable maximum strike price	EUR/MWh

Bias Parameters (Asymmetric Information)

Symbol	Description	Unit
K_λ^i	Price expectation bias for player i	—
K_P^i	Production expectation bias for player i	—
$\lambda_{t,\omega}^i$	Biased price realization for player i	EUR/MWh
$P_{t,\omega}^{G,i}$	Biased production realization for player i	MWh

Utility and Game Theory

Utility Functions

Symbol	Description	Unit
u_i	Utility of player i	EUR
d_i	Disagreement point of player i	EUR
π_i	Revenue function of player i	EUR
A_i	Risk aversion parameter for player i	—
α	CVaR confidence level	—

Nash Bargaining

Symbol	Description	Unit
τ	Bargaining power parameter	—
τ_G	Generator bargaining power	—
τ_L	Load bargaining power	—
U	Utility possibility set	—
B	Bargaining set	—

Risk Measures

Symbol	Description	Unit
$\text{CVaR}_\alpha(\cdot)$	Conditional Value-at-Risk at confidence level α	EUR
$\text{VaR}_\alpha(\cdot)$	Value-at-Risk at confidence level α	EUR
$E[\cdot]$	Expected value operator	—
$\text{Var}[\cdot]$	Variance operator	—
ρ	Correlation coefficient	—

Stochastic Process Parameters

Ornstein-Uhlenbeck Process

Symbol	Description	Unit
X_t	Stochastic process at time t	—
θ	Mean reversion rate	—
$\mu(t)$	Time-varying long-term mean	—
μ_0	Initial mean level	—
μ_1	Annual trend parameter	—
σ	Volatility parameter	—
ϵ_t	Random increment	—

Abbreviations

Abbreviation	Full Form
PPA	Power Purchase Agreement
BL	Baseload
PAP	Pay-as-Produced
CVaR	Conditional Value-at-Risk
VaR	Value-at-Risk
NBS	Nash Bargaining Solution
OU	Ornstein-Uhlenbeck
CfD	Contract for Difference
GoO	Guarantee of Origin
TSO	Transmission System Operator
DSO	Distribution System Operator
VRES	Variable Renewable Energy Sources
NPV	Net Present Value
CAPEX	Capital Expenditure
COD	Commercial Operation Date

Abbreviation	Full Form
LCOE	Levelized Cost of Electricity

1 Introduction

1.1 Context and motivation: Merchant Exposure of VRES in Electricity Markets

The renewable energy market in Denmark has and is undergoing a substantial transformation, notably by attempting to shift [1] away from traditional government-supported support schemes, such as feed-in tariffs and contracts for difference (CfDs), and increasingly embracing full merchant exposure. This evolution places significant reliance on market revenues and long-term Power Purchase Agreements (PPAs), pivotal for securing financing and mitigating investment risk. Without these former state support mechanisms, renewable energy projects must now secure income through market revenues or corporate PPAs to maintain economic viability. Such agreements provide essential financial stability, enabling developers to attract funding from banks by guaranteeing stable revenue streams over extended periods [2].

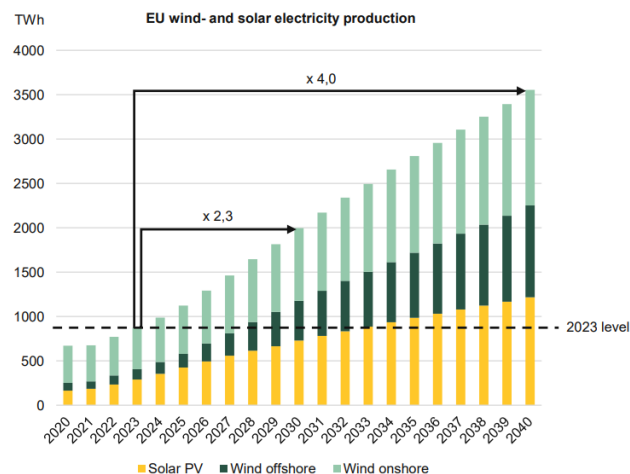


Figure 1.1: Expected growth of renewable energy in Europe [3]

In the day-ahead market, the increased penetration of variable renewable energy sources (VRES), particularly wind and solar, has introduced significant challenges. These challenges won't dissipate in the future with an expected growth of 2.3 times the renewable energy production in Europe between 2023 and 2030 1.1. Price cannibalization due to VRES growth is becoming increasingly evident, whereby renewable producers depress market prices during periods of high generation. The capture prices for wind energy specifically in Denmark (DK1 and DK2) show a widening gap compared to baseload prices, i.e., average electricity price, reflecting this price cannibalization [3]. Capture

rates, indicating the average price earned relative to the market's average, are forecast to decline substantially as a product of this phenomenon. This situation presents risks for standalone VRES installations in negotiations for PPAs as lower capture prices also have an adverse effect on the pricing. Consequently, the policy goals of increased VRES share are put at risk.

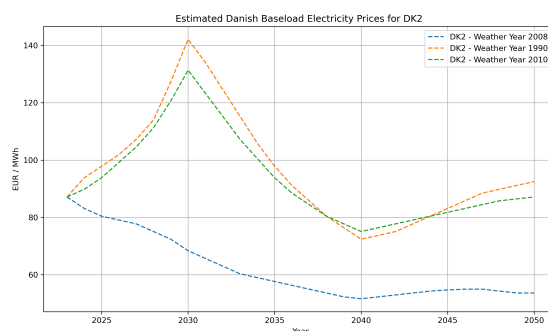


Figure 1.2: Prediction for baseload prices in DK2 under 2023 basis of assumption[4]

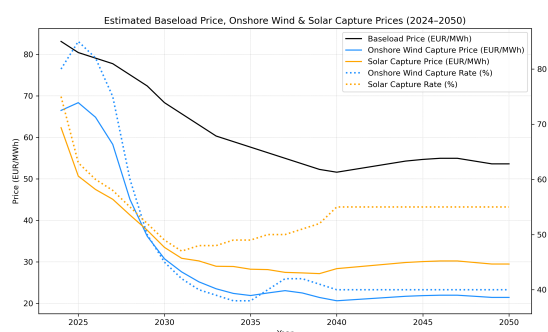


Figure 1.3: Estimated Capture prices in DK2 [3],[4]

1.1.1 Long Term Electricity Pricing

Energinet's projections highlight a notable decline in Denmark's baseload electricity prices from around 80 EUR/MWh in 2023 to below 50 EUR/MWh by 2050 as observed in 1.2, particularly pronounced through the transition phase toward 2030 due to expected renewable deployment and reductions in conventional generation capacity [4]. The expected increase in electrification across society, through industries, transportation, and heating means, will increase demand, and prices will also increase during periods of renewable energy shortages. This is also exemplified in 1.2, where different weather years are considered along with the potential effects of variable weather conditions on electricity prices. Furthermore, as renewables often, cannot schedule their production and rely on the weather, the production average electricity price, capture price, will differ from the average electricity price. This has been a particular focus in recent years, placing several solar projects at risk due to depressed electricity prices resulting from the increasing share of solar energy in the grid [5]. In Denmark, solar capture prices

are predicted to continue declining from current levels of around 75% to near 55% by 2030 [3] and onshore is expected to fall even more dramatically from around 80% to 40% in 2040. All in all, electricity prices are projected to become increasingly uncertain, as the Danish & European societies strives towards carbon neutrality in its electricity generation. These developments underscore the growing need for risk management and flexible contractual structures to maintain the long-term financial attractiveness of renewable energy investments. In this context, there is an increasing need for financial mechanism to hedge against risk such as term contracts and PPAs, ensuring projects surpass their Levelized Cost of Electricity (LCOE) and are bankable [6].

1.2 Power Purchase Agreements (PPAs): Challenges and Opportunities

PPAs are bilateral contracts between two parties, typically the buyer (the consumer) and the seller (the electricity producer), wherein the price of electricity is locked in and hedged for both parties in tenors of 5-20 years. Electricity generators enter into such agreements to ensure stable revenue streams, rather than depending on the often volatile wholesale electricity market. This is especially important for renewable energy generators whose revenue is subject to fluctuations in both production and market prices. Revenue uncertainty significantly impacts bankability for renewable energy projects which are characterized by their high capital expenditures (CAPEX). Thus, PPAs serve as a critical mechanism for securing financing and driving additionality in new renewable energy development [2]. On the other side, the buyer's, PPAs act as a hedge against rising electricity prices and serve as a tool to obtain renewable energy certificates [7]. These certificates enable consumers to claim the use of renewable electricity in their carbon footprint which is often linked to environmental, social, and governance (ESG) goals. In recent years, PPAs have gained traction among corporations and heavy industries as a price hedging instrument and to meet sustainability targets [5].

Following the energy crisis triggered by post-pandemic demand shocks and geopolitical tensions, particularly the Russian invasion of Ukraine, there has been a global re-assessment of energy security. The European Commission responded by launching the REPowerEU plan to eliminate the EU's dependence on Russian fossil fuels before 2030 [8]. A key measure within this plan has been accelerating and streamlining authorization processes for PPAs, aiming to remove bureaucratic barriers and support the deployment of renewable energy. As such, PPAs are increasingly recognized as corporate procurement tools and as instruments of public energy policy in enhancing supply and fostering investment in RES [8].

Several PPA structures have been developed in order to satisfy different risk profiles and goals between seller and buyers [9, 10], and further innovation is at the same time happening with the acceleration of integration of BESS projects seen across Europe [5]. The two most common types of PPAs are denoted as the 'Pay as Produced' (PaP) and 'Baseload' (BL), and will be the focus of this thesis. PaP denotes a scheme in

which the producer is paid for a fixed fraction or all of the production at a fixed price. In the baseload the producer has to supply a fixed amount of power at each settlement period, with failure to provide it resulting in a buy-back at the market price. PPAs are a matter of risk mitigation and transferring them between the buyer and seller, making their negotiation complicated as both parties want to ensure a profitable outcome.

PPA Negotiation

Settling on the pricing scheme of Power Purchase Agreements (PPAs) constitutes a critical aspect, directly influencing their attractiveness and feasibility for both energy producers and consumers, and is a lengthy process that typically takes 2-12 months [10]. The pricing mechanisms used in PPAs reflect the broader market conditions, future expected market dynamics, and risk considerations inherent in renewable energy projects.

In general, a fixed-price arrangement is established in which electricity is purchased at a constant price over the contract duration, providing certainty and enabling risk management for both parties. There does, however, exist a variety of alternative pricing structures, including inflation indexing, hybrid structures, and cap-and-floor mechanisms, designed to align further pricing within the risk appetite of both buyer and seller [9][10]. The thesis focuses solely on a fixed price settlement scheme where a price is settled upon and will remain fixed across the whole duration of the tenor.

Strike Price Negotiation

Settling on the strike price is a non-trivial part of the PPA negotiation, and often requires several rounds of back-and-forth bids between both parties for these to be satisfied. The parties will bring their expectations about the market outlook, including future electricity prices, as well as forecasts concerning expected production and consumption. For the price forecasts, the forward curve serves as a general benchmark. The forward curve represents the market's expectations of future electricity prices and is a key input in the pricing and valuation of PPAs. Derived from traded futures or bilateral forwards, these curves indicate what the market would pay for electricity in future delivery periods. While estimating accurate forward curves is a complex process in itself, it falls outside the scope of this thesis.

Contract Amount Negotiation

The contract volume usually aligns with the expected production capacity of a given plant. Nevertheless, even with accurate production forecasts, negotiation participants remain concerned not only with expected net earnings but also with risk, the potential for actual earnings to deviate adversely from expectations. Thus, negotiations inherently aim to balance expected profitability against acceptable risk levels, tailored to each participant's risk preferences.

This naturally leads to the question of how the division of risks and benefits in PPA negotiations can be represented in a formal model, a gap that Nash bargaining theory is well-suited to address in the context of PPA negotiations.

1.3 Research Questions

Building on the identified challenges and opportunities, this thesis aims to answer the following research question:

How to predict an (equitable/fair) negotiation outcome in pricing and contract volume in PPAs?

To address this research question, we focus on developing an analytical framework for modeling PPA negotiations. Providing realistic representations of the stakeholders and solution methods to compute the resulting negotiated contract prices and volumes, assuming both are seeking a successful negotiation. In particular, we aim to answer the following sub-questions:

1. How can a 'fair' strike price and contract volume for Baseload and Pay as Produced PPAs be determined?
2. In what ways do differences in risk aversion and bargaining power of the generator and load influence the negotiated contract outcomes?
3. How do key stochastic parameters, such as production variability, price expectations, and capture rates, affect the negotiated contract outcomes under both symmetric and asymmetric information settings?

1.4 Literature Review and Remaining Gaps

1.4.1 PPA Design and Pricing: Limitations of Existing Models

Methods in the literature for estimating optimal PPAs include: Gabrielli, Aboutaleb, and Sansavini [11] is an investment decision problem optimizing the installed capacity of a portfolio of geographically and technologically diverse assets formulated as a two-stage stochastic program. The model maximises expected cash-flow from the PPAs while minimising downside exposure via Conditional Value-at-Risk, tracing an efficient frontier of risk–return trade-offs. Their case study shows that diversification across solar and wind can cut financial risk with only a fractional loss in expected returns. In Mendicino et al. [12] the authors formulate using an extensive expression of the LCOE combining both renewable-generation costs and the corporate off-taker's "green-branding" benefit as a methodology of determining the strike price and contractual length.

These models provide a starting point, but they lack an actionable model that represents how the negotiation process involves stakeholders with specific preferences and strategic decision-making that deviates from a well-intentioned model. In particular, Gabrielli, Aboutaleb, and Sansavini [11] assumes a strike price has been agreed upon, and Mendicino et al. [12] assumes a predetermined fraction of production has been agreed upon that will go to the PPA, while also assuming that using LCOE models is a method to determine a fair price between both parties.

1.4.2 Modeling Strategic Decision-Making

One framework for analyzing strategic decision-making and predicting outcomes of situations between parties is game theory, often synonymously associated with John Nash's 'Nash Equilibrium' in non-cooperative game theory. It is a mathematical approach to analyzing these systems and quantifying the impact of a decision maker's actions, while also considering how one's own decision affects that of another decision maker. It can be explained more plainly as follows: Game theory is a theory of investigating the mutual influence of strategies [13]. While Nash's popularization of the Nash Equilibrium was introduced in 1951 [14], the general formal definition of game theory was given by John von Neumann and Oskar Morgenstern in 1944, who also introduced the concept of cooperative game theory, which was also detailed in Nash's first paper [15]. The primary distinction between cooperative and non-cooperative game theory lies in agreements and enforcement mechanisms in resource allocation problems.

In cooperative games, players can form binding contracts and coalitions, with enforceable agreements that focus on distributing gains through cooperation among the participants. This approach is beneficial for analyzing negotiations, coalition formation, and issues such as fair surplus allocation [13]. An industry that has researched the implementation of cooperative game theory extensively is telecommunications, in allocating flows within the limited bandwidth accessible [16]. Cooperative theory emphasizes collaboration and negotiated outcomes, making it an interesting framework to utilize within PPAs. A methodology utilizing cooperative game theory is the Nash Bargaining Solution proposed in Nash's first seminal paper Nash [15].

Cooperative Game Theory in Energy Systems

Cooperative game theory also has its applications in energy markets, which have grown substantially in size and complexity due to market liberalizations and the increasing complexity between agents acting with game-theoretic approaches. Recent research has applied both cooperative and non-cooperative frameworks to energy trading problems. Tushar et al. [17] show that a cooperative approach in peer-to-peer energy trading can achieve better outcomes compared to traditional non-cooperative models. Their analysis shows that cooperation among distributed energy sources allows better utilization of their resources and higher social welfare (system efficiency). Similarly, Fan, Ai, and Piao [18] presents a Nash bargaining cooperative framework for energy trading between demand response aggregators and distribution companies, showing that cooperative bargaining enables joint determination of trade volumes and payments while maximizing social welfare. The increase in utility using cooperative approaches is also supported by Zhang et al. [19] who model an integrated electricity-heat-hydrogen system using Nash bargaining theory. Their results demonstrate that cooperative bargaining strategies achieve higher economic benefits and resource utilization compared to non-cooperative equivalents. Hu et al. [20] restructures an energy trading model of an aggregator of several smaller renewable assets and flexible loads as an asymmetric Nash-bargaining game, posing the real-world fact that parties do not contribute equally. Bargaining power is tied to their individual contribution to the value function, as such

their payoff automatically reflects the actual value added. This contribution-based design both enlarges the total cooperation surplus and provides a higher allocation to the most helpful participants, showing how asymmetric bargaining can deliver efficiency and 'fairness'.

Cooperative Game Theory in PPAs

For bilateral PPA negotiations, cooperative game theory and especially Nash bargaining theory offers a suitable framework. Yu, Tesfatsion, and Liu [21] establish the first direct application of Nash bargaining to financial bilateral contracts in wholesale electricity markets. Yu, Tesfatsion, and Liu [21] presents a Nash bargaining model for negotiating bilateral contracts between a deterministic baseload producing generator and a load-serving entity, incorporating risk preferences through Conditional Value at Risk and information asymmetries through bias in the expectation of prices. They formalize the "Barter Set," which is the set of feasible contract outcomes, laying the theoretical foundation upon which this thesis builds. Their work shows how negotiated outcomes systematically depend on relative risk aversion and biases in price expectations.

More contemporary work has focused on including flexibility in Nash bargaining. Fina, Heidrich, and Krogstie [22] focuses on energy communities with photovoltaic generation and battery systems, showing how Nash bargaining can determine optimal fixed prices while accounting for production uncertainty and the addition of risk components in expectation. Their framework combines long-term production forecasting with historical price data to optimize both buyer revenues and costs utilizing BESS in only fulfilling contractual obligations.

Kandpal, Backe, and Granado [23] develops a framework for negotiating 24/7 PPAs that incorporates demand flexibility through battery storage. Their approach combines Nash bargaining theory with machine learning methods, demonstrating how demand flexibility can reduce strike prices while enhancing demand-generation matching.

Chen and Hou [24] develops a Nash bargaining solution in which several buyers bargain with a developer in signing a PPA. They model the bargaining process as a separate negotiation, agent-based negotiation, and joint negotiation, deriving closed form solution for the contract price, size and welfare split. Their simulations show that equal bargaining power and aligned risk preferences are critical for collective deals to be successful.

Garcia et al. [25] and Garcia et al. [26] proposes an alternative cooperative bargaining method in bilateral contracts between a risk neutral seller and buyer using the Raiffa-Kalai-Smorodinsky Bargaining Solution. The approach is compared to the Nash Bargaining Solution, showcasing that by changing the axiom of 'Independence of Irrelevant Alternatives' used in Nash bargaining to the monotonicity axiom improves revenues for both players considering a risk-neutral utility function.

In summary, cooperative game theory, and in particular Nash bargaining, provides a promising mathematical framework to model PPA negotiations, providing an analytical solution method to derive fair and mutually beneficial contract outcomes based on

changing market conditions such as prices, and production. However, current studies lack accounting for the interplay between multiple stochastic parameters and the effect of risk-aversion that exist between the parties in determining the optimal strike price and contract volume. Additionally, despite pay as produced having been the most prevalent PPA structure in practice [5], no existing research has combined Nash bargaining with PAP contracts. The previous approaches [21, 27, 23, 25] have focused on baseload or shaped contracts, leaving a gap with pay as produced. Current literature predominantly also focuses on short-term (hourly) optimization, with little research examining an annual granularity over long-term horizons (10-20 years).

1.5 Contributions

This thesis makes the following contributions to the modeling and analysis of Power Purchase Agreement (PPA) negotiations using cooperative game theory:

1. **Extension of barter set frame to stochastic production and capture rates in baseload PPAs:** Building upon the deterministic production formulation of Yu, Tesfatsion, and Liu [21], a stochastic production, price, and capture rate formulation is made of the barter set considering a modified risk-return function. This allows the Nash bargaining solution to explicitly account for production variability, price cannibalization effects, and risk aversion.
2. **Formalization of the barter set for pay as produced contracts:** While PAP structures are historically the most prevalent corporate PPA type in practice, they have not previously been modeled within a Nash bargaining framework. The barter set, viable strike price bounds, and the conditions of the PAP considering stochastic production and prices are derived.
3. **Integration of risk aversion, bargaining power, and asymmetric information into a stochastic optimization model:** A mathematical framework combining Nash bargaining, risk-return utility functions, and asymmetry of information is developed. This framework captures the joint effects of risk aversion, negotiation power asymmetry, and biased expectations on negotiated strike prices and contract volumes.
4. **Comprehensive sensitivity analysis across multiple parameters:** Quantifying the elasticity of optimal contract terms with respect to changes in price levels, production, consumption, and capture rates under both symmetric and asymmetric expectations. This analysis reveals which parameters most influence bargaining outcomes, providing insights into how parameters affect contract design and change negotiation strategies.

Collectively, these contributions advance the application of cooperative game theory in electricity markets by 1) expanding the analytical scope of the barter set, 2) introducing a stochastic, risk-aware formulation for prevalent PPA structures, and 3) linking theoretical negotiation outcomes to practical sensitivities relevant for industry decision-making.

1.6 Assumptions

The main assumptions the thesis is based on are listed below. These will affect the results to some extent, and the possible effects of the assumptions will be discussed in later sections.

- The market structure in use will be the Nordic structure, specifically DK-2 is the chosen region, and historical data will be sampled from this region
- The Day-Ahead (Wholesale Market) is the only merchant market considered in this thesis. Intraday and balancing market is not considered, as such modelling cost/benefits that are incurred in these markets are not accounted for.
- The producer is assumed always to have access to bid and sell in the market across the entire time horizon and scenarios. No curtailment from the distribution system operator (DSO) or the transmission system operator (TSO) will occur.
- The renewable energy producer is assumed to have a marginal cost of 0 and to be a pure-price taker in the market.
- Granularity of the optimization is assumed to be yearly, as such concepts like seasonality or intermediate deviations leading to systematic arbitrage is not possible.
- Discount rate and estimated inflation are not accounted for in the optimization or future revenues.

1.7 Thesis Outline

The remainder of this thesis is organized as follows. Chapter 2 reviews the structure, valuation, and negotiation of PPAs. Chapter 3 outlines the fundamentals of Nash bargaining theory, while Chapter 4 applies this framework to PPA negotiations and formalizes the barter set for baseload and pay as produced contracts. Chapter 5 presents the case study and stochastic optimization framework, and Chapter 6 describes the datasets and base case configuration. Chapter 7 reports the base case and experimental results, and Chapter 8 concludes with key findings and recommendations for future research.

2 Review of the Power Purchase Agreement Market

With the power market in recent years involving significantly more renewable energy generation, price cannibalization has become a significant issue in Europe, affecting not only assets operating purely in the merchant market but also the pricing of PPAs. Pay as produced contracts are priced lower as the capture rate declines since the buyer's hedge requirements diminish. For the Baseload contract, the seller's risk profile worsens with declining capture rates and increasing volatility. This occurs because the hours in which the seller must buy back missing production are generally also those with high electricity prices, given that renewable energy production is lower [28].

The European Power Purchase Agreement market has developed significantly over the last decade with the rapid expansion of renewable energy leading to an increasingly complex environment of new structures and mechanisms that meet the risk needs of both buyers and sellers [5]. This chapter focuses on describing the content of PPAs and also their ongoing evolution. These changes ultimately impact how PPAs are negotiated. The beginning will focus on the elements of PPAs and the ongoing developments in the industry, including new structures and environmental aspects. Then, the industry methodology for valuing a PPA, considering the inherent risks associated with it and the general negotiation procedure between a seller and an off-taker, is explained, providing context for the usage of cooperative game theory in later chapters.

2.1 Overview of the standard PPA structures

PPAs can be structured in several ways according to various attributes, the key ones being: generation technology type, physicality of delivery, pricing structure, tenor, and the counterparty (off-taker). Other attributes include balancing, energy certificates, insurance and the commencement of delivery (COD) of the contract [10, 29].

In general, PPAs can be split into two categories: *Physical PPAs* and *Financial/Virtual PPAs* [2]. Physical PPAs refer to the production that is delivered to the offtaker. It can be split into two further categories: on-site and off-site. On-site projects are smaller projects, such as PV on rooftops, where the point of connection between generation and consumption occurs at the same location. If the PPA is off-site, the generation is located at some physical distance removed from the consumption and needs to be transmitted across transmission lines. Generally, the PPA also requires that the contract take place within the same geographic area. In order for the off-taker to 'attain' the produced generation from the off-site PPA, the PPA needs to be 'sleeved' by signing a contract with an energy provider who buys the production on the market and sells it back to the offtaker, and then balance potential deficits or surpluses that happen in the market [10]. A virtual PPA, on the other hand, is a pure financial product where

there is no physical delivery of electricity, and can also be cross-border [10]. The PPA functions as a 'Contract for Difference', where parties agree on a fixed strike price. The energy produced from the project is sold on the wholesale market, and depending on the difference between the agreed-upon strike price and the market price, cash flows are exchanged between both parties. A virtual PPA enables the offtaker to obtain energy certificates, similar to physical ones, but from different projects across borders. The virtual PPA serves as a price hedge for the seller and a source of certificates for the buyer[8]. A key contention in negotiating such a contract is determining which wholesale price to use when both parties are not located within the same geographical region. Still, delinking the physicality of delivery from production increases the potential pool of offtakers, which in turn helps the seller in achieving financial close [30]. The offtaker, or counterparty, in most PPA negotiations can be split into two categories: Corporations & Industries and utilities, often also called corporate or utility PPA. The main purpose of the utility PPA is hedging price volatility, whereas the corporate focus is more on the energy certificates for their environmental goals. Utility PPAs were previously the main offtaker, but with the advent of more ambitious carbon reduction goals, corporations are becoming the primary counterparty. Previously, corporate offtakers were only one entity ie one company negotiating a PPA for themselves, however multi-buyer PPAs are emerging [2][5]. These can include corporations with lower overall energy demand and lower credit ratings. They either sign separate PPAs with the same renewable project or buy in as a conglomerate. This way, smaller companies lacking the conventionally required collateral can participate in the PPA market [10].

The technology type of a PPA constitutes a fundamental design element, as it largely determines both production stability and generation profile. Traditionally, PPAs are categorized by a single technology type, such as Solar PPAs, Wind PPAs, or biomass and hydropower. However, with the rise of price cannibalization in electricity markets, multi-technology PPAs have emerged as a means of mitigating revenue volatility [31]. By combining multiple technologies into a generation portfolio, the resulting production profile can be shaped to more closely align with the buyer's consumption pattern [10].

2.1.1 Delivery Schemes

The delivery scheme specifies both the contract's delivery schedule and the corresponding electricity volume. The most common and straightforward arrangement is the pay as produced (PAP) scheme, in which a predetermined fraction of generation is delivered as it is produced and paid for at a fixed price. Under PAP, the buyer bears the majority of the risk: if the seller's production does not meet consumption needs, the buyer must procure the shortfall at the prevailing market price. Conversely, if production exceeds consumption, the buyer is still obliged to purchase the excess and resell it at market prices [10]. When production occurs during hours with market prices below the agreed rate, the buyer effectively pays a premium for electricity. A sub-variant, the As-Forecast scheme, requires the buyer to purchase a forecasted quantity determined before generation and communicated in advance. This arrangement allows the buyer to adapt operational processes to the expected delivery, potentially improving efficiency through

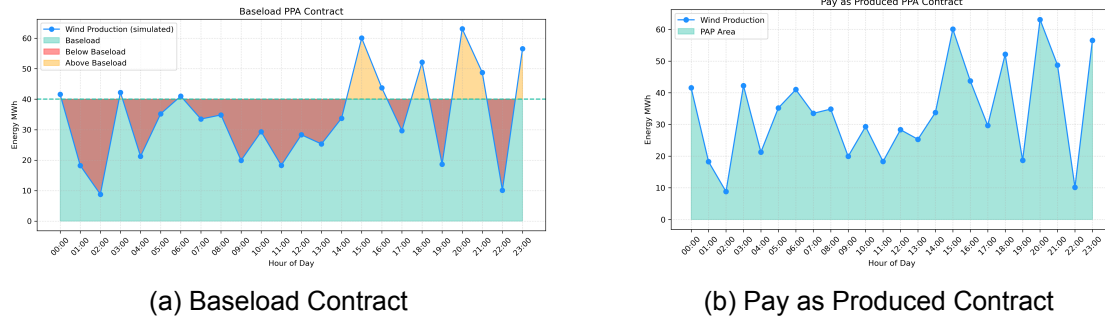


Figure 2.1: Delivery Profiles of Baseload and Pay as Produced

demand response measures [10].

Another common delivery scheme is the Baseload (BL) contract, in which a fixed contract volume is agreed upon and must be supplied every hour. If the seller fails to meet this obligation, they must procure the shortfall at the prevailing market price. While baseload contracts are often specified on an annual basis, variants exist that define volumes on a monthly scale to accommodate seasonal variations in consumption. A related variant is the Shaped PPA, where the seller delivers a predefined hourly volume profile rather than a constant amount. This structure shifts risk toward the seller, who must match the agreed shape and buy back any shortfall at market prices. However, depending on the specific profile, the buyer may also bear significant risk [10]. In contrast, the Pay-as-Consumed scheme operates as the inverse of the pay as produced model. Here, the buyer purchases only the amount of electricity they consume, irrespective of the seller's generation. This arrangement requires the seller to balance generation with consumption, a task typically feasible only when supported by a diversified portfolio of generation assets. Consequently, the Pay-as-Consumed scheme places the majority of balancing and market risk on the seller [10].

2.1.2 Pricing Structure

The pricing structure refers to how the electricity is priced through the contract. The pricing structure can be categorized into fixed or variable electricity prices over the tenor. The fixed price is the simplest structure, where the price remains fixed for the entire duration of the contract. A variation of the fixed-price 'escalation' PPA exists, where the initial price either increases or decreases in response to inflation or a fixed percentage [10]. By adjusting to the nominal price, the developers are positively affected as it benefits their bankability, where net present value (NPV) matters substantially.

The other scheme, 'Variable Pricing', tracks and follows the market price, though often at a discount. By following the market price, the seller might benefit if prices are rising, but also be at risk if prices fall. Variable pricing schemes can include a price 'floor', 'cap' or a 'collar' (both combined). The floor guarantees that the price cannot fall below a specific value; conversely, a 'cap' ensures that the price cannot exceed a value, provid-

ing security within the range of what the PPA price might be. This structure functions much like a CfD. A 'reverse' collar operates the opposite way, if the price is between the strike price and the floor, then the buyer has to remunerate the seller for the difference. Conversely, if the price is between strike and cap, the seller has to pay the buyer [10]. Variable pricing is less common for corporate PPAs and is instead meant for buyers with higher electricity operating costs. Pricing structures with caps, floors, collars, and claw-back arrangements are less common for corporate PPAs than their fixed counterpart and provide less price certainty, which might be important in securing financing for a new renewable energy asset [9].

2.1.3 Duration and Commencement Date

A PPA can be signed for both short-term periods of 2–5 years or long-term periods of 5–20 years. Generally, shorter PPAs command higher prices due to increased exposure to price and volume risks. Long-term PPAs, by contrast, are often pursued to secure project financing, as stable and predictable revenue streams are essential for obtaining investment capital. In this context, the seller bears the implicit burden of securing a PPA to ensure bankability. In contrast, the offtaker typically has less need to lock in costs two decades in advance. While short-term markets can be volatile and uncertain, these risks are mitigated through long-term contracts, which also reduce the relative financial impact of a single poor weather year. For these reasons, most PPAs used to finance new projects fall within the 10–20 year range [10]. The Commencement Date (COD) marks the start of the contract, typically aligning with the beginning of production.

2.1.4 Insurance and Penalties

Insurance

Insurance requirements are a standard feature in PPAs, designed to safeguard both the seller and the off-taker against unforeseen events that could impact performance and financial stability. PPAs can feature different forms of insurance to cover unforeseen events, such as damages to the renewable asset [10].

Penalties

PPA contracts frequently incorporate penalty clauses to address situations where the seller fails to deliver the contracted amount of energy [12]. Such penalties may be enforced regardless of whether the shortfall arises from curtailment by the transmission system operator (TSO) and must therefore be considered during contract structuring [32]. Additional penalties can apply if the agreed Commencement Date (COD) is not met due to project delays [9].

In this context, force majeure clauses are critical for addressing events beyond the reasonable control of the parties, such as extreme weather events or armed conflict, that may prevent fulfillment of contractual obligations.

2.1.5 Trading of PPAs

After a PPA is signed, the offtaker or asset owner typically needs to hedge the price exposure using traded contracts on an exchange such as the European Energy Exchange (EEX) or Nord Pool. However, there is no liquid market for contracts extending 10 years

or more. EEX has started a 10-year future market with increasing liquidity [33]. Most trading volume, and therefore liquidity, is concentrated in contracts with terms of 1 to 3 years ahead. A market is considered *liquid* if large trades can be made without significantly moving the price, and if it is easy to find a buyer or seller. In contrast, an *illiquid* market means that even relatively small trades can move the price, and executing a large trade may require paying a premium due to limited participation. To manage this limitation, traders use a strategy known as *stack-and-roll*. It involves hedging the total volume (exposure) by stacking the entirety into blocks and then continuously rolling the remainder into the following year. For instance, a 20 MW PPA with a 10-year term corresponds to a total exposure of 200 MW in one-year contract equivalents. In practice, it may only be feasible to hedge up to two years ahead, which would involve splitting the exposure into two 100 MW blocks using futures. As delivery commences for the first block, part of the volume (e.g., 20 MW) is delivered, while the remaining exposure (80 MW) is *rolled* forward into year 3, establishing a new hedge for the smaller residual volume. This rolling process is repeated annually until year 10, when the final exposure is delivered. This approach is illustrated in Figure 2.2.

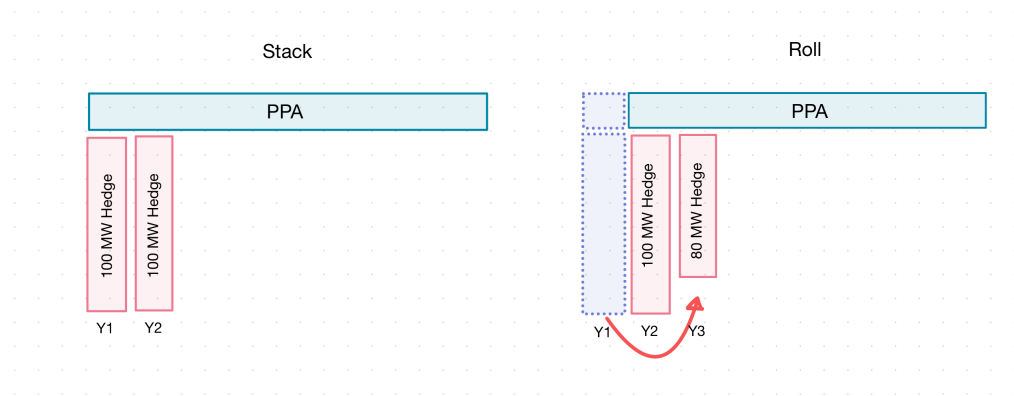


Figure 2.2: Stack'n'Rolling Hedging visualized

The difference between a *forward* and a *future* lies in the way they are traded. Forwards are privately negotiated over-the-counter (OTC) contracts between two parties, allowing for flexible terms. Futures are standardized contracts traded on exchanges, such as EEX, with the exchange acting as the counterparty and requiring daily margin settlement. Despite these differences, both instruments are used to hedge future price risks and are settled based on the difference between the contract price and the market price at the time of delivery or maturity.

While stack-and-roll is a practical solution to hedging long-term exposure in short-term markets, it is not without limitations. The strategy assumes that the prices of short-term assets (the liquid part) will move in line with the longer-term (illiquid parts) of the forward curve. In reality, this is rarely the case. The short end of the power forward curve tends to be more volatile and reacts more strongly to market events, such as weather or

outages. The long end moves more gradually and reflects the long-term expectations of supply/demand, policy goals, and the trajectory of future costs. As a result, the two segments of the curve are not perfectly correlated [34].

This mismatch introduces what is known as *basis risk*, meaning that the changes in the value of the hedge might not fully match the changes in the value of the actual PPA exposure. Because short-term contracts tend to fluctuate more, the volume hedged may need to be adjusted to avoid covering too much or too little risk. Additionally, every time the hedge is extended into a new year, practical costs are incurred, such as less favorable pricing or fewer available trading options. These factors, taken together, mean that stack-and-roll cannot provide a perfect hedge throughout the entire lifetime of a long-term PPA.

2.1.6 Emerging Structures

With the challenges of price volatility, cannibalization risk, and intermittent renewable production outlined in the introduction, a range of innovative 'hybrid' PPA structures have emerged in recent years to address these market needs [28]. In general, a 'hybrid' PPA is an umbrella term for co-locating the asset with storage or combining with other generation technologies. Mixed technology PPAs all share the strength of more flexibility in power deliveries, which results in delivery profiles that are more suitable for price hedging [5]. A study by Pexapark and Eurelectric found that an optimal mix of wind and solar in a PPA can lead to an hourly matching of 60% to 70% for a baseload profile. At the same time, this availability can approach 100% when battery storage is also included in the project [35]. The interest in hybrid PPA profiles can be reflected in the volumes of mixed-technology PPAs, which saw a 219% increase in volumes in 2024 compared to 2023 [5]. Another trend in PPAs includes multi-buyer PPAs, where buyers are aggregated to include corporations with smaller energy demands. By aggregating demand of smaller companies, it opens up further demand and alleviates the credit risk smaller corporations might have.

While this thesis does not explicitly analyze these new structures, their increasing adoption indicates a shift in what is becoming important in PPA negotiations.

Shaped PPA

A Shaped PPA is a power purchase agreement where the seller delivers a predefined delivery profile that mimics the offtaker's consumption pattern, such as a Fixed-Hourly Profile or Baseload structure [28]. In a hybrid setting, the technologies of RES and battery is combined in-order to fulfill that shape and is paid as a pay as produced. Pricing can be variable, depending on the hours delivered, and may also vary by month. While the payment structure is PAP, if delivery is missed, the missing position needs to be settled at market price. As such the profile and volume risk is assumed by the seller, but can use its storage to shape output accordingly, and in return receive a price premium [28].

Blended PPA

Another hybrid structure is the 'blended' renewables and storage PPA, where the renewable and storage components operate under the same pay as produced (PAP) contract.

In contrast to shaped delivery, this structure allows the buyer to optimize the whole system operation, including determining the charge and discharge schedules over the PPA's duration [28]. It provides the buyer a good amount of flexibility in choosing when they want energy. Still, it puts the seller/optimizer at risk of potential opportunity costs since they are letting go of their 'full flexibility'. A related structure is the tolling agreement, where the buyer pays a fixed rate per MW or MWh in exchange for dispatch rights over the asset under predefined terms [28].

24/7 PPA

The '24/7' structure is the ultimate 'pay as consumed' model, where the goal is to match renewable energy supply with consumption on an hourly basis, rather than on an annual or monthly average. Achieving high hourly matching typically requires portfolios that combine multiple renewable assets and storage (hybrid or blended setups). The 24/7 model not only enhances the effectiveness of PPAs as a price hedging tool but also supports corporate decarbonization goals and provides stronger claims for renewable energy consumption reporting [35, 5]. This method also puts the greatest risk on the seller, and in return, a higher price premium.

2.1.7 Environmental and Regulation

While previously PPAs were signed by utilities and a few corporations as price hedges and to address certain long-term energy costs, Environmental, Social, and Governance (ESG) goals have become an increasingly important factor in the uptake of corporate PPAs[36]. With the introduction of RE100 in 2014, a group of some of the world's largest companies has pledged to meet 100% of their energy requirements from renewable sources. The global focus on ESG reporting has made PPAs a key part of helping corporations prove their sustainability[36]. In the future, by 2029, due to the Corporate Sustainability Reporting Directive (CSRD), 50.000 companies within the EU will be required to report on the origins of their energy usage [37].

Current Framework

Carbon accounting for corporations follows the GHG protocol [38], where corporations account for their Scope 2 emissions. Scope 2 emissions refer to the emissions resulting from the generation of heat and electricity, and account for roughly one-third of the world's emissions[38]. Accordingly, emissions must be accounted for using 'location-based' and 'market-based' emissions. In location-based emissions, the consumption of electricity is multiplied by the grid average emission factor, and in market-based emissions, the emissions are directly related to the procurement of electricity [38]. Under a market-based emissions system, the procurement of electricity through a PPA can offset emissions by purchasing energy certificates. In Europe, these energy certificates go by the name 'Guarantees of Origin' (GOs) and can be bought directly (unbundled) or bundled with a PPA. Lastly, one can use a 'supplier' emission factor, relying on the fact that it is a utility with a high share of renewables.

Guarantees of Origin (GOs) constitute part of the revenue provided through a PPA, as they are certificates that can be bought as proof of renewable energy production, and it is these specific certificates that allow utilities and corporations to prove what

kind of electricity they consume [38]. Getting these certificates through a PPA serves as an additional incentive to signing, in addition to the price hedging. A Guarantee of Origin (GO) is a certificate under the Renewable Energy Directive (RED II) that attests the renewable origin of energy [39]. GOs are transferable and can be traded in the market, providing a mechanism for renewable energy producers to demonstrate the renewable origin of their energy and for consumers to choose to support renewable energy [39]. PPAs are usually bundled with GOs, which means that a transfer of GOs happen between the seller and buyer, or GOs can be bought directly from the market [38]

Another important aspect, beyond verifying the origin of production, is the concept of additionality, which is often a key corporate objective in sustainability strategies. Additionality refers to the direct impact of a given decision that would not have occurred otherwise [2]. In the context of renewable energy procurement, it denotes the expansion of renewable generation capacity through purchases that actively promote new projects, rather than relying solely on existing installations [2].

The lack of additionality is a central critique of Guarantees of Origin (GOs). While GOs enable companies to claim “emission-free” electricity consumption, the validity of this claim is subject to several nuances [40, 41]. Since GOs are accounted for on an annual basis, any generation within the year can be used to offset the total annual consumption of a corporation. Furthermore, because GOs are fully tradable across Europe, they do not need to originate from the same geographical area as the consumption. As a result, companies can claim 100% renewable electricity usage even during hours when they are in fact relying on fossil fuel generation.

While unbundled GOs remain the most popular mechanism for purchasing “green” electricity, they are also the least effective in ensuring genuine decarbonization, drawing criticism for their spatial and temporal mismatches. In response, Energinet, in collaboration with Reel and Granular Energy, launched a pilot project in Denmark for hourly GOs, referred to as Granular Guarantees of Origin (GGOs) [42]. GGOs require that renewable energy production and consumption be matched within the same hour, rather than merely on an annual basis. This approach is intended to reflect both temporal and spatial alignment, creating incentives for investments in technologies that enable a power system capable of relying entirely on renewable electricity in every hour of the year. There is a consensus that bundled GOs with PPAs create more additionality than purchasing unbundled GOs [40]. The reasoning is that PPAs are often essential for renewable energy project developers to secure project financing, and the renewable power plant likely would not be built without a PPA in place [2]

2.2 Valuation methodology of PPAs in Industry

The valuation of a PPA can be complex and time-consuming, ultimately depending on negotiations between the seller and buyer. However, several methodologies exist for determining the benchmark price of a PPA. Arguably, the most important data points are the relationship between expected PPA prices and the predicted (and actual) Levelized

Cost of Electricity (LCOE) of a project [6]. The LCOE is the average net present cost of electricity generation for a generation plant across its entire lifetime. It is primarily a concern for the developer as it often underpins the minimum return expectations for any project being constructed. Put simply, if a project were to receive its LCOE for all its generation, the net present value of the project would net zero at the end of its lifetime, serving as a minimum threshold. The expected price as previously is the expectation of the price forward curve, which depends on the changes in power generation mix, weather, and carbon prices.

2.2.1 Capture Price

The yearly capture price λ^{CR} is the annual average price the renewable project experiences during its production, as explained in Equation 2.1 with λ_t is the hourly price, P_t^G denoting the hourly production, and T is all the hours of the year. Below in Equation 2.2, the average yearly capture rate is calculated by comparing the capture price to the average baseload price.

$$\lambda^{CR} = \frac{\sum_{t \in T} \lambda_t \cdot P_t^G}{\sum_{t \in T} P_t^G} \quad (2.1)$$

$$CR = \frac{\lambda^{CR}}{\frac{\sum_{t \in T} \lambda_t}{T}} \quad (2.2)$$

Where the price data is derived from and used to calculate the capture rate may be up for negotiation, as the influence from it is substantial, currently using different data the average capture rate for wind and solar lies between 60% and 70% [5]. By predicting what the capture price will be in the long term, the *Expected Value* of the PPA can be estimated.

2.2.2 Pricing Guarantees of Origin

Part of the revenue from a PPA is derived from the bundled GOs, and is generally included in the PPA valuation. Typically, handling of the GOs involves integrating them into the PPA price as a premium, but they can also be purchased independently. Pricing of the GOs within the contract can follow either a dynamic pricing scheme, where it tracks the GO market price, or a fixed price for the certificates. The GO market has experienced significant uncertainty in its pricing due to the rapidly changing landscape of electricity demand, expansion of renewable energy, and regulatory changes. On the other hand, GO values might decrease in the long term as grid electricity approaches net-zero emissions [37]. On the other hand, stricter regulations like the CSRD on matching production and consumption could drive prices up due to the added demand.

The uncertainty of GO pricing is exemplified in 2024 where prices started at 5 EUR/MWh in the beginning of the year, but gradually decreased to around 1 EUR/MWh at the end of the year, with a trading average price between 2-3 EUR/MWh, depending on the

tenor of the future. However, GO prices are expected to increase in the future, as more companies are expected to report the imminent CRSD by 2029 [37].

When modeling future PPA revenue streams, GO sales should be included in the holistic evaluation of the contract, but this isn't easy due to GOs being a different market.

2.2.3 Risks, Costs and Premiums in PPAs

From the baseload price, multiple risks will then be factored in, typically resulting in a final PPA price below the capture price. Especially for a PAP PPA, some of these risks significantly reduce the price. In contrast, other PPAs may have fewer or more risks, depending on their type, and also include premiums that offset these risks in the final price. The final offset in what is the expected price is called the *Market Price of Risk*. The risks that are usually included in the pricing of PPAs are listed below taken from [28, 29] and visualized in Figure 2.3.

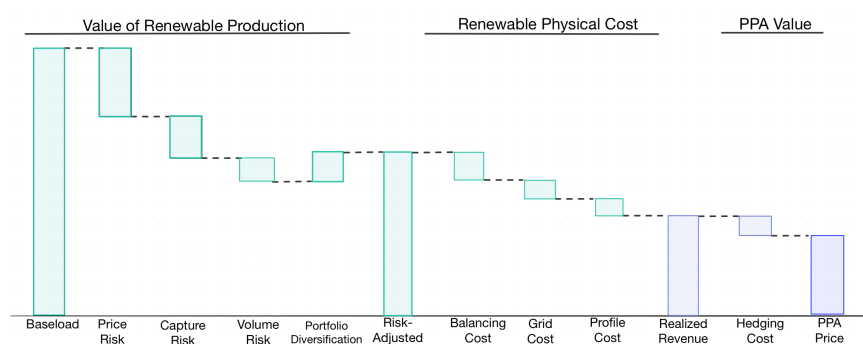


Figure 2.3: Waterfall of the decrease in price value due to risk

Price Risk

Price risk refers to the uncertainty regarding the future price of electricity during the production period. This uncertainty primarily arises from volatility in the wholesale and futures markets. The less liquid the market, the more difficult it becomes to forecast future prices accurately.

Capture Risk

Capture risk refers to the cannibalization risks in the day-ahead and the spot market that occur when an abundance of renewable energy with the same production pattern is produced, thereby reducing the electricity price and, consequently, reducing revenues.

Volume Risk

Volume risk refers to the long-term (monthly or yearly) variation between expected and actual production resulting from weather dependency that may fluctuate from year to year.

Shape Risk

Shape risk refers to the risk that arises when the actual hourly generation profile of a renewable energy asset does not align with the fixed hourly pricing structure in the PPA,

applicable to BL and shaped.

Portfolio Diversification

Portfolio diversification in the context of PPAs refers to combining multiple types of renewable assets of different technologies, geographies, and time profiles to reduce overall risk and improve revenue stability.

Balancing Cost

Balancing costs refer to the degree of imbalances derived from scheduled production and the forecast error (real production). The degree of repayment/penalty that might be incurred depends on the specific balancing market and whether it involves a portfolio or a single asset traded.

Grid Costs

Grid costs include grid usage and connection fees that must be paid to the grid operators.

Profile Costs

Profile costs arise because the output of a renewable energy plant (e.g., solar or wind) does not match the hourly shape of demand or market prices. This mismatch between when energy is generated and when it's most valuable leads to lower revenues.

Hedging Costs

Hedging Cost applies to trading the PPA, which primarily is done by trading houses and utilities. The cost accrues through 'stack'n'roll hedging' as previously described, which requires back-to-back deals, possibly resulting in losses when the price you buy and sell is not consistent, and administrative costs of these transactions.

Credit Risks

The risk is related to counterparty default or failure to fulfill contractual payment obligations. This includes the possibility that either the buyer cannot pay for delivered electricity or the seller cannot pay compensation for nondelivery.

The final price of a PPA is then the *Expected Price* offset by the *Market Price of Risk*.

$$\text{PPA Price} = \text{Expected} - \text{Market Value of Risk} \quad (2.3)$$

2.2.4 Risk differences between BL and PAP

pay as produced (PAP) and Baseload (BL) PPAs redistribute risks in fundamentally opposite directions. Under PAP structures, the buyer predominantly bears the risks associated with price, cannibalization, and balancing. Sellers in a PAP arrangement are shielded from these market risks due to the guaranteed purchase of their entire output at the agreed price, transferring these risks to the buyer. This significant shift in risk to buyers generally necessitates substantial risk discounts, making PAP agreements typically priced lower compared to other PPA structures [10, 43].

Conversely, BL contracts shift significant risk onto the seller. Sellers under BL PPAs must manage and accurately forecast production, as failure to deliver the fixed hourly volumes results in obligations to purchase the shortfall at potentially high market prices.

Particularly problematic are instances of low renewable production, typically correlated with high market price periods, exacerbating seller risks due to high imbalance costs [28]. This inherent profile or shape risk significantly impacts the financial stability of projects under baseload agreements, as sellers are obliged to actively engage in short-term markets to mitigate shortfalls, introducing additional trading and hedging costs [28, 8].

The valuation methodologies between PAP and BL contracts reflect these distinct risk allocations. PAP valuations are relatively straightforward, primarily driven by forecasting production volumes and capture rates to calculate risk-adjusted revenues [5]. Baseload valuations are inherently more complex, incorporating forward price curves, hourly capture factors, and risk-adjusted revenue projections to account for imbalances, profile costs, and hedging requirements. Co-located energy storage or hybrid renewable systems become particularly valuable in baseload arrangements, significantly reducing profile risks by aligning production profiles more closely with contractual obligations [28, 35].

Thus, while PAP PPAs offer simplicity and clear risk management advantages for producers, BL PPAs offer improved financial predictability and hedge effectiveness for buyers. The market is increasingly recognizing BL contracts' enhanced bankability due to their predictable revenue streams, though at the cost of significantly increased complexity and risk management requirements for sellers [5, 8].

2.3 PPA Negotiation Process

PPA negotiations represent complex bilateral contracting processes characterized by substantial customization requirements, extended negotiation periods, and significant economic stakes. Unlike standardized commodity transactions, these agreements require extensive bilateral coordination to align generator revenue requirements with off-taker risk tolerance and renewable energy procurement objectives [10]. This negotiation complexity, combined with the strategic interdependence between parties seeking mutually beneficial outcomes, makes cooperative game theory a suitable model of modeling.

2.3.1 PPA Negotiation and Cooperative Game Theory

The structural characteristics of PPA negotiations align with cooperative game theory's foundational assumptions, making Nash bargaining an optimal analytical framework.

First, PPAs constitute legally enforceable long-term contracts with substantial penalty mechanisms for non-performance, satisfying cooperative game theory's binding agreement requirement [15]. Contract enforcement mechanisms distinguish PPAs from informal agreements or spot market transactions, where commitment credibility is less a concern. Second, PPA negotiations create positive-sum outcomes through risk reallocation rather than zero-sum value extraction. Generators obtain revenue certainty, enabling project financing, while offtakers achieve price stability and renewable energy compliance [18]. This mutual benefit structure ensures negotiations focus on surplus

division rather than potential competitive value destruction. Third, unlike anonymous market mechanisms, PPA negotiations involve extensive bilateral communication, due diligence processes, and information sharing regarding production forecasts, financial positions, and operational constraints [10]. This direct communication facilitates the complete information assumption underlying Nash bargaining theory.

2.3.2 Nash Bargaining Solution Concept

Among cooperative solution concepts, Nash bargaining provides the most appropriate framework due to its foundations in aiming at delivering a proportional fair solution to both parties.

3 Nash Bargaining Fundamentals

This chapter introduces the game-theoretic framework for analyzing Power Purchase Agreement (PPA) negotiations in Section 4. We describe fundamental game-theoretic concepts as well as the core axioms of Nash Bargaining Solutions.

3.1 Fundamental Game-Theoretic Concepts

Game theory provides a mathematical framework for analyzing strategic interactions between rational decision-makers [13, 44]. We begin by establishing the components that characterize any strategic interaction:

- **Players:** Decision-making entities seeking to optimize their objectives
- **Strategies:** The set of feasible actions available to each player
- **Payoffs:** Functions mapping strategy combinations to numerical outcomes
- **Information Structure:** Each player's knowledge about strategies and payoffs
- **Order of play:** The sequence in which decisions are made

These components must satisfy certain conditions for game-theoretic analysis [13]:

- Finite number of players ($n \geq 2$).
- Every player has a finite set of strategies.
- Interdependent payoffs, each player's outcome depends on all players' choices
- Common knowledge of the game structure, knowing strategies of other players.
- Decision making is driven by rationality in the players' attempts to maximize their own gains while also considering their interests. The payoff from a strategy must be numerical or quantifiable.

Games can be categorized along several dimensions that influence the selection of appropriate solution concepts [44, 45]. Broadly speaking, based on the assumptions regarding player rationality, game theory can be classified into two main categories: *classical* and *evolutionary*. Classical game theory assumes players possess perfect rationality, consistently choosing options that maximize their utility. In contrast, evolutionary game theory assumes limited rationality where players adapt strategies based on experience and limited information [13].

Classical game theory can be further subdivided into *non-cooperative* and *cooperative* categories, distinguished by the presence or absence of explicit collaboration among players. Additionally, the classification considers the information available to the players: games can be of *complete information*, where players have full knowledge of others'

characteristics, or *Bayesian-incomplete information*, where such knowledge is uncertain. Lastly, the temporal structure of player decisions leads to a distinction between *static games*, in which players act simultaneously, and *dynamic games*, involving sequential decision-making over multiple stages [13, 45].

3.2 Nash Bargaining Theory

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3.2.1 Game Definition

Definition 1 (Bargaining Problem) A bargaining problem is a pair $(U, \mathbf{d})$ where:

- $U \subseteq \mathbb{R}^2$ is a compact, convex set representing all feasible utility allocations
- $\mathbf{d} = (d_1, d_2) \in U$ is the disagreement point representing each player's utility if negotiations fail

For a bargaining problem to be well-defined, the set U must satisfy the following feasibility condition:

$$U \cap \{\mathbf{x} \in \mathbb{R}^2 : x_j > d_j \text{ for } j = 1 \text{ or } j = 2\} \neq \emptyset \quad (3.1)$$

This condition ensures that at least one feasible allocation exists where some player can be made better off than at the disagreement point.

3.2.2 Pareto Frontier and Bargaining Set

The concept of Pareto efficiency is central to bargaining theory. An allocation is Pareto efficient if no alternative exists that improves one player's utility without reducing the other's.

Definition 2 (Pareto Frontier) The Pareto frontier Ω^e consists of all allocations in U where it is impossible to improve one player's utility without reducing the other's utility. Formally, an allocation $\mathbf{u} = (u_1, u_2)$ belongs to the Pareto frontier if and only if there exists no other feasible allocation $\mathbf{v} = (v_1, v_2)$ in U such that:

- $v_1 \geq u_1$ and $v_2 \geq u_2$ (both players are at least as well off)
- At least one inequality is strict (at least one player is strictly better off)

In mathematical notation, this can be written as:

$$\Omega^e = \{\mathbf{u} \in U : \nexists \mathbf{v} \in U \text{ with } \mathbf{v} \geq \mathbf{u} \text{ and } \mathbf{v} \neq \mathbf{u}\} \quad (3.2)$$

For two-player bargaining, the Pareto frontier can be represented by a function h that maps player 1's utility to the maximum achievable utility for player 2:

$$u_2 = h(u_1) \quad (3.3)$$

This function is strictly decreasing (higher u_1 implies lower u_2) and concave, reflecting the fundamental trade-off: any gain for one player must come at the expense of the other when operating on the efficient frontier. Given the feasibility constraint, we define the bargaining set:

Definition 3 (Bargaining Set) The bargaining set for problem $(U, \mathbf{d})$ is:

$$B(U, \mathbf{d}) = U \cap \{\mathbf{x} \in \mathbb{R}^2 : \mathbf{x} \geq \mathbf{d}\} \quad (3.4)$$

The set B contains all feasible agreements within the utility set U that provide each player at least their disagreement point.

3.3 The Nash Bargaining Solution

Nash [15] proposed selecting a unique point from the bargaining set by maximizing the product of gains:

Definition 4 (Nash Bargaining Solution) *The Nash bargaining solution to problem $(U, \mathbf{d})$ is:*

$$f^* = \arg \max_{(u_1, u_2) \in B} (u_1 - d_1)(u_2 - d_2) \quad (3.5)$$

The maximization problem stated above has a unique solution as the maximand $(u_G - d_G)(u_L - d_L)$, which is referred to as the Nash product. This optimization problem has a unique solution because: For any bargaining problem $(U, \mathbf{d})$ satisfying the feasibility condition, the Nash product is continuous and strictly quasi-concave on the bargaining set B .

Together, these properties ensure that there is a 'Nash Bargaining Solution' (NBS) f^* with a unique agreement $f^* \in (U, \mathbf{d}) \equiv (u_G^*, u_L^*) \in B$ that maximizes the Nash product while upholding four axioms that are deemed necessary for a 'fair' bargaining result.

Visually, the solution can be shown as seen in Figure 3.1. The blue curve is the Pareto frontier, showing $u_L \equiv h(u_G)$ where the increase in u_G leads to a deficit in the u_L . The black curve shows the Nash bargaining solution, where the maxima connecting the Pareto frontier and NBS is the optimal utility (u_G^*, u_L^*) for both players that is Pareto efficient.

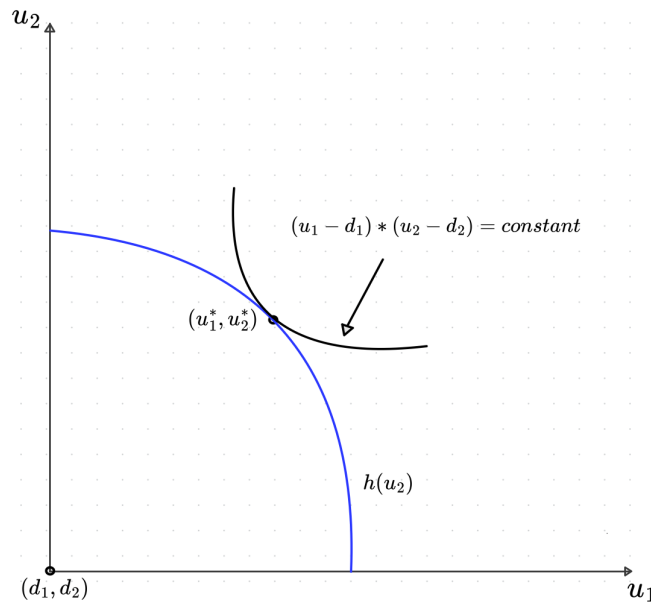


Figure 3.1: Nash Bargaining Solution for an arbitrary problem

3.4 Axioms of Nash Bargaining

The solution to Nash Bargaining is based on some foundational building blocks known as axioms. An axiom is a postulate we assume to be always true. In the context of a bargaining process, Nash characterized four axioms considered to be essential for any fair and efficient bargaining process[45]: invariance under positive linear affine transformations; symmetry; independence of irrelevant alternatives; and Pareto efficiency.

Axiom 1 (Invariance to Affine Transformations) *For any positive affine transformation $T_i(u_i) = a_i u_i + b_i$ with $a_i > 0$, if $\mathbf{u}^*$ solves $(U, \mathbf{d})$, then $T(\mathbf{u}^*)$ solves $(T(U), T(\mathbf{d}))$.*

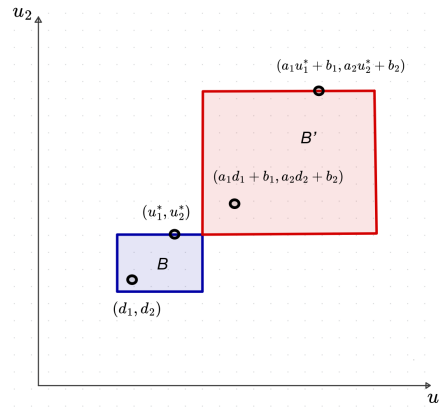


Figure 3.2: Axiom 1: Invariance under positive linear affine transformations

If the utility is rescaled using a linear transformation T , then the optimal solution does not change. The numerical values will be changed, but the relative position of u_G, u_L remains the same in B .

Axiom 2 (Pareto Efficiency) *The solution $\mathbf{u}^*$ lies on the Pareto frontier of U .*

Any agreed point must be Pareto efficient, ensuring that no other agreement can make one player better off without making the other worse off. There is no more value to barter over along the Pareto frontier.

Axiom 3 (Symmetry) *If the bargaining problem is symmetric (i.e., $(u_1, u_2) \in U \iff (u_2, u_1) \in U$ and $d_1 = d_2$), then $u_1^* = u_2^*$.*

Axiom 4 (Independence of Irrelevant Alternatives) *If $B' \subseteq B$, $\mathbf{u}^* \in B'$, and $(B', \mathbf{d})$ is a valid bargaining problem, then $\mathbf{u}^*$ is also the solution to $(B', \mathbf{d})$.*

Of all the axioms, this one is the most controversial as it implies that if one party concedes a large proportion of the barter set, i.e. concedes its own boundaries to be smaller/bigger then it does not matter since the NBS is still included in the new set.

These axioms ensure that the solution depends only on the economic fundamentals of the bargaining situation, not on arbitrary modeling choices.

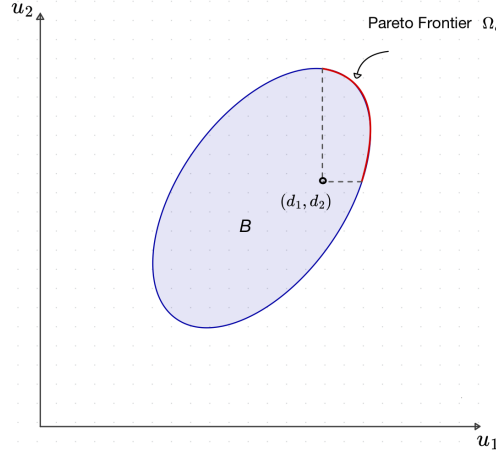


Figure 3.3: Axiom 2: Pareto Efficiency

The NBS is also generalizable for a multi-person $n > 2$ game with the objective function shown in 3.6. Each player i much like the standard two-player game needs to attain $u \geq d$.

$$\max_{(u_i, u_i) \in B} \prod_{i \in N} (u_i - d_i) \quad (3.6)$$

3.5 Fairness of Nash Bargaining Solutions

The Nash Bargaining constitutes one way of interpreting fairness, but what other ways of interpreting a fair solution exist, and how do they differ? This section aims to explain the differences between some bargaining rules. Each bargaining rule follows each a different set of axioms, and as such there does not exist one single bargaining rule that satisfies all axiomatic properties.

Utilitarian Bargaining Rule: Consider first an utilitarian bargaining rule that focuses on system-wide fairness[16], i.e., the solution focuses on maximizing social welfare between players. In the context of cooperative game theory, the utilitarian solution maximizes the sum of utilities:

$$\max_{(u_G, u_L) \in B} (u_G + u_L) \quad (3.7)$$

where B represents the bargaining set defined in equation 3.4. A common example of an utilitarian rule implemented in energy systems today is the day-ahead market, with the 'merit order curve' constituting the utilitarian bargaining rule/principle. The merit-order acts as a method for a central operator in deciding which generators to schedule where, with the goal of social-welfare (utility) maximization across the system. The sum of utilities is however neutral towards inequalities making the utilitarian solution achieved at the expense of some players [16].

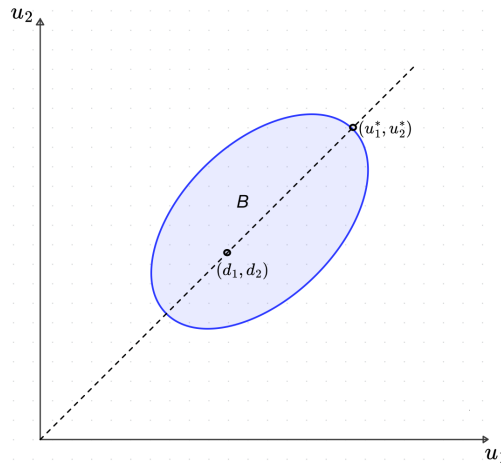


Figure 3.4: Axiom 3: Symmetry

Nash Bargaining and Proportional Fairness: Nash bargaining is also generalized as a 'proportional' rule [16], which favors bargaining in the logic that a negotiation should continue until the proportional gain from agreement no longer exceeds the proportional loss either party would face using the Nash bargaining solution 3.5 using the axioms 1-4 described.

This rule, as already described, balances the gains of each player relative to their disagreement point; however, a main critique and ponderous question remains. If a player concedes a large portion of their bargaining boundaries or opens up further options, should a solution not follow this movement of change in preference? The 'Independence of Irrelevant Alternatives' axiom dictates that it does not. Consider this example Harbig [46]: Assume player 2's preference relations are $C \gg B > A$ while player 1's is $A \gg B > C$, where the symbol $\gg$ denotes a considerably stronger preference than $>$. If option C was not available, the players would agree on A. However, the availability of option C enables player 2 to advocate for their second-best option B, arguing that their best option is forfeited. This may lead to the players choosing B in the new situation, contradicting the principle of the independence of irrelevant alternatives

3.6 Extensions: Asymmetric Nash Bargaining Problems

The standard Nash bargaining solution assumes symmetric bargaining positions between negotiating parties, reflected in the symmetric Nash product $(u_G - d_G)(u_L - d_L)$. However, real-world negotiations frequently involve asymmetric conditions where parties possess different negotiation advantages due to market position, information access, financing constraints, or regulatory environments. It can also be compared to the fact that the parties might not be able to share information, given potential confidentiality issues. This section extends the Nash bargaining framework to incorporate such asymmetries through two mechanisms: differential bargaining power and asymmetric information structures.

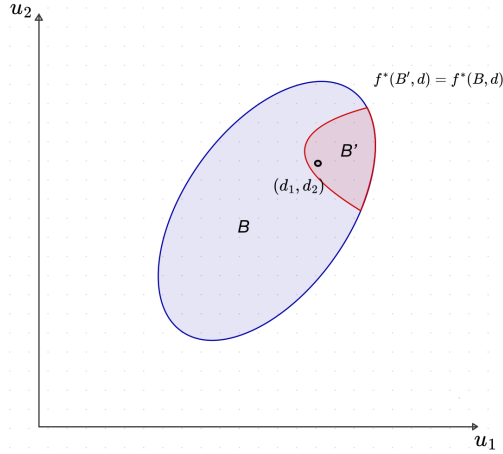


Figure 3.5: Axiom 4: Independence of irrelevant alternatives

3.6.1 Asymmetric Bargaining Power

The asymmetric Nash bargaining solution modifies the standard Nash product by introducing bargaining power parameters τ_G and τ_L that reflect each party's relative negotiation strength 3.8:

$$\max_{u_G, u_L \in B} (u_G - d_G)^\tau \cdot (u_L - d_L)^{(1-\tau)} \quad (3.8)$$

Where $\tau_G, \tau_L > 0$ represent the bargaining power weights of the generator and load, and $\tau_G + \tau_L = 1$ to ensure the allocated utility is the gained utility from the barter set. For each value of $\tau \in [0, 1]$, an asymmetric f_τ satisfies axiom 1,2 and 4. If $\tau = 0.5$, then the asymmetric NBS is identical to the NBS of 3.5, and also satisfy axiom 3 of symmetry. As τ increases, player G's utility increases while player L's decreases. The players are bargaining over the partition of transferable utility after each has agreed to give each other their disagreement point d they each would respectively receive. And then the surplus allocation is allocated between the fraction of τ^G and $1 - \tau_G$.

The bargaining power τ can capture several real-world asymmetries in the context of PPA negotiations: 1) A developer might have multiple potential offtakers, as such the τ_G will be higher than τ_L . 2) Conversely, the developer might be facing financing constraints and needs to find an offtaker quickly to reach financial close with the bank. This would reduce the bargaining power τ_G , and the offtaker is put in a better position.

An interesting extension of the Nash Bargaining problem is that its solution is extended under certain criteria to the Rubinstein 'Alternating Offers' framework. In this framework, two players give each other alternating bids on a bargaining problem. The bids might contain a discount rate per bid, thus decreasing the surplus value of the contract for each iteration of bidding. This alternating model, under certain conditions (explain

more), matches the asymmetrical Nash bargaining solution, where the discount rate is comparable to the bargaining power τ [45]. In game theoretical terms, the value of τ is associated with the concept of risk aversion[45].

3.6.2 Asymmetric Information

In incomplete information settings, parties hold different beliefs about uncertain parameters that influence contract value. Unlike the standard Nash bargaining assumption of complete information, real negotiations often involve significant uncertainty about future results, with each party potentially having different perspectives.

The asymmetry is modeled in this case by allowing each player to hold a subjective probability measure, following the approach of [21]. The information asymmetry is modeled as bias factors for each player, denoted as K^G and K^L for the generator and load, respectively. The bias factors capture the factors that are related to 'interpretation' or development of parameters involved in the bargaining process. In the context of a PPA negotiation, several potential information asymmetries exist. Each player might have their own proprietary weather model or buy a different forecast model, which they use to evaluate the value of a contract. Even with the same model, they might interpret it differently and draw differing conclusions. The asymmetry of information is interpretable as the belief a player has in their outcome.

4 PPA Negotiations as a Nash Bargaining Problem

4.1 PPAs in a Cooperative Game Framework

Power Purchase Agreement negotiations exhibit key characteristics that we can model within a cooperative game framework:

Players: PPA negotiations (often) involve exactly two players[10], not considering outside outside players or mediators:

- Generator (G): A renewable energy producer seeking revenue certainty to ensure project bankability.
- Load-serving entity (L): An offtaker (industrial & corporate consumer or utility) seeking to hedge electricity price risk.

Unlike spot electricity markets, where anonymity categorizes participants, PPAs involve identified counterparties in direct negotiations [10].

Strategies: The strategy space in PPA negotiations encompasses contract parameters such as strike price and volume size, rather than a fixed quantity.

- Strike price: $S \in [S^{\min}, S^{\max}]$ where bounds reflect the boundary
- For baseload contracts (MW): Contract volume $M \in [0, M^{\max}]$, where $M^{\max}$ often represents the capacity.
- For pay as produced: Production share (%) $\gamma \in [0, 1]$

Information and Timing

- *Complete information:* Parties share production forecasts, price expectations, and risk tolerances during negotiations[10].
- *Dynamic bargaining:* Both parties normally propose and revise terms iteratively; negotiations are as such generally not a simultaneous game. However, for simplicity and implementation of Nash bargaining, we will assume that it is a **simultaneous game**.

Payoffs: Both parties evaluate outcomes through risk-adjusted utility functions.

4.2 Utility Functions in PPA Negotiations

Utility is, in economic terms, a measure of a person's satisfaction or value that is derived from a particular outcome or allocation of resources. In the context of Nash Bargaining in our PPA negotiation, utility serves as a numerical expression of preference regarding

their *Return-Risk* function, which quantifies the net benefits perceived by each player in relation to their expected earnings and risk exposure. The following sections introduce the specific utility functions specific to the revenue structures of the BL and PAP PPA. For readability, scenarios ω are omitted in the following definitions.

4.2.1 Revenue functions

Buyer and Seller Perspectives

The seller (G) is assumed to own a single wind farm, and it is assumed to have zero outages or curtailments during operation throughout the entire tenor T , and all bids in the day-ahead market are accepted. The production P_t^G is assumed to be intermittent between 0 and its capacity. The cost derived from the difference between day-ahead bidding and its actual production in the balancing market is not considered in the revenue function, and its marginal cost of production is assumed to be 0. The granularity of the timesteps is per year; as such, the prices λ_t and P_t^G are the aggregated sums of prices and production of that year. To account for the potential cannibalization of renewables, the annual average capture rate is denoted CR^G for the wind farm. The risks faced by wind farms are 1) the volume risk from the intermittency of renewables, 2) the prices risk from the variability in the day-ahead market, and 3) the cannibalization risk that emerges when the wind farm produces at the same time as other renewables, depressing the average baseload price.

For the buyer (L), it submits demand bids to purchase energy at the day-ahead market at the market price. The demand is assumed to be always met without any outages. Similarly to the generator, the deviation between day-ahead and its realization is not considered. P_t^L denotes the annual gross consumption per year. CR^L denotes the load's annual capture rate; the reasoning for including this is that the load may be either inflexible or flexible in response to prices.

Baseload

In the baseload, players G and L negotiate an annual baseload contract for the amount M (MWh) and the strike price S , which must be fulfilled every hour throughout the entire tenor. A CfD is attached where if the market price differs from the strike price S , then the advantaged party must compensate the disadvantaged party accordingly. The revenue function for G is shown in 4.1, and for L 4.2. Scenarios are omitted for now for clarity.

$$\pi_G(M, S) = \sum_{t=1}^T \{CR_t^G \lambda_t P_t^G + (S - \lambda_t)M\} \quad (4.1)$$

$$\pi_L(M, S) = \sum_{t=1}^T \{-P_t^L CR_t^L \lambda_t + (\lambda_t - S)M\} \quad (4.2)$$

Pay as Produced

In the Pay as Produced, players G and L negotiate the percentage γ of the total production from the wind farm, that the buyer will buy at the strike price S . In the seller's

revenue function 4.3, revenue is shifted from market price to strike price. However, as delivery happens as production occurs, the market value of the wind production needs to be considered in buyers revenue function 4.4 by the difference in $\gamma P_t^G (CR_t^G \lambda_t - S)$

$$\pi_G(\gamma, S) = \sum_{t=1}^T \{(1 - \gamma)(CR_t^G \lambda_t P_t^G) + \gamma P_t^G S\} \quad (4.3)$$

$$\pi_L(\gamma, S) = \sum_{t=1}^T \{-P_t^L CR_t^L \lambda_t + \gamma P_t^G (CR_t^G \lambda_t - S)\} \quad (4.4)$$

4.2.2 Risk-Averse Utility Functions

To incorporate risk aversion into the *return-risk* utility of the players, we use the Conditional Value-at-Risk (CVaR), a risk measure widely applied in both financial and electricity markets. CVaR evaluates the expected losses that occur beyond a predefined threshold, specifically the Value-at-Risk (VaR), at a given confidence level $(1 - \alpha)$. While VaR represents the maximum loss not exceeded with confidence α , CVaR considers the *expected shortfall*, i.e., the average of the worst-case losses exceeding the VaR threshold.

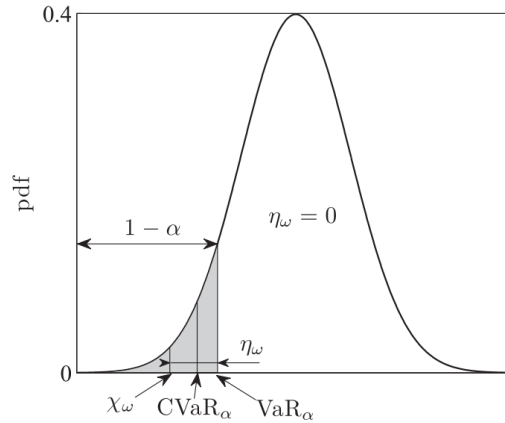


Figure 4.1: Visual representation of Conditional Value at Risk [47]

Figure 4.1 illustrates this concept graphically on an arbitrary probability density function (PDF) of potential outcomes. The VaR corresponds to the quantile threshold at $(1 - \alpha)$, whereas CVaR accounts for the area under the curve to the left of the VaR cutoff. As such, CVaR not only incorporates the likelihood of extreme losses but also captures their severity, making it a more informative and conservative risk metric. CVaR will henceforth be used in evaluating the left-tail of the net earnings, shifting the CVaR to be the right-tail of a loss distribution would be equivalent.

The resulting *Return-Risk* utility function combines each player's expected revenue with their risk aversion, modeled via CVaR, as defined in Equations 4.5 and 4.6. $E(\pi)$ de-

notes the expectation of G and L's net earnings, and $CVaR_\alpha$ denotes the CVaR associated with their net earnings, conditional on any given confidence interval α . A_G and A_L denote the *risk aversion factor* for each player, respectively, that determines their tradeoff between expected net earnings and risk exposure. Higher values of A_G and A_L will favor focusing on worst-case scenarios, whereas lower values reflect a more risk-neutral strategy.

$$u_G = (1 - A_G)\mathbb{E}(\pi_G(M, S)) + A_G CVaR_\alpha(\pi_G(M, S)) \quad (4.5)$$

$$u_L = (1 - A_L)\mathbb{E}(\pi_L(M, S)) + A_L CVaR_\alpha(\pi_L(M, S)) \quad (4.6)$$

From the utility function, the disagreement points for G and L are defined in 4.7 and 4.8. Under these circumstances, their utility function reflects their baseline utility with full participation in the market.

$$d_G = (1 - A_G)\mathbb{E}(\pi_G(0, 0)) + A_G CVaR_\alpha(\pi_G(0, 0)) \quad (4.7)$$

$$d_L = (1 - A_L)\mathbb{E}(\pi_L(0, 0)) + A_L CVaR_\alpha(\pi_L(0, 0)) \quad (4.8)$$

In this particular bartering problem this disagreement point seems the most reasonable, however the definition or value of the disagreement point could be different from this.

4.3 Nash Bargaining Solution for PPA Negotiations

We have now framed the PPA negotiation process as a Nash Bargaining problem and expressed the utility functions of the seller and buyer in PPA negotiations. Following the definitions in Section 3.3, we can find the solutions of this Nash Bargaining problem, i.e. strike price and contract amount by solving the following optimization problems:

4.3.1 Baseload Contract Model

The baseload model is presented in 4.9 through 4.15. The decision variables are the contract volume, M (in MW), and the strike price, S (EUR/MWh). The auxiliary decision variables associated with the CVaR constraint are ζ and η_ω . ζ is the value-at-risk at the given confidence interval, and η_ω is the shortfall of each scenario, that is, or exceeds η . d_G, d_L represents the disagreement points where each player fully merchant; selling or buying at market prices. This represents their alternatives if negotiations fail. The time is indexed as t within the time set T , and scenarios are indexed as ω in the scenario set W . The objective function 4.9 calculates the Nash Product from the utility functions 4.10 - 4.11, where the disagreement points are defined as in 4.7 - 4.8. Equations 4.12–4.13 enforce the Conditional Value at Risk constraints for both players G and L. The contract size is bounded 4.14 between the minimum contract size M^R and its maximum M^U , which is assumed to be the maximum capacity of the wind farm. The strike price is bounded between 4.15 and the minimum strike price S^R and the maximum strike price S^U , which is the lowest and highest value either player would find acceptable. The auxiliary variables η, ζ are bounded in through 4.16 and 4.17.

$$\max_{M, S, \zeta^G, \zeta^L, \eta_\omega^G, \eta_\omega^L} (u_G - d_G) \cdot (u_L - d_L) \quad (4.9)$$

s.t.

$$u_G = (1 - A_G) \sum_{\omega \in W} \pi_\omega \sum_{t \in T} \{CR_{t,\omega}^G \lambda_{t,\omega} P_{t,\omega}^G + (S - \lambda_{t,\omega})M\} - A_G \left(\zeta_G + \frac{1}{1 - \alpha} \sum_{\omega \in W} \pi_\omega \eta_\omega^G \right) \quad (4.10)$$

$$u_L = (1 - A_L) \sum_{\omega \in W} \pi_\omega \sum_{t \in T} \{-P_{t,\omega}^L CR_{t,\omega}^L \lambda_{t,\omega} + (\lambda_{t,\omega} - S)M\} - A_L \left(\zeta_L + \frac{1}{1 - \alpha} \sum_{\omega \in W} \pi_\omega \eta_\omega^L \right) \quad (4.11)$$

$$\eta_\omega^G \geq \zeta_G - \sum_{t \in T} \{CR_{t,\omega}^G \lambda_{t,\omega} P_{t,\omega}^G + (S - \lambda_{t,\omega})M\} \quad \forall \omega \in W \quad (4.12)$$

$$\eta_\omega^L \geq \zeta_L - \sum_{t \in T} \{-P_{t,\omega}^L CR_{t,\omega}^L \lambda_{t,\omega} + (\lambda_{t,\omega} - S)M\} \quad \forall \omega \in W \quad (4.13)$$

$$M^R \leq M \leq M^U \quad (4.14)$$

$$S^R \leq S \leq S^U \quad (4.15)$$

$$\eta_\omega^G \geq 0, \quad \eta_\omega^L \geq 0 \quad \forall \omega \in W \quad (4.16)$$

$$\zeta_G \in \mathbb{R}, \quad \zeta_L \in \mathbb{R} \quad (4.17)$$

4.3.2 pay as produced Contract Model

While the baseload structure provides revenue certainty through fixed volumes, it transfers shape risk to the generator who must deliver regardless of production conditions. The pay as produced (PAP) alternative fundamentally alters this risk allocation by linking payments directly to actual generation, requiring modifications to both decision variables and revenue functions. The PAP model maintains the same optimization structure as the baseload case (equations 4.18 through 4.23) but substitutes the contract volume M with the production share γ , representing the fraction of total generation sold under the PPA. This share is bounded between minimum and maximum participation levels in 4.23. The revenue functions are updated according to 4.3 and 4.4, while the CVaR constraints 4.21–4.22 incorporate these modified revenue streams. Strike price bounds and CVaR auxiliary variables remain unchanged from the baseload formulation.

$$\max_{\gamma, S, \zeta^G, \zeta^L, \eta_\omega^G, \eta_\omega^L} (u_G - d_G) \cdot (u_L - d_L) \quad (4.18)$$

s.t.

$$u_G = (1 - A_G) \sum_{\omega \in W} \pi_\omega \sum_{t \in T} \{ (1 - \gamma) CR_{t,\omega}^G \lambda_{t,\omega} P_{t,\omega}^G + \gamma P_{t,\omega}^G S \} - A_G \left(\zeta_G + \frac{1}{1 - \alpha} \sum_{\omega \in W} \pi_\omega \eta_\omega^G \right) \quad (4.19)$$

$$u_L = (1 - A_L) \sum_{\omega \in W} \pi_\omega \sum_{t \in T} \{ -P_{t,\omega}^L CR_{t,\omega}^L \lambda_{t,\omega} + \gamma P_{t,\omega}^G (CR_{t,\omega}^G \lambda_{t,\omega} - S) \} - A_L \left(\zeta_L + \frac{1}{1 - \alpha} \sum_{\omega \in W} \pi_\omega \eta_\omega^L \right) \quad (4.20)$$

$$\eta_\omega^G \geq \zeta_G - \sum_{t \in T} \{ (1 - \gamma) CR_{t,\omega}^G \lambda_{t,\omega} P_{t,\omega}^G + \gamma P_{t,\omega}^G S \} \quad \forall \omega \in W \quad (4.21)$$

$$\eta_\omega^L \geq \zeta_L - \sum_{t \in T} \{ -P_{t,\omega}^L CR_{t,\omega}^L \lambda_{t,\omega} + \gamma P_{t,\omega}^G (CR_{t,\omega}^G \lambda_{t,\omega} - S) \} \quad \forall \omega \in W \quad (4.22)$$

$$\gamma^R \leq \gamma \leq \gamma^U \quad (4.23)$$

4.4 Barter Set in PPA Negotiations

This section establishes the mathematical framework for determining the negotiation space that is mutually beneficial in PPA contracts through Nash bargaining. We first develop the theoretical foundation through a series of lemmas that extend on [21] to incorporate stochastic production and pay as produced contracts. These mathematical properties then allow us to define viable strike price boundaries and characterize the conditions under which feasible barter sets exist.

4.4.1 Mathematical Foundation of the Barter Set

The construction of the barter set relies on several properties that are extensions to the framework from [21] to incorporate stochastic production, capture rates, and pay as produced contract. The following lemmas establish the theoretical foundation for the bargaining analysis.

Lemma 1 (Utility Space Mapping under Strike Price Variation) *The mapping of the utility curves under varying contract amounts follows the following:*

1. **Baseload:** For any fixed contract volume $M \in [M^R, M^U]$, varying the strike price S from S^R to S^U maps into a straight line in the utility possibility set U with constant slope -1 .

2. **pay as produced:** For any fixed production share $\gamma \in [\gamma^R, \gamma^U]$, varying the strike price S from S^R to S^U maps into a curved line in U with variable slope dependent on γ and the CVaR derivatives.

The constant slope of -1 for baseload contracts simplifies the analysis compared to [21], where the slope was $-(1 + A_L)/(1 + A_G)$. For pay as produced contracts, the interaction between stochastic production and contract parameters creates dependent utility curves. Detailed derivations are provided in appendices C and D.

Lemma 2 (Concavity of Utility Curves) *If the strike price is fixed at S^R (or S^U), then the set of points (u_G, u_L) traced out in U as the contract amount varies determines a concave curve V_1 (or V_2):*

1. **Baseload:** As M varies from M^R to M^U
2. **pay as produced:** As γ varies from $0 \equiv \gamma^S$ to $1 \equiv \gamma^U$

This concavity establishes that the Pareto frontier made by $h(u_G) = u_L$ is a concave function. The incorporation of a new utility function 4.5 - 4.6 and stochasticity of parameters modifies but preserves the concavity structure (see appendices E.1 and F.1).

Lemma 3 (Monotonicity of Utilities with Respect to Contract Volume) *The relationship between strike price boundaries and utility changes as the contract amount varies:*

1. **Baseload:**
 - If $S^R < S^{R*}$ (as defined in Equation (4.24)), then with strike price fixed at S^R , as the contract amount M increases: u_G decreases and u_L increases.
 - If $S^U > S^{U*}$ (as defined in Equation (4.25)), then with strike price fixed at S^U , as the contract amount M increases: u_G increases and u_L decreases.
2. **pay as produced:**
 - If $S^R < S^{R*}$ (as defined in Equation (4.31)), then with strike price fixed at S^R , as the production share γ increases: u_G decreases and u_L increases.
 - If $S^U > S^{U*}$ (as defined in Equation (4.32)), then with strike price fixed at S^U , as the production share γ increases: u_G increases and u_L decreases.

This monotonicity establishes the direction of the utility changes along the curves V_1 and V_2 , ensuring that the viable strike price boundaries S^{R*} and S^{U*} represent the limits beyond which no mutually beneficial contract is possible. When the strike price is below S^{R*} , increasing contract amount benefits the load at the generator's expense. Conversely, when the strike price exceeds S^{U*} , increasing contract amount benefits the generator at the load's expense. The proof, incorporating stochastic production and capture rates, is provided in Appendices E.2 - F.2.

Lemma 4 (Barter Set Existence Conditions) *A non-empty, bounded barter set exists when the slope of the utility frontier crosses specific thresholds:*

1. **Baseload:** The slope satisfies $\frac{du_L}{du_G}|_{M^R, S^R} < -1$ and $\frac{du_L}{du_G}|_{M^U, S^R} > -1$
2. **pay as produced:** The slope satisfies variable threshold conditions dependent on the CVaR derivatives with respect to S

These conditions ensure the existence of mutually beneficial agreements by guaranteeing that the marginal rate of substitution between utilities crosses the critical threshold. Complete derivations are in Appendices E.3 and F.3.

4.4.2 Viable Strike Prices

Building on the mathematical foundation established by the lemmas, we can now define the concept of viable strike prices. These represent the boundaries within which mutually beneficial agreements can exist. Unlike the minimum or maximum values input into an optimization model, which represent the absolute limits a developer or buyer might consider, the viable strike price range $[S^{R*}, S^{U*}]$ reflects the price interval where the barter set can be constructed. Prices outside this range cannot provide mutually beneficial agreements, as they would leave at least one party worse off than their disagreement point.

The viable boundaries are determined by identifying where the marginal utility with respect to contract volume reaches its maximum, leveraging the concavity property established in Lemma 2.

Baseload

For baseload contracts, the viable strike prices can be determined analytically. In contrast to [21] where worst-case outcomes of λ could be drawn directly from its distribution, our framework incorporates the interaction between prices λ_t , capture rates CR_t , and stochastic production/consumption P_t^G/P_t^L .

Finally, we define S^{R*} and S^{U*} that define the boundaries where a barter can be constructed:

$$S^{R*} = \min \{S_{G1}^{R*}, S_{G2}^{R*}, S_L^{R*}\} \quad (4.24)$$

$$S^{U*} = \max \{S_{G1}^{U*}, S_{G2}^{U*}, S_L^{U*}\} \quad (4.25)$$

With $S_{G1}^{R*}, S_{G2}^{R*}, S_L^{R*}$ defined as the following:

$$S_L^{R*} = \frac{(1 - A_L)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_L\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_L(M, S^U) \leq \text{VaR}_\alpha(\pi_L(M, S^R))\right]}{T} \quad (4.26)$$

$$S_{G1}^{R*} = \frac{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_G\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S^U) \leq \text{VaR}_\alpha(\pi_G(M, S^R))\right]}{T} \quad (4.27)$$

$$S_{G2}^{R*} = \frac{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_G\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S^U) \leq \text{VaR}_\alpha(\pi_G(M, S^R))\right]}{T} \quad (4.28)$$

$S_{G1}^{U*}, S_{G2}^{U*}, S_L^{U*}$ are omitted as structurally they are similar just evaluated at a different point.

Where each threshold is evaluated at the strike price level being considered (S^R or S^U), with:

- S_{G1}^{R*}, S_{G1}^{U*} : Evaluated for $M > \max_\omega \sum_{t=1}^T P_{t,\omega}^G \cdot CR_{t,\omega}^G$
- S_{G2}^{R*}, S_{G2}^{U*} : Evaluated for $M \leq \min_\omega \sum_{t=1}^T P_{t,\omega}^G \cdot CR_{t,\omega}^G$
- S_L^{R*}, S_L^{U*} : Evaluated consistently across all cases

The derivation of these conditions is provided in Appendix E.2.

Pay as Produced

For pay as produced contracts, the viable strike prices must be determined numerically due to the multiplicative interaction between γ , S , and stochastic production P_t^G . Unlike the baseload case where the translation-equivariant property of CVaR facilitates an analytical solution, this interaction prevents direct analytical resolution.

The CVaR derivatives with respect to contract share, evaluated at $\gamma = 0$, capture how the initial allocation affects each party's risk exposure:

$$\left. \frac{d\text{CVaR}(\pi_L(\gamma, S))}{d\gamma} \right|_{\gamma=0} = \mathbb{E} \left[\sum_{t=1}^T P_t^G (CR_t^G \lambda_t - S) \mid \pi_L(\gamma, S) \leq \text{VaR}_\alpha(\pi_L(\gamma, S)) \right] \quad (4.29)$$

$$\left. \frac{d\text{CVaR}(\pi_G(\gamma, S))}{d\gamma} \right|_{\gamma=0} = \mathbb{E} \left[\sum_{t=1}^T P_t^G (S - CR_t^G \lambda_t) \mid \pi_G(\gamma, S) \leq \text{VaR}_\alpha(\pi_G(\gamma, S)) \right] \quad (4.30)$$

The viable boundaries are then determined by solving:

$$SR^* = \left\{ S^* : (1 - A_L) \cdot \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot (CR_t^G \lambda_t - S) \right] + A_L \cdot \frac{d}{d\gamma} \text{CVaR}(\pi_L(\gamma, S)) \Big|_{\gamma=0} = 0 \right\} \quad (4.31)$$

$$SU^* = \left\{ S^* : (1 - A_G) \cdot \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot (S - CR_t^G \lambda_t) \right] + A_G \cdot \frac{d}{d\gamma} \text{CVaR}(\pi_G(\gamma, S)) \Big|_{\gamma=0} = 0 \right\} \quad (4.32)$$

The methodology for deriving these conditions is established in Appendix F.2.

4.4.3 Barter Set Characterization

With the mathematical foundation established by the lemmas and the viable strike price boundaries defined, we can now characterize the conditions under which feasible barter sets exist. The analysis assumes that the lowest permissible strike price satisfies $S^R < S^{R*}$, the highest satisfies $S^U > S^{U*}$, and the contract volume lies in $0 \leq M \leq \bar{P}^G$.

Under these assumptions, we construct two curves V_1 and V_2 by fixing the strike price at its boundaries (S^R and S^U respectively) and varying the contract size. From the lemmas 1,2,3 and 4 in combination with theorem 2 provided in [21] results in the barter sets exhibiting distinct geometric features:

- The baseload barter set B forms a right-triangle region due to the constant slope of -1
- The pay as produced barter set forms a curvilinear triangle due to the variable slope

Three cases characterize the possible configurations of the barter set, determined by whether the conditions in Lemma 4 are satisfied:

Case 1: Bounded Barter Set

For baseload contracts, a bounded barter set exists when both conditions in Lemma 4 are satisfied:

$$\frac{du_L}{du_G} \Big|_{M^R, S^R} < -1 \quad (4.33)$$

$$\frac{du_L}{du_G} \Big|_{M^U, S^R} > -1 \quad (4.34)$$

Expanding these conditions:

$$\frac{(1 - A_L)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) - TS + \frac{d\text{CVaR}(\pi_L(M^R, S^R))}{dM}}{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T (-\lambda_t)\right) - A_G \frac{d\text{CVaR}(\pi_G(M^R, S^R))}{dM} + TS} < -1 \quad (4.35)$$

The slope crossing -1 between M^R and M^U ensures that a mutually beneficial agreement exists for some $M \in [M^R, M^U]$. The reference slope of -1 arises from Lemma 1.

For pay as produced contracts, the conditions involve a variable reference slope dependent on the derivative with regard to the strike price S , making the condition look as follows:

$$\left. \frac{du_L}{du_G} \right|_{\gamma^R, S^R} < \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (4.36)$$

$$\left. \frac{du_L}{du_G} \right|_{\gamma^U, S^R} > \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (4.37)$$

Expanding the right-hand side $\frac{\frac{du_L}{dS}}{\frac{du_G}{dS}}$:

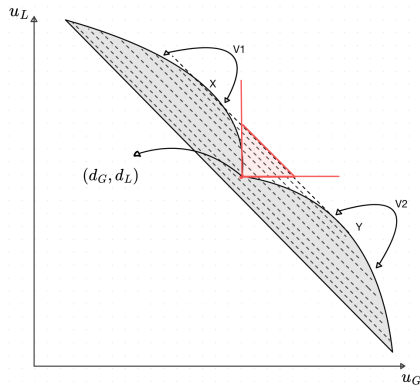
$$\left. \frac{du_L}{du_G} \right|_{\gamma^R, S^R} < - \frac{(1 - A_L)\gamma\mathbb{E}(\sum_t (-P_t^G)) + \frac{d}{dS}\text{CVaR}_\alpha(\pi_L)}{(1 - A_G)\gamma\mathbb{E}(\sum_t P_t^G) + \frac{d}{dS}\text{CVaR}_\alpha(\pi_G)} \Big|_{\gamma^R, S^R} \quad (4.38)$$

$$\left. \frac{du_L}{du_G} \right|_{\gamma^U, S^R} > - \frac{(1 - A_L)\gamma\mathbb{E}(\sum_t (-P_t^G)) + \frac{d}{dS}\text{CVaR}_\alpha(\pi_L)}{(1 - A_G)\gamma\mathbb{E}(\sum_t P_t^G) + \frac{d}{dS}\text{CVaR}_\alpha(\pi_G)} \Big|_{\gamma^U, S^R} \quad (4.39)$$

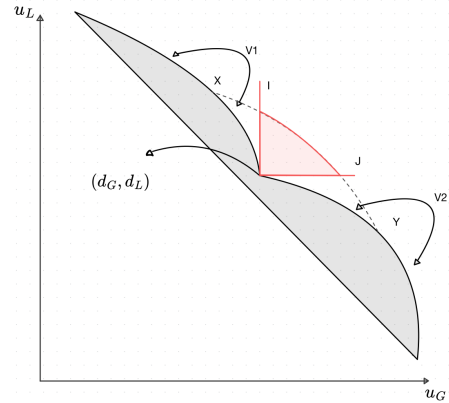
Expanding the left-hand side the following

$$\frac{(1 - A_L)\mathbb{E}\left(\sum_{t=1}^T P_t^G(CR_t^G \lambda_t - S)\right) + A_L \frac{d\text{CVaR}(\pi^L(\gamma^R, S^R))}{d\gamma}}{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T P_t^G(S - CR_t^G \lambda_t)\right) + A_G \frac{d\text{CVaR}(\pi^G(\gamma^R, S^R))}{d\gamma}} < \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (4.40)$$

The resulting barter set forms a right triangle (baseload) or curvilinear triangular (pay as produced) region with the disagreement point at one vertex. The Nash bargaining solution resides within this region (see Figures 4.2a and 4.2b).



(a) Barter Set (Red Triangle) - Case 1
Baseload



(b) Barter Set (Red Triangle) - Case 1
Pay as Produced

Figure 4.2: Case 1. X and Y denote the maxima of the curves V_1 and V_2 where I and J denote the horizontal and vertical lines drawn from the disagreement point.

Case 2: No Barter Set

When condition (4.33) or (4.38) fails to hold, no mutually beneficial agreement exists beyond the disagreement point. The utility set U contains only the point where $u_G = d_G$ and $u_L = d_L$ (see Figure 4.3).

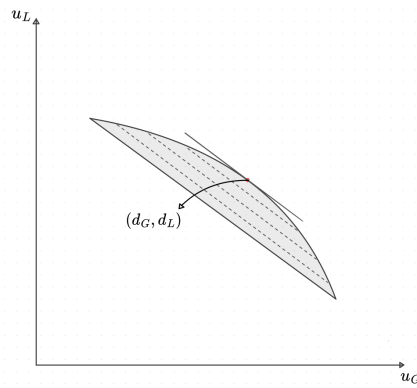
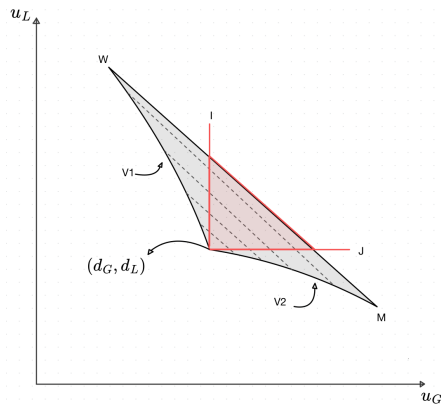


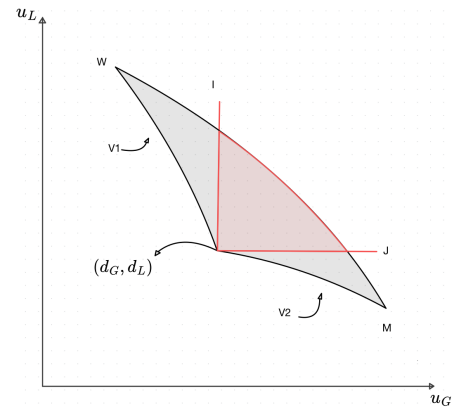
Figure 4.3: Barter Set - Case 2. No feasible contract is viable given the conditions.

Case 3: Unbounded Barter Set

When the initial condition holds but the second fails (i.e., (4.34) or (4.39) is not satisfied), the slope of the utility frontier never crosses the threshold. The concave curves V_1 and V_2 have not reached their maxima at the upper contract boundary, resulting in an unbounded barter set that extends to the maximum feasible contract volume (see Figures 4.4a and 4.4b).



(a) Barter Set (Red Triangle) - Case 3 Baseload



(b) Barter Set (Red Triangle) - Case 3 Pay as Produced

Figure 4.4: Case 3. W and M denote the maxima of the the curves V_1 and V_2 at the upper boundary. I and J denote the horizontal and vertical line drawn from the disagreement point.

5 Methodology

This chapter describes the methodology used to answer the primary research question: how cooperative game theory can be applied in designing and pricing PPAs. The analysis employs a synthetic case study of a wind farm developer seeking a PPA contract to ensure project bankability, providing practical boundaries for the modeling framework. The methodology combines Nash bargaining for contract negotiations, stochastic optimization for uncertainty modeling, Monte Carlo simulation for scenario generation. This approach aims to quantify the impacts of risk aversion, negotiation power, and market parameters on the optimal contract design. The chapter first presents the synthetic case study to establish realistic parameters. It then develops the mathematical model used for computational optimization before describing the Monte Carlo simulation approach, the stochastic process of future price generation, and finally the scenario reduction to make the model tractable.

5.1 Case Study Design

To illustrate the thesis, a synthetic case is presented of a developer seeking to develop a wind farm in Ringsted municipality, comprising 10x3 MW turbines, through a PPA contract with local industries. Ringsted has a high share of Asphalt, Construction & Building Materials, providing 20% of private employment [48]. Ringsted has committed to achieving carbon neutrality by 2045 [49] and had an annual electricity consumption of 700 TJ (194 GWh) in 2023 [50]. The PPA aims to help ensure the municipality reaches its climate neutrality goals and allows the industries to hedge their electricity price. The wind farm will only consist of wind turbines, and the PPA contract will be an off-grid, behind-the-meter arrangement in terms of production. Several companies exist within the municipality, but will be assumed to be a single stakeholder for negotiation on behalf of all of them.

5.2 Mathematical Modeling Framework

Having established the practical context through the Ringsted case study, we now develop a computational model based from the mathematical models presented through 4.3.1 and 4.3.2 applying Nash Bargaining to analyze the negotiation dynamics between the wind farm developer and the industries. The framework addresses the research questions through a stochastic optimization model incorporated with the Nash Bargaining approach that analyzes both BL and PAP PPA structures. This model provides insight into how different risk profiles, negotiation power dynamics, and market parameters influence contract outcomes. The model will be using an annual granularity to focus on long-term outcomes rather than operational optimization. While this detracts from seasonal patterns and short-term arbitrage opportunities, it captures the fundamental risk-return trade-offs given that the contracts can last over 10-20 years, and the focus of a PPA is on long-term hedging. The models in 4.3.1 and 4.3.2 are formulated as

Non-Linear Programs (NLP) with decision variables, constraints, objective functions, parameters, and scenarios, implemented in Python using GurobiPy and solved by Gurobi. The model is an NLP due to bi-linear terms of the decision variables in the objective function and CVaR constraints. The complete nomenclature of variables, sets, and parameters is provided in the appendix.

5.2.1 Linearize Objective Function

In the objective functions 4.9 and 4.18, the resulting mathematical models are non-convex with several bilinear and trilinear terms, which Gurobi cannot solve. It is possible to make them solvable by taking the logarithmic identity and rewriting the objective into the following 5.1.

$$\max_{u_G, u_L \in B} \ln(u_G - d_G) + \ln(u_L - d_L) \quad (5.1)$$

$$\text{s.t. } u_i - d_i \geq \epsilon, \quad \forall i \in \{G, L\} \quad (5.2)$$

$$\epsilon \geq 0 \quad (5.3)$$

To ensure the model is feasible using the logarithm, an additional constraint 5.2 ensures the difference is strictly greater than ϵ .

The final model remains a non-linear problem (NLP) because the objective function contains a bilinear expression within u_G and u_L , involving the product of S and M ; however, the model is still not convex as the logarithmic expression of u_G and u_L contains a constant. The model is now solvable in Gurobi. However, due to the inclusion of 5.1 - 5.3, Gurobi treats the model as a 'Non-Convex Mixed Integer Non Linear Problem (MINLP)'. Some small optimality gaps exist within the model of around 0.005%, where the logarithmic constraint is violated by 10^{-5} . Given these minor violations, the trends and results provided by the model are interpreted overall to be meaningful.

5.2.2 Asymmetric Nash Bargaining

Incorporating the negotiation power of the generator and load is done through 5.4 - 5.5. If one party gets more negotiation power, then their utility is more valuable than that of their counterparty. At the extremes $\tau_G = 0$ or $\tau_L = 0$, it effectively means their value function in the objective function disappears; however, 5.2 is still active as for an acceptable contract, the counterparty needs to fulfill ($u^* \geq d$) according to Nash Bargaining Theory.

$$\max_{u_G, u_L \in B} \tau_G \cdot \ln(u_G - d_G) + \tau_L \cdot \ln(u_L - d_L) \quad (5.4)$$

$$\tau_L = 1 - \tau_G \quad (5.5)$$

5.2.3 Asymmetric Information

The asymmetric information implementation is implemented in the optimization framework through biased parameter expectations in the two following parameters.

Price Bias

Below the biased price realization are shown, where K_{λ}^i represents player i 's systematic forecast bias relative to objective expectations $\mathbb{E}[\lambda_{t,\omega}]$. Positive bias indicates higher expectations, and a negative bias reflects lower expectations.

- Generator's price realization: $\lambda_{t,\omega}^G = \lambda_{t,\omega} + K_{\lambda}^G \cdot \mathbb{E}[\lambda_{t,\omega}]$
- Load's price realization: $\lambda_{t,\omega}^L = \lambda_{t,\omega} + K_{\lambda}^L \cdot \mathbb{E}[\lambda_{t,\omega}]$

Production Bias

Below are shown the biased production realizations, where K_P^i represents player i 's systematic forecast bias relative to objective expectations $\mathbb{E}[P_{t,\omega}^G]$.

- Generator's production realization: $P_{t,\omega}^{G,G} = P_{t,\omega}^G + K_P^G \cdot \mathbb{E}[P_{t,\omega}^G]$
- Load's production realization: $P_{t,\omega}^{G,L} = P_{t,\omega}^G + K_P^L \cdot \mathbb{E}[P_{t,\omega}^G]$

Model Implementation

The biased parameters replace the baseline values in the optimization models for BL and PAP:

1. **Utility Functions:** In equations 4.10 and 4.19 the price and production terms $\lambda_{t,\omega}, P_{t,\omega}^G$ are replaced with $\lambda_{t,\omega}^G, P_{t,\omega}^{G,G}$. Similarly, in 4.11 and 4.20, the price terms are replaced with $\lambda_{t,\omega}^L$; only 4.20 is the biased generator production $P_{t,\omega}^{P,L}$ applied.
2. **CVaR Constraints:** In the BL CvaR constraints 4.12 - 4.13 and the PAP constraints 4.21 - 4.22 use the biased price expectation $\lambda_{t,\omega}^G, \lambda_{t,\omega}^L$. The production biases $P_{t,\omega}^{G,G}, P_{t,\omega}^{G,L}$ is also applied uniformly, except in 4.13.

5.3 Monte Carlo Simulation Method

This section describes the Monte Carlo processes used to generate the scenarios for the mathematical model. The parameters are all generated through Monte Carlo simulations with adjustments to more accurately represent the distribution.

5.3.1 Monte Carlo Process

Monte Carlo simulation is a numerical technique used to evaluate the expected value and distribution of outcomes for systems characterized by uncertainty and randomness. In the context of the thesis, Monte Carlo enables the sampling of the input parameters, baseload prices, wind production, consumption, and capture rates based on their empirically derived probability distributions from historical data. By generating a large number of random scenarios and calculating the outcome for each, it becomes possible to approximate the probability distribution of the revenue and utility. Through Monte Carlo simulation, the expected value and CVaR can be quantified. Each parameter is sampled randomly and independently from its sampling method, and combined into a scenario ω that is assumed to be 'Independent Identically Distributed' (I.D.D), each with equal probability $1/N$.

5.3.2 Production & Consumption

The production $P_{t,\omega}^G$ and consumption $P_{t,\omega}^L$ scenarios are simulated by assuming that the distribution in the future will continue to follow the empirical distribution based on historical data. The empirical distribution is found by fitting a distribution using the *Distfit* [51] package to the yearly historical production and consumption data, and sampling from that distribution.

5.3.3 Capture Rates

The capture rate scenarios are generated utilizing the coefficient of variation between price λ and production/consumption P^G/P^L .

$$C^R = 1 + \rho \frac{\sigma_\lambda}{\mu_\lambda} \frac{\sigma_P}{\mu_P} \quad (5.6)$$

The capture rate coefficient is calculated using equation 5.6, where the relative uncertainty depends on three parameters derived from historical data: price coefficient of variation $\frac{\sigma_\lambda}{\mu_\lambda}$, production/consumption coefficient of variation $\frac{\sigma_P}{\mu_P}$, and the correlation between λ and P (ρ). To account for the variability of capture rates, the correlation coefficient ρ is treated as stochastic rather than fixed. Historical yearly correlations between electricity prices and production/consumption are calculated, which provides a distribution that is then transformed using the Fischer-Z transformation [52]. This transformation converts the bounded correlation values $(-1, +1)$ into a normal distribution, allowing for sampling. The mean (μ_{z_ρ}) and standard deviation (σ_{z_ρ}) of these transformed correlations across years are estimated to characterize the correlation distribution $\mathcal{N}(\mu_{z_\rho}, \sigma_{z_\rho})$.

For each simulation run across multiple years, the model is the following:

1. **Samples correlation uncertainty:** Sample random noise $\mathcal{N}(0, \sigma_{z_\rho})$ and add it to the mean Z-transformed correlation μ_{z_ρ}
2. **Back-transforms to correlation space:** Uses $\tanh(\cdot)$ to convert back to valid correlation values $(-1, +1)$
3. **Apply the capture rate formula:** $CR = 1 + \rho \left(\frac{\sigma_\lambda}{\mu_\lambda} \right) \left(\frac{\sigma_P}{\mu_P} \right)$

5.3.4 Prices

Electricity prices are modeled using an Ornstein-Uhlenbeck (OU) process with a time-varying long-term mean to capture the mean-reverting behavior characteristic of electricity markets [53, 54]. The mean-reverting behavior fits well for electricity prices, as even with periods of despite temporary price spikes, it has reverted towards historical mean levels. Predicting long-term prices is an incredibly complex problem needing to account for changing the mean-reverting behavior and sudden volatility spikes, the OU process supplied here is a simplification of a complex topic.

The continuous-time OU process is given by 5.7 where X_t represents the stochastic process at time t , μ is a fixed long-term mean, θ is the mean reversion rate, σ is the

volatility and dW_t represents a Gaussian increment from Brownian motion.

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t \quad (5.7)$$

To account for the expected change in average electricity price, the long-term mean is allowed to vary across time using the following equation 5.8.

$$\mu(t) = \mu_0 + \mu_1 \cdot t \quad (5.8)$$

where μ_0 is the initial mean level, μ_1 represents the annual trend, and t is time in years.

Parameters are estimated using maximum likelihood estimation on the discrete-time approximation. The regression specification:

$$\Delta X_t = a + bX_{t-1} + c \cdot t + \epsilon_t \quad (5.9)$$

yields parameter estimates: $\theta = -b/\Delta t$, $\mu_0 = -a/b$, $\mu_1 = -c/b$, and $\sigma^2 = \text{Var}(\epsilon_t)/\Delta t$, where $\Delta t = 1/12$ represents monthly time steps.

Monthly price scenarios are generated using the discrete-time approximation:

$$X_{t+\Delta t} = X_t + \theta(\mu_0 + \mu_1 \cdot t - X_t)\Delta t + \sigma\sqrt{\Delta t}\epsilon_t \quad (5.10)$$

where $\epsilon_t \sim \mathcal{N}(0, 1)$ are independent standard normal increments.

Annual average prices are calculated as $\lambda_{t,\omega} = \frac{1}{12} \sum_{i=1}^{12} X_{t,i}$ where $X_{t,i}$ represents the i -th monthly price in year t for scenario ω . To ensure non-negative prices, any negative values are set to zero, reflecting the practical lower bound in electricity markets.

This modeling approach captures the fundamental mean-reverting and trending behavior of electricity prices but does not explicitly model price spikes or seasonal patterns beyond linear trends. The assumption of constant volatility may not also fully capture the time-varying volatility observed in electricity markets [53]. Despite these limitations, the OU framework should provide a good foundation for long-term price scenario generation suitable for PPA valuation analysis [55].

5.4 Scenario Reduction

The Monte Carlo simulation generates N scenarios representing uncertain market conditions over the PPA contract period. However, solving the bargaining optimization problem becomes computationally intensive for large N due to the nonlinear objective function and CVaR constraints. To maintain computational feasibility while preserving the essential characteristics of the uncertainty distribution, we employ K-means clustering for scenario reduction.

5.4.1 K-Means

K-Means is an unsupervised machine-learning method of clustering data into K clusters based around some centroids. The final centroids are computed by reducing the variance in each cluster. This allows the scenarios to capture the variance structure of the initial distribution.[56] In the case of the bargaining model, we want to ensure that the tails are sufficiently represented due to the CVaR constraint.

$$\arg \min_{S_k} \sum_{k=1}^K \sum_{\omega \in S_k} \|\mathbf{x}_\omega - \mathbf{c}_k\|_2^2 \quad (5.11)$$

where S_k represents cluster k and $\mathbf{c}_k$ is the cluster centroid.

The clustering operates on the joint feature space $\mathbf{x}_\omega = [\pi_\omega^G, \pi_\omega^L]^T$ representing generator and load revenues respectively when no contract is signed, where:

- $\pi_\omega^G = \sum_{t \in T} CR_{t,\omega}^G \lambda_{t,\omega} P_{t,\omega}^G$ incorporates prices, wind production, and capture rates
- $\pi_\omega^L = \sum_{t \in T} P_{t,\omega}^L CR_{t,\omega}^L \lambda_{t,\omega}$ incorporates prices, consumption, and load capture rates

Rather than using cluster centroids directly, we select the actual Monte Carlo scenario closest to each centroid in terms of Euclidean distance. The scenario is assigned the probability of the centroids' probability.

For each cluster k :

1. Compute the cluster centroid: $\mathbf{c}_k = \frac{1}{|S_k|} \sum_{\omega \in S_k} \mathbf{x}_\omega$
2. Select the representative scenario: $\omega_k^* = \arg \min_{\omega \in S_k} \|\mathbf{x}_\omega - \mathbf{c}_k\|_2^2$
3. Assign probability weight: $p_k = \frac{|S_k|}{N}$

K-means was primarily selected for its computational efficiency when handling large scenario sets, as it scales linearly with the number of observations. This approach seeks to maintain the correlation between the input parameters of revenues, as using K-means across all the input parameters and using the centroids of those would represent the mean values of the cluster, potentially missing some edge cases (tail events). The amount of cluster was determined by doing an elbow-test on different amount of clusters.

6 Data

This chapter describes the data used in this analysis, defining the base values and the sensitivity ranges used in answering the research questions.

6.1 Technical and Economic Data

This section describes the technical data of the wind farm and the boundaries associated with the physical project.

Table 6.1: Wind Farm Technical Specifications [57]

Parameter	Value	Source/Justification
Turbine Model	Siemens SWT-3.0-101	Representative of modern onshore turbines
Number of Turbines	10	Case study design constraint
Total Capacity	30 MW	Municipal-scale project
Hub Height	90 m	Modern on-shore turbine height
Project Lifetime	25 years	Standard for wind projects

The turbines used in estimation are a Siemens SWT 3.0 101 turbine with a hub height of 90 m, placed slightly outside Ringsted. The estimated LCOE 40 EUR/MWh will serve as the minimum strike price S^R that the seller is willing to accept, as any lower strike price would increase the chance of a negative NPV at the end of the asset's lifetime given the remaining period it would have to be fully merchant. The capacity 30 MW will serve as the maximum contract volume M^U ; the seller will not be able to hedge above its capacity. The lower bound M^R for contract size is 0. The maximum strike price that the buyer will accept will be a constant above their capture price $S^U = 1.2 \cdot CP^L$; the rationale behind this choice is that the buyer, at worst, wants to hedge against extreme prices.

6.2 Historical Data

This section describes the data used to estimate the distributions used for scenario generation.

6.2.1 Production and Load Profiles

The wind farm production is extracted from Renewables Ninja [58, 59] using weather data from Jan. 2020 - Dec. 2024, with the wind turbine specifications provided earlier, providing hourly production data. 2025 data is not included as the 'full' weather year is not available. Production statistics are provided in table 6.2

The load consumption data is extracted from Energinet's dataservice [60], providing hourly consumption data for each municipality, divided into industries, public sector,

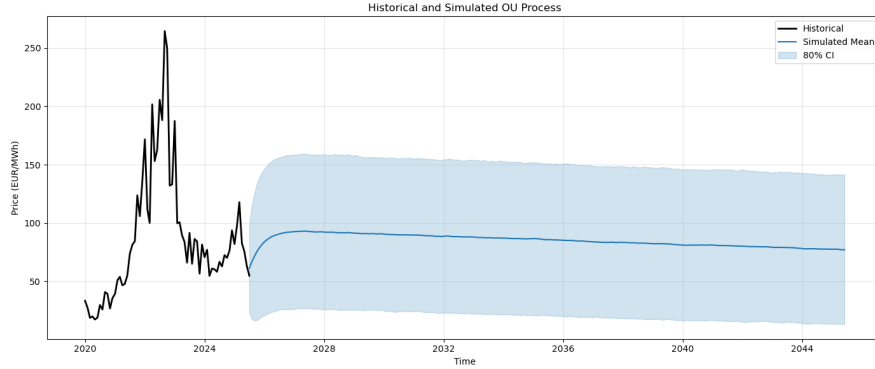


Figure 6.1: Price forecasts model using OU process

and private units. The industry consumption profile of Ringsted "329" is extracted from Jan. 2021 to Jun. 2025, as data availability starts from 2021. Consumption statistics are provided in table 6.2

Table 6.2: Production data Dec. 2020- Dec. 2024. Consumption data Dec. 2021 - Jun. 2025

	Unit	Median	Mean	Std
Wind Production P^G	GWh/year	100.00	97.90	4.65
Load Consumption P^L	GWh/year	151.24	129.83	34.49

6.2.2 Price Profile

Day-ahead electricity prices for the DK2 bidding zone are obtained from Energinet's Data Service [61], covering the period January 2020 to June 2025 Table 6.4. Using the OU process the following Figure 6.1 is made to model the future baseline prices, with the following parameters from Table 6.3. The average price is quite extreme in comparison to the 'expected' development presented in the introduction [3, 4], however, to investigate the holistic impact of prices, the accuracy of the prediction has less value given the synthetic nature of the case study. If this were a 'real' commercial project, accuracy would be important.

Table 6.3: Ornstein-Uhlenbeck estimated parameters

Parameter	θ Per year	μ_0 EUR/MWh	μ_t EUR/MWh per year	σ^2 EUR/MWh
Value	1.9655	91.95	-1.06	101.42

Table 6.4: Price and Capture Statistics from historical data 2020-2025

	Unit	Median	Mean	Std
Price λ	EUR/MWH	70.01	94.39	98.005
Correlation $\rho(\lambda, P^G)$	$\rho/year$	-	-0.48	0.12
Correlation $\rho(\lambda, P^L)$	$\rho/year$	-	0.14	0.12

6.2.3 Capture Rate Profiles

The capture rate for the wind production is calculated using the aforementioned DK2 day-ahead prices and the "OnshoreWind>50kW" category in DK2 between Jan. 2020-Jun. 2025, also sourced from the dataset, because the production data is dependently generated and does not reflect the actual 'realized' production rate of an onshore wind farm by using this dataset.

To simulate the load's stochasticity in realized prices, a yearly capture rate is constructed using day-ahead prices linked to consumption data from industries in the municipality in the period Jan. 2021 - Jun. 2025 [60]. While it is unknown whether the loads of the industries within the municipality are flexible or inflexible, it is a fair assumption that they should be investigated to account for this. The methodology for deriving the 'load capture rate' is based on the same capture rate as the wind farm.

6.2.4 Model

In total, 100.000 discrete scenarios are simulated using Monte Carlo. The simulated scenarios are then filtered by removing scenarios that contain a price not between P1 and P99. The scenarios are then reduced by K-means down to 5.000 representative scenarios, with each scenario having its own probability based on the number of scenarios assigned to the cluster. The model is solved using Gurobi version 12.0.03 on a desktop having a Ryzen 5600x CPU and 16 GB of RAM. The final model is formulated as a non-convex MINLP in Gurobi. The optimality gap lie between 0.004 % - 0.008 % depending on the risk aversion and negotiation power used. The average runtime per model is 5-7 seconds.

A summary of the statistics of the data are shown in Table 6.5 , their distribution have been attached in Appendix I The values shown in hours, should be seen as the expected value if the annual value were to be split across all hours in all the years.

6.3 Base Case description and sensitivities

This section establishes the baseline parameter configuration and describes the experimental framework designed to investigate the research questions systematically. The base case provides a neutral starting point that enables clear attribution of effects when individual parameters are varied in the sensitivity analysis.

Table 6.5: Statistical summary of the parameters used in optimization after scenario reduction to $K = 5000$. Revenues statistics shows the tenor average across scenarios, while the rest are statistic of all $[t, \omega]$

Parameter	Expectation	Std	Min	P5	P50	P95	Max
Price [EUR/MWh]	84.57	33.48	15.34	32.96	82.58	143.43	182.34
Capture Price Wind [EUR/MWh]	50.48	21.51	6.46	19.02	48.46	89.42	159.49
Capture Price Load [EUR/MWh]	88.07	34.97	15.53	34.28	89.51	149.70	200.44
Production [GWh/year]	94.56	6.93	64.50	83.12	94.57	105.91	127.00
Production [MWh]	10.79	0.79	7.36	9.49	10.80	12.09	14.50
Load [GWh/year]	144.24	5.34	122.48	135.58	144.15	153.18	169.06
Load [MWh]	16.47	0.61	13.98	15.48	16.46	17.49	19.30
Capture Rate [%]	59.70	8.80	29.34	46.00	59.22	74.96	107.44
Load Capture Rate [%]	104.14	3.11	90.07	98.90	104.19	109.18	116.63
Revenue G [Mio EUR]	95.48	10.87	55.84	78.03	95.23	113.64	143.61
Revenue L [Mio EUR]	-254.06	27.34	-366.45	-299.58	-253.69	-209.79	-147.05

6.3.1 Base Case Specifications

The baseline scenario assumes symmetric conditions between the wind farm developer (generator, G) and municipality (load, L) to isolate the effects of parameter variations. The key base case parameters are:

Risk Preferences

Both parties exhibit moderate risk aversion $A_G = A_L = 0.5$, representing a balanced preference between expected returns and worst-case risk protection.

6.3.2 Comparisons

As a way to compare the solutions in the base case and for the sensitivity analyses, an additional case is provided with a predetermined strike price set at the expected price $\mathbb{E}(\lambda_{t,\omega})$ for baseload and the expected capture price for the wind farm $\mathbb{E}(CR_{t,\omega}^G \lambda_{t,\omega})$ in the PAP while still optimizing for the optimal contract amount M/γ . These serve as a straightforward baseline negotiation outcome if risk isn't included in the valuation.

Bargaining Power

Equal negotiation strength is assumed with $\tau_G = \tau_L = 0.5$, meaning neither party has a structural advantage in the negotiation process. The negotiation power in reality is difficult to quantify and depends on the market.

Information Structure

Under the base case, both parties share identical expectations regarding the uncertain parameters included in the negotiation: Price, production, consumption, and the respective capture rates.

Contract Parameters

The analysis considers both baseload and pay as produced contract structures with a contract tenor of 20 years, reflecting a typical long term PPA duration to support project financing.

This base case will serve as a reference point for all subsequent sensitivity analyses, enabling direct comparison of how each factor alters negotiated outcomes.

6.3.3 Analysis Framework

To address the research questions systematically, five experiments are designed to isolate different aspects of the Nash bargaining framework:

Analysis 1: Risk Aversion Impact

Investigation of how varying risk aversion parameters A_L, A_G affect optimal strike prices, contract volumes, and revenue outcomes. This experiment maintains equal bargaining power $\tau_G = \tau_L = 0.5$. The analysis provides insight into how risk aversion affects the contract structures, and helps answer Questions 1 and 2.

Analysis 2: Negotiation Power Analysis

Analysis of how asymmetric bargaining power τ influences contract terms and surplus distribution in the base case, but also under different risk aversion combinations. This analysis systematically shifts negotiation advantage from complete generator dominance $\tau_G = 1$ to load dominance $\tau_L = 1$, directly addressing Research Question 2 regarding power asymmetries.

Analysis 3: Asymmetric Information Analysis

Examination of contract outcomes when players hold different expectations about key parameters, transitioning from complete to incomplete information. This experiment addresses Research Question 3 by analyzing how expectation biases affect negotiated outcomes:

1. Price expectation bias factor K_λ^G, K_λ^L are applied in the optimization models.
2. Production expectation bias factor K_P^G, K_P^L are applied in the optimization models.

The new biased price or production replaces the baseline in the respective utility function and CVaR, depending on the perspective from which the bias is taken. The expectation differences should allow for meaningful interpretation of how small differing forecast uncertainties affect the negotiation outcome in terms of the negotiated strike price and amount.

Analysis 4: Parameter Sensitivity: Symmetric and Asymmetric

Sensitivity analysis examining how the input parameters affect the contract terms on an absolute scale. Parameters are varied uniformly across both parties to assess the general impact in the symmetric, and asymmetrically to identify its absolute impact. Two methods of varying the parameters are investigated in the symmetric case: 1) Varying the expectation, and 2) varying the volatility (Standard deviation) of the parameter, the asymmetric will only be varying the expectation. The distribution is not resampled with the bias, instead the expectation of the distribution is shifted through 6.1 without changing its standard deviation. Similarly, the standard deviation is altered without changing the expectation through 6.2. Where in both cases X represents a distribution and 0.01 is a 1% shift in the distribution.

$$\tilde{X} = X + 0.01 \cdot \mathbb{E}(X) \quad (6.1)$$

$$\tilde{X} = \mathbb{E}(X) + (1 + 0.01) \cdot (X - \mathbb{E}(X)) \quad (6.2)$$

The parameters and their sensitivity analysis are the following for the symmetric analysis $[-40\%, 40\%]$:

- **Price levels and variability:** Variation in expectation and standard deviation in the baseload prices $\lambda_{t,\omega}$.
- **Wind production levels and variability:** Variation in both expectation and standard deviation of wind production $P_{t,\omega}^G$.
- **Consumption levels and variability:** Variation in both expectation and standard deviation of consumption $P_{t,\omega}^L$.
- **Capture Rate levels:** Variation in expected value of the capture rate of the wind production and load consumption $CR_{t,\omega}^G, CR_{t,\omega}^L$.

The parameters and their sensitivity analysis are the following for the asymmetric analysis, the range of values lie in $[-10\%, 10\%]$:

- **Price levels:** Variation in expectation in the baseload prices $\lambda_{t,\omega}$.
- **Wind production levels:** Variation in expectation of wind production $P_{t,\omega}^G$.

The analysis is aggregated into a single comparison by evaluating the change in strike price and contract amount (x), and assessing the elasticity between these parameters and the changed parameters (y), see Equation 6.3. The elasticity is defined as the sensitivity of one variable to another. A Log-linear model is used to estimate the elasticity of the parameters to the strike price and contract amount.

$$\epsilon_{x,y} = \frac{\% \text{change in } x}{\% \text{change in } y} \quad (6.3)$$

The analysis is presented through base-case risk aversions of $A_G = A_L = 0.5$ and other risk combinations of risk aversions, as they can likely lead to higher or lower elasticities. Due to the number of parameters and combinations of risk-aversions, the scenario size is reduced to 500. This scenario size still maintains the same probability structure of revenues, see Appendix K.

6.3.4 Sensitivity Parameter Table

The sensitivity ranges are applied as specified in 6.6, enabling an analysis of how the parameter uncertainty propagates through the modeling frame, influencing the optimal design solution.

The complete list of the sensitivity analysis parameters and their ranges are shown in Table 6.6. Appendix contains sample distributions of the base case input parameters.

Table 6.6: Parameters and Sensitivity Ranges

Parameter	Notation	Unit	Base Value	Sensitivity Range
Risk Aversion	A_G, A_L	-	0.5, 0.5	[0, 0.1, 0.25, 0.5, 0.75, 0.9, 1]
Negotiation Power	τ	%	0.5	[0, 0.25, 0.5, 0.75, 1.0]
Price Scenarios	$\lambda_{t,\omega}$	EUR/MWh		60% - 140% of $\mathbb{E}(\lambda_{t,\omega})$, $\sigma(\lambda_{t,\omega})$
Wind Production	$P_{t,\omega}^G$	GWh		60% - 140% of $\mathbb{E}(P_{t,\omega}^G)$, $\sigma(P_{t,\omega}^G)$
Load Consumption	$P_{t,\omega}^L$	GWh		60% - 140% of $\mathbb{E}(P_{t,\omega}^L)$, $\sigma(P_{t,\omega}^L)$
Wind Capture Rate	$CR_{t,\omega}^G$	%		60% - 140% of $\mathbb{E}(CR_{t,\omega}^G)$
Load Capture Rate	$CR_{t,\omega}^L$	%		60% - 140% of $\mathbb{E}(CR_{t,\omega}^L)$
Price Bias	$\lambda_{t,\omega}^C, \lambda_{t,\omega}^L$	EUR/MWh	0	90% - 110% of $\mathbb{E}(\lambda_{\omega,t})$
Production Bias	$P_{t,\omega}^{G,G}, P_{t,\omega}^{G,L}$	GWh	0	90% - 110% of $\mathbb{E}(P_{t,\omega}^G)$

7 Results

This chapter presents the results of the experiments conducted to answer the research questions, as outlined in the methodology. The results are presented in the order as they are described.

7.1 Base Case Results

Section 7.1 establishes baseline Nash bargaining outcomes to address Research Question 1: determining fair strike prices and contract volumes for both baseload and pay as produced PPAs. The base case assumes symmetric information, equal bargaining power $\tau_G = \tau_L = 0.5$, and moderate risk aversion $A_G = A_L = 0.5$ for both parties. These results serve as the reference point for all subsequent sensitivity analyses and demonstrate how the Nash bargaining framework deviates from naive expected-price approaches.”

The base result of both contract BL and PAP are presented in Table 7.1. As a comparison to the calculated results, the expectation of the baseline price $\mathbb{E}(\lambda_{t,\omega})$ and capture price of wind $\mathbb{E}(CR_{t,\omega}^G \lambda_{t,\omega})$ are set as a fixed value for strike price and optimize still optimizing the contract amount M and γ . The Δu reports the difference in gained utility from the disagreement points (d_G, d_L) with the Nash product reporting $\Delta u_G \cdot \Delta u_L$.

The BL contract outcome for S and M is higher than the expected price model, as this does not account for the risk aversion of each player. As such, in the Nash Bargaining framework the most ‘fair’ result using the expected price 84.57 EUR/MWh provides a solution that benefits u_L more than u_G given Δu_L is 12.31 and Δu_G is 6.33. On the other hand, optimizing for both S and M results in an equal share of Δu_G and Δu_L as a result of the negotiation, leading to an overall higher strike price at 89.15 EUR/MWh and Nash product. The base PAP contract reports a higher strike price of 50.83 EUR/MWh than the expected capture price of 50.48 EUR/MWh, while maintaining the same contract share of $\gamma = 100\%$. The minor difference between strike prices can be explained in that the value delivered through a PAP is diminished by the capture rate CR^G . The differences are also not symmetrical as the utility gain is not a fixed value swap like the BL because the production delivered from the PAP is stochastic. From both Nash Bargaining Solutions, what can be observed is that the risk aversions of both players are incorporated into the negotiation outcome, making the contract proportionally fair for both of them, considering their disagreement points.

The graphical representation of the Barter Set B is shown in Figure 7.1 showing how the optimal solution lies within the barter set of the baseload contract. Here, the two curves V_1 and V_2 are drawn from the disagreement point with the lower and upper strike price bounds S^R and S^U . The proportional gain satisfies 4 for both lines at the same contract size of 12.14 MWh. The straight line drawn between the two points has a

Table 7.1: Base results with $A_G = A_L = 0.5$

Model	Strike Price S (EUR/MWh)	Contract Amount M/γ (MWh / %(MW))	Δu_G	Δu_L	Nash Product
Baseload	89.15	12.14	10.00	10.01	97.19
$\mathbb{E}(\lambda_{t,\omega})$	84.57	7.47	6.33	12.31	77.94
Pay as Produced	50.83	100 % (30 MW)	9.80	9.60	97.63
$\mathbb{E}(CR_{t,\omega}^G \lambda_{t,\omega})$	50.48	100 % (30 MW)	9.14	10.62	97.19

slope of -1 as defined in 1 and together with the vertical and horizontal line drawn from the disagreement point, the grey triangle forms the Barter set B . The dashed-purple line is the utility curve of u_G and u_L mapped with the optimal strike price $S^* = 89.15 \text{ EUR/MWh}$, varying the contract amount M . At a low contract amount M , the solution benefits the generator more; however, as M is increased, the marginal gain begins to benefit the load more. At the optimum (red-dot), the ratio between u_G and u_L crosses the boundary.

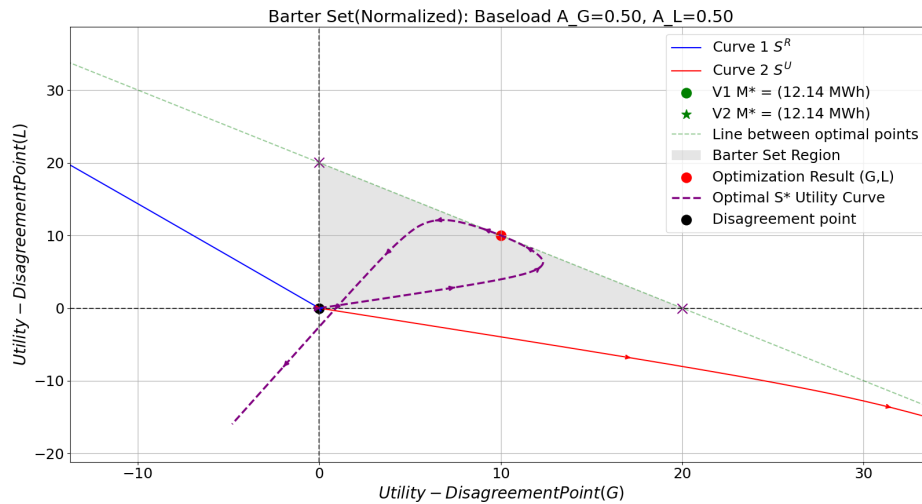


Figure 7.1: Baseload base case barter set with solution.

Graphically, the PAP solution is plotted in Figure 7.2. Here, visually, the curves V_1 and V_2 satisfy the initial condition of F.3, but the unique optimum of the Nash Bargaining solution has not been reached. If γ could go over 1, then eventually it would cross the boundary set by 4. Visually, this can also be seen through the purple line outlining the utility $\mathbf{u} = (u_G, u_L)$ with the optimal strike price S^* as it has not plateaued yet.

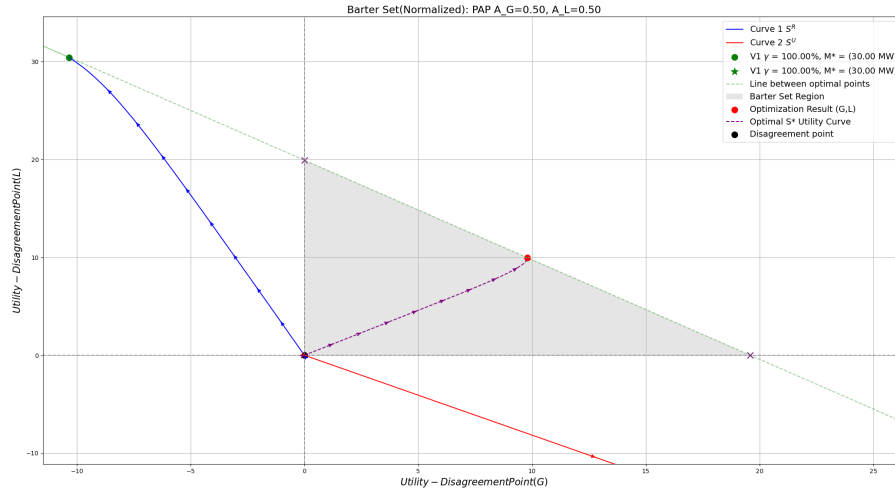


Figure 7.2: Case 3: PAP Base Case. The Barter set is practically a triangle as 1 for PAP is very close to -1 .

7.2 Risk-Aversion

Section 7.2 directly addresses Research Question 2 by examining how differences in risk aversion between the generator and load influence negotiated contract outcomes. The analysis systematically varies risk aversion parameters while maintaining equal bargaining power to isolate the pure effect of risk preferences on strike prices and contract volumes

7.2.1 Baseload Interactions

Figure 7.4 presents the Nash bargaining outcomes for baseload contracts across varying risk aversion parameters with strike price S on the left table and contract amount M on the right. With the risk aversions varied between $[0, 1]$.

Starting with Strike Price, looking at the table from bottom-to-top, with a fixed A_G : increasing the risk aversion of the load A_L will systematically increase the strike price S , with the lowest value at 78.63 EUR/MWh and the highest 95.18 EUR /MWh. With each increase in A_L they become more risk-averse and reflect an increased willingness to pay a premium for price certainty.

Conversely going left-to-right with a fixed A_L , the generator's risk aversion shows a non-systematic relationship with strike price. For $A_L < 0.5$, increasing A_G leads to lower strike prices as the generator becomes more willing to accept a price discount. However, when $A_L \geq 0.5$, this relationship becomes irregular as the strike price no longer follows a steady decrease; instead, the strike price increases with some high values of risk aversion from both parties. In these circumstances, the CVaR dominates.

Plotting the disagreement points as a function of risk aversion provides the following Figure 7.3. Both lines of the generator and load lie on each other as the change in rate between risk aversions is the same given the utility functions Equation 4.5 and Equation 4.6, however on an absolute scale they significantly differ between high and low. The generator has a difference of 22 Mio EUR between risk-neutral ($A_G = 0$) and fully risk averse ($A_G = 1$), whereas the load has a difference of 60 Mio EUR. This makes the value A_L more 'worth' in terms of opening space for negotiation compared to a change A_G in absolute terms.

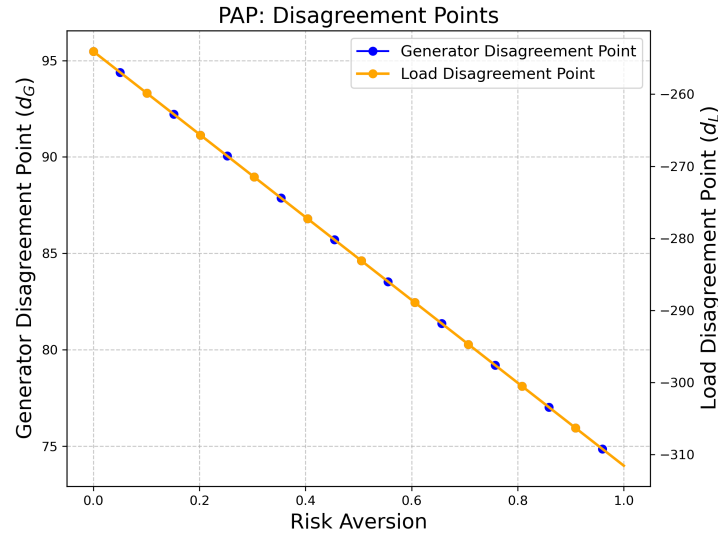


Figure 7.3: Disagreement points as function of risk aversions A_G, A_L

Examining the contract amount with the same structure reveals additional interesting dynamics. If the relative risk aversion favors A_G , eg, ($A_G < A_L$), then the contract amount is higher as the generator is willing to take on more volume and is less concerned with the downside risk as it bargains a higher strike price than expected. The generator is even willing to assume a burden that is beyond its expected production as seen in Table 6.6 as it expects a potential revenue upside from the fixed transaction, even with the buyback. Since the load seeks stability, the contract volume becomes high, and strike prices adjust to reflect the load's risk aversion, which drives up the price. Flipping the relative risk-aversion to $A_G > A_L$, the opposite happens as the generator is looking to hedge low revenue, but the load is less concerned with its downside risk. Since the generator wants stability and cannot gain an upside, it wants to ensure it can deliver keeping the amount low. So the strike price adjusts to the generator's risk aversion.

Looking at the diagonal of both tables, it is observed that the contract amount is unchanged, but the strike price is increased as reflected through their absolute risk aversion.

The bottom left, where $A_G = A_L = 0$ is blank because no feasible contract exists

and is due to the chosen revenue and utility functions. Without risk considerations, the baseload PPA reduces to a zero-sum financial swap where the gradient is $du_L/du_G = -1$ throughout the feasible region, providing no space for mutually beneficial agreement. This mathematical point has a neat practical implication: Some degree of risk aversion from at least one party is necessary for the PPA to create value. For $A_G = 0.1, A_L = 0$, the proportional gain does not satisfy condition 4.33, the generator isn't risk-averse enough to comply with A_L .

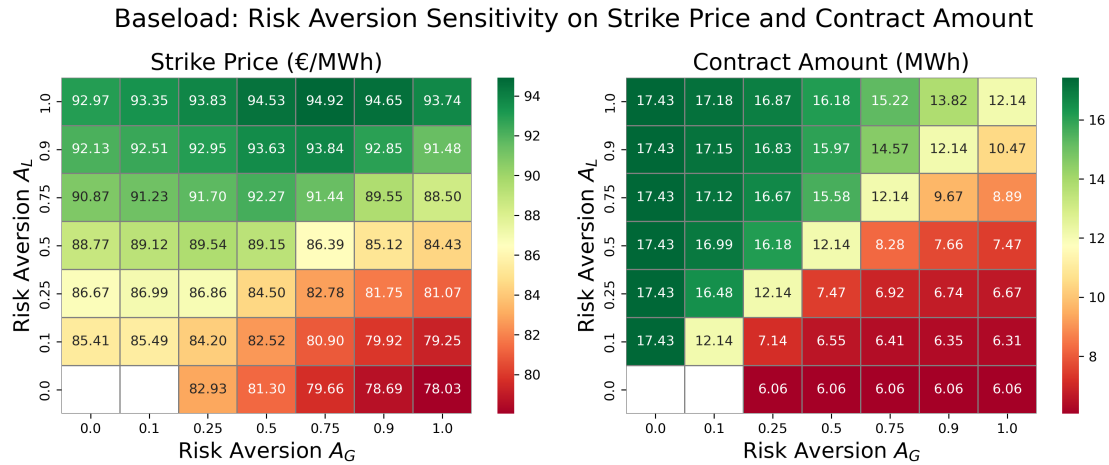


Figure 7.4: Baseload Risk Sensitivity

Baseload Risk Transfer

Figure 7.5 shows the earnings distributions across contract tenors for varying risk aversion parameters, with generator risk aversion fixed at $A_G = 0.5$. The analysis progresses from the disagreement scenario (no contract) through contracts with predetermined strike prices to optimized Nash bargaining solutions with increasing load risk aversion A_L .

The earnings analysis reveals a systematic risk transfer pattern driven by relative risk aversion. Under the no-contract scenario, the generator faces a revenue range of 55-145 Mio EUR (median: 96.5 Mio EUR), while the load experiences costs ranging from -365 to -150 Mio EUR (median: -256 Mio EUR). These baseline distributions exhibit standard deviations of 14.5 Mio EUR and 36.4 Mio EUR, respectively.

As the load's risk aversion increases from $A_L = 0$ to $A_L = 1$, the contract systematically transfers price risk from load to generator. At low load risk aversion $A_L = 0.1$ the generator's earnings distribution compresses to 80-110 Mio EUR. Conversely, the load achieves moderate risk reduction with costs constrained to -320 to -195 Mio EUR.

The most pronounced risk redistribution occurs at maximum load risk aversion $A_L = 1$. Here, the load achieves near-complete hedging with costs tightly bounded between -295 and -270 Mio EUR. This extensive risk protection comes at the cost of transferring substantial earnings volatility to the generator, whose revenue distribution expands to

55-185 Mio EUR. The generator's earnings variance is increased, but given their risk aversion, they are satisfied with the increased volatility.

The Nash bargaining framework incorporating risk aversion generates substantially different contract terms than fixed strike prices based on expected values, which fail to account for risk-averse load preferences.

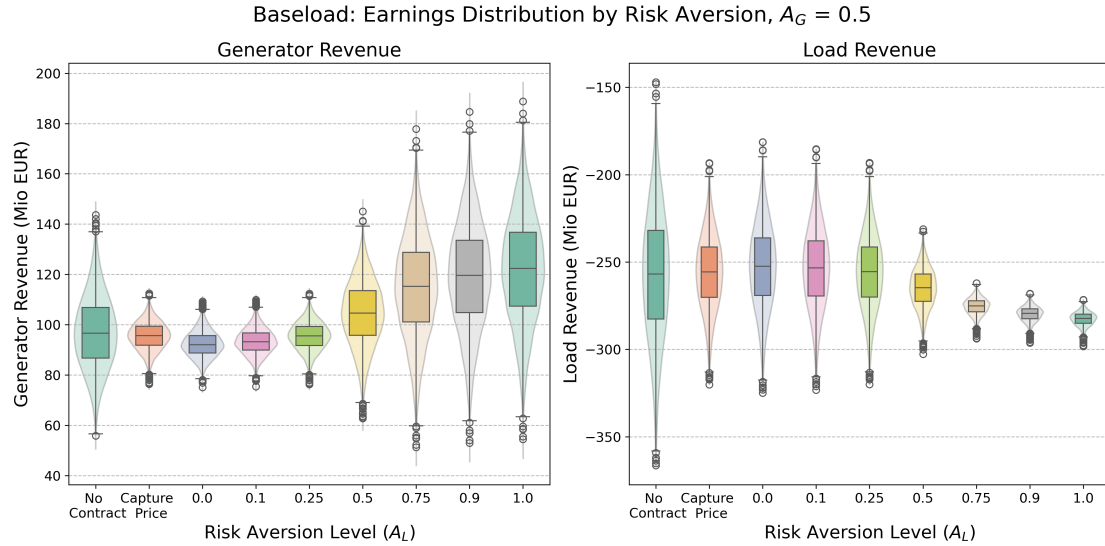


Figure 7.5: Baseload earnings given fixed risk aversion $A_G = 0.5$

7.2.2 Pay as Produced Interactions

Figure 7.6 shows the Nash bargaining outcome of the PAP contract with strike price S on the left table and contract amount $\gamma \cdot \bar{P}^G$ (MW) on the right as the risk aversions are varied between $[0, 1]$. Much like the BL contract, going from bottom-to-top: with a fixed A_G and systematically increasing A_L increases the strike price S . Going from left-to-right, keeping risk aversion A_L fixed, the strike price S consistently decreases. Maintaining that all else is equal, as each player becomes more risk-averse, the negotiated strike price moves in a direction that favors the other. Interestingly, the strike price S is generally kept below the capture price of wind 50.48 EUR/MWh if the risk aversion favors the load. If the relative risk aversion favors the generator, the strike price increases above the capture price.

The contract amount remains mostly constant ($30MW \equiv \gamma = 1$) across all combinations, except when the load is completely risk-neutral since the 'ideal' Nash Bargaining solution still has not been reached. There is still some proportional gain to be had if γ could be larger than 1.

PAP Risk Transfer

Looking over the earnings in Figure 7.7 for PAP contracts reveals a fundamentally different risk allocation pattern compared to the baseload. The earnings distributions demonstrate asymmetric risk mitigation across increasing load risk aversion levels. For

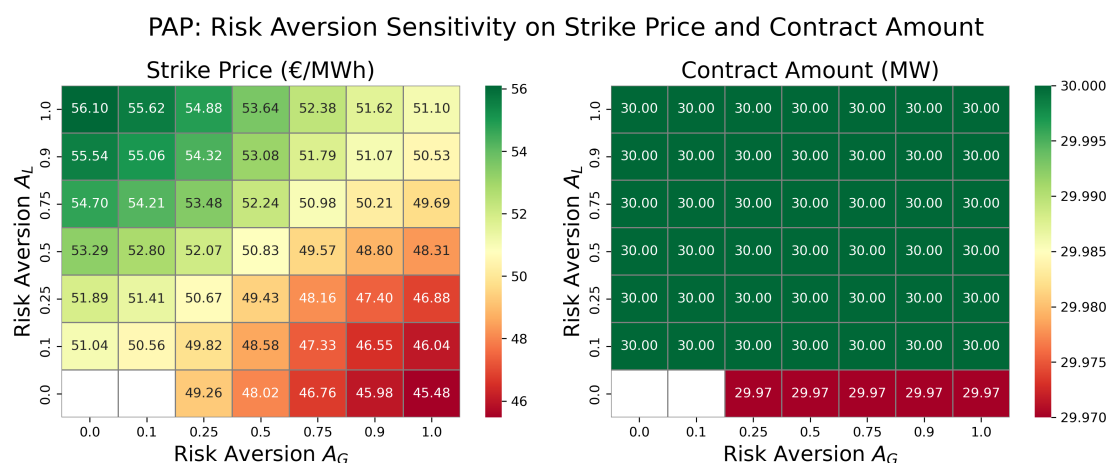


Figure 7.6: PAP Risk Sensitivity

the generator, PAP contracts deliver immediate and substantial variance reduction regardless of contract terms. At $A_L = 0.1$, generator earnings stabilize to 91.9 million EUR with standard deviation compressed to just 1.6 million EUR. This near-complete risk elimination persists consistently across all risk aversion scenarios. Conversely, the load experiences markedly limited risk reduction benefits. Even at maximum load risk aversion $A_L = 1$, the load achieves costs of -260 million EUR with a standard deviation of 23 million EUR. This modest improvement contrasts sharply with the generator's near-complete risk elimination and highlights the contract's asymmetric hedging effectiveness. The generator achieves substantial volatility reduction by eliminating price risk on its production. However, the load's risk reduction is limited because they remain exposed to the correlation between generator production and market prices through the capture rate effect. This fundamental asymmetry explains why PAP contracts typically trade at discounts relative to expected capture prices, and why they are being phased out.

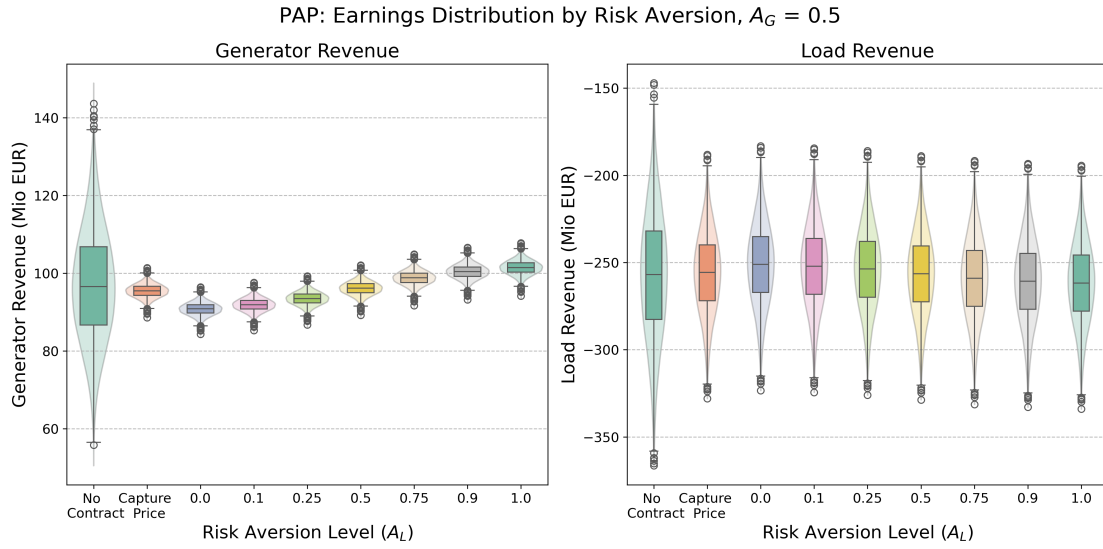


Figure 7.7: Pay as produced earnings given fixed risk aversion $A_G = 0.5$

7.2.3 Nash Product as an indicator of risk allocation

Lastly, in the analysis of risk aversions, we consider the Nash Product of the two contracts in Figure 7.8. Providing a physical meaning to the Nash product, which is the product of gained utilities, is challenging and, to an extent, a matter of interpretation. In this interpretation, we aim to demonstrate that the differing risk allocation between BL and PAP is reflected in this value. For both contract types, low risk aversion leads to a small Nash product as there is a limited space to negotiate. In the case of BL it produces the highest Nash products with low risk aversion of A_G and high A_L in the upper left corner 7.8a. In this space, the load is so risk-averse and the generator risk-neutral that the generator is willing to take on all the consumption of the load at a high price. This opens up a large space of potential negotiation outcomes as the generator is willing to transfer the risk the load experiences to itself. The Nash product reflects this.

On the other hand, PAP's highest Nash products exist in the upper right corner in Figure 7.8b where the risk aversion of both is high. This also corresponds to intuition as from a risk perspective, the PAP provides overall less risk mitigation to the load and there is less gain for the generator at low risk aversion, even with the highest strike prices in the table. Instead, at high aversions of A_G , they start valuing their worst-case outcome of the capture price, emphasizing their willingness to bargain. As such, the PAP from the perspective of risk aversion through Nash Bargaining shows that this contract is more oriented towards risk-averse generators and loads.

Visually, the bartering space can be seen in 7.9a and 7.9b. These plots show the bartering space for a select combination of risk aversions A_G, A_L , where it has been normalized by subtracting the disagreement point (d_G, d_L) from (u_G, u_L) . The highest Nash products correspond to the largest bartering set, as discussed. Inside this space

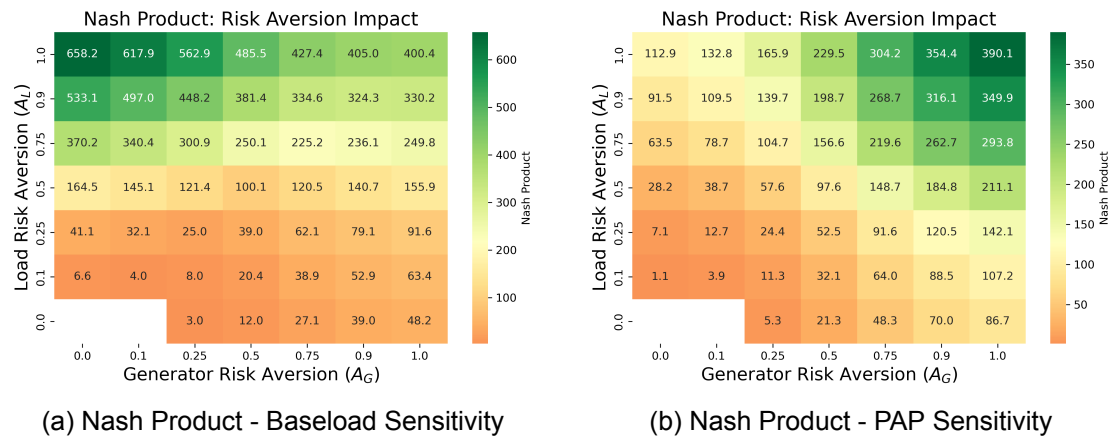
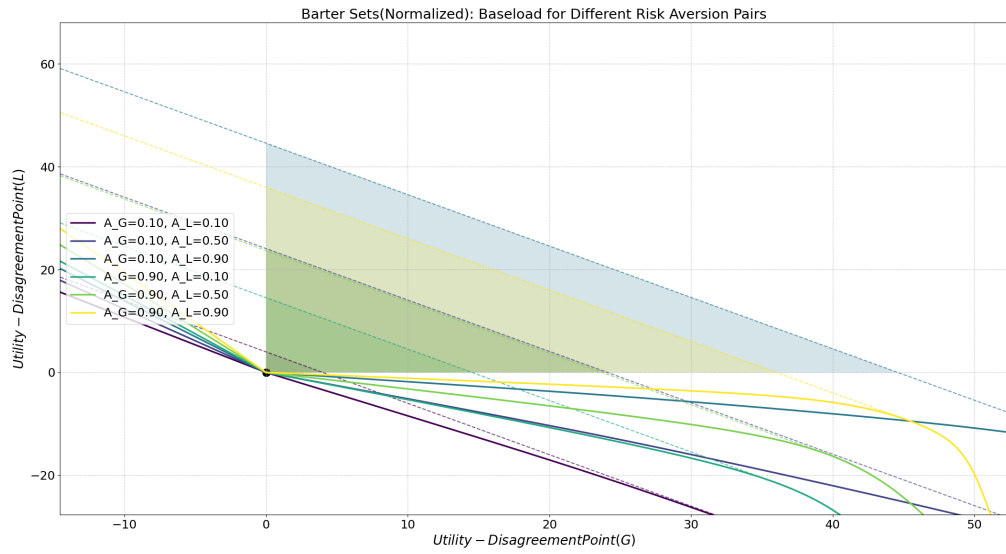
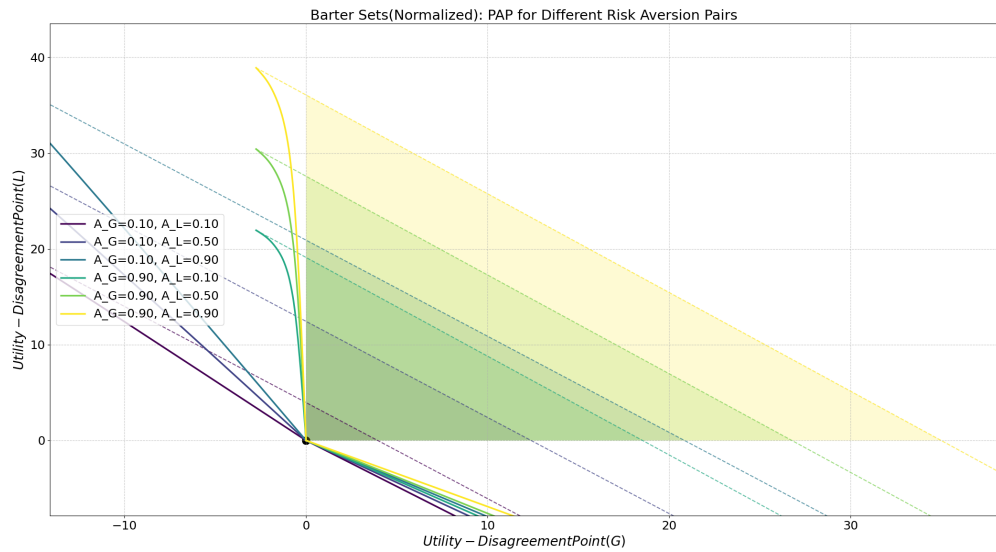


Figure 7.8: Nash Product - Sensitivities

are all viable potential contracts. As indicated by the Nash product, the bartering set B increases in size with higher values. The optimal solution presented by the model is the 'predicted' unique optimum; however, in a real situation, any of the solutions within this space would provide benefit to both players, albeit not 'fairly'.



(a) Barter sets of combinations of risk aversions in baseload. The biggest triangle(blue) is $A_G = 0.1, A_L = 0.9$, smallest (purple) is $A_G = 0.1, A_L = 0.1$.



(b) Barter sets of combinations of risk aversions in pay as produced. The biggest triangle(yellow) is $A_G = 0.9, A_L = 0.9$, smallest (purple) is $A_G = 0.1, A_L = 0.1$. Mutual high risk aversions have larger barter sets (Nash Products).

Figure 7.9: Barter set visualization in response to risk aversions A_G and A_L

7.3 Asymmetric Nash Bargaining - Negotiation Power

Section 7.3 continues addressing Research Question 2 by analyzing how asymmetric bargaining power τ influences contract terms and surplus distribution. Different combinations of risk aversions are included, also to reveal how utility risk aversion interacts with bargaining power across the barter set. The negotiation power ranges from being fully allocated to the generator ($\tau_G = 1$) to the load dominating ($\tau_L = 1$), following the expression $\tau_G = 1 - \tau_L$. Changing the negotiation power implies that the player with higher power will gain more than the counterparty.

7.3.1 Baseload - Negotiation Power

Figure 7.10 shows how the strike price decreases given that the negotiation power is systematically shifted from τ_G to τ_L in the baseload. With enough bargaining power, the strike price is negotiated to be 84.44 EUR/MWh, just below the expected price 84.57 EUR/MWh. The colors are grouped by sharing the same fixed risk aversion of A_L . The dashed line in Figure 7.10 indicates the expected value of prices (left) and production (right). Having a lower risk aversion than the other results in a strike price in your favor. Going from left-to-right, with the load having no bargaining power, having a high risk aversion leads to a very high strike price, even above the expected value with full bargaining power $\tau_L = 1$. The decrease in S can be attributed to the size of the Nash product 7.8 and the resulting barter set. Higher Nash product leads to a higher potential span of prices. Another interesting observation is that the contract amount does not change with a change in the negotiation power; it remains constant across all combinations. This is consistent with the Nash bargaining solution of the baseload contract as the parameters of the problem are not changing, the allocation between the utilities is just shifting by the strike price. As such the negotiated outcome moves between the intersections of the horizontal and vertical lines drawn from the disagreement on the straight line drawn between the maxima of the curves V_1 and V_2 that has the slope -1 in Figure 7.1, whose size is shown through the Nash product.

Given that the strike price is the only moving variable in the baseload solution, the overall earnings distributions in Figure 7.11 maintain their shape while shifting vertically as negotiation power transfers between parties. This fundamental characteristic of Nash bargaining manifests clearly in the numerical results. When the generator holds full negotiation power $\tau_G = 1$, average revenues reach 114.33 million EUR with a standard deviation of 12.93 million EUR. As power shifts to the load, generator revenues decrease monotonically to 94.33 million EUR when $\tau_G = 0$, while maintaining the same standard deviation of 12.94 million EUR. Table of values is provided in Appendix H.

The load experiences the precise inverse pattern, confirming the zero-sum nature of surplus distribution in Nash bargaining. Under generator dominance τ_G , load costs increase to 274.84 million EUR annually, while complete load negotiation power τ_G reduces costs to 254.83 million EUR. When $\tau_G = 0$ the generator's utility equals exactly their disagreement point $u_G = d_G$, as all negotiated surplus has been allocated to the load, this still satisfies the Nash bargaining feasibility requirement $u_G \geq d_G$

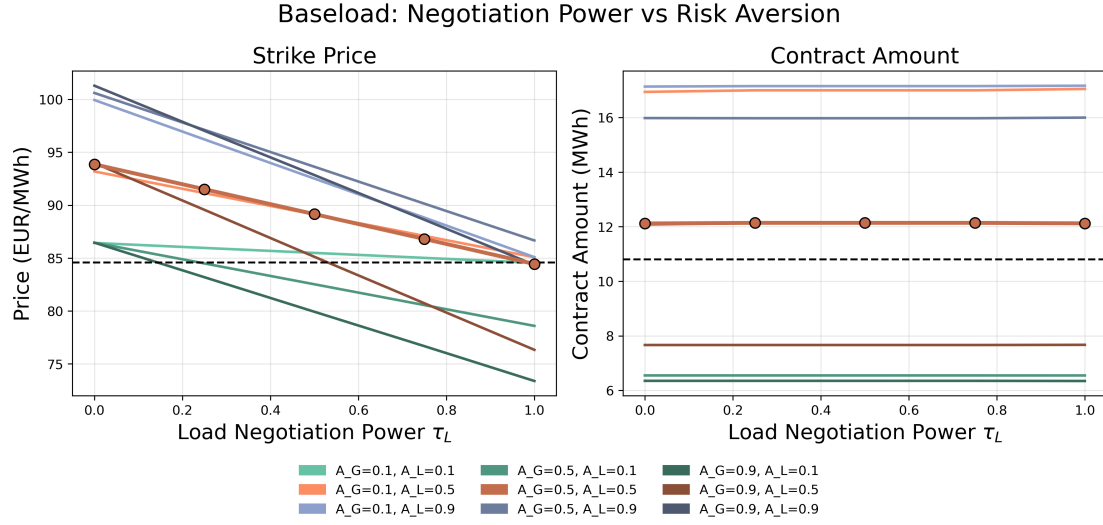


Figure 7.10: Negotiation power τ varied across risk aversions. Left plot is the strike price in EUR/MWh. The right plot shows the contract amount M . Dotted line represents the base case

7.3.2 Pay as Produced - Negotiation Power

Figure 7.12 shows how the strike price decreases from 56.10 EUR/MWh to 45.56 EUR/MWh given that the negotiation power is systematically shifted from τ_G to τ_L in the PAP. The dashed line indicates the expected value of capture prices (left) and the contract share γ (right). The figure shows a similar relationship where bargaining power increases or decreases the strike price depending on who has more, but now the contract amount can vary slightly. The risk aversions now follow an apparent path in that the absolute value of risk aversion determines the endpoints, given full negotiation power, and high risk aversions lead to significantly varying prices.

In most circumstances, the contract share is not affected given that the barter set is unbounded given 4, even with the lemma 1 and Equation 4.31 - 4.32. However, as observed in Figure 7.6 at $A_L = 0$ the barter set is bounded, and we can observe that the strike price now influences the contract share γ in 7.12. The change can be attributed to the fact that given a high strike price, the load would prefer a smaller γ , and on the flip side, at lower strike prices, the generator would prefer a lower contract amount γ given the change in negotiation powers. This is similar to the relationship shown with risk aversion.

An interesting observation is that the strike price reaches the capture price 0.53 faster at lower load bargaining powers τ_L , assuming the same risk-aversion as the BL contract. Given the risk profile of the PAP, the result is not unsurprising as the load bears the shape and the effect of capture rates depresses the value gain for the load.

When the generator holds full negotiation power $\tau_G = 1$, average revenues reach 108.48

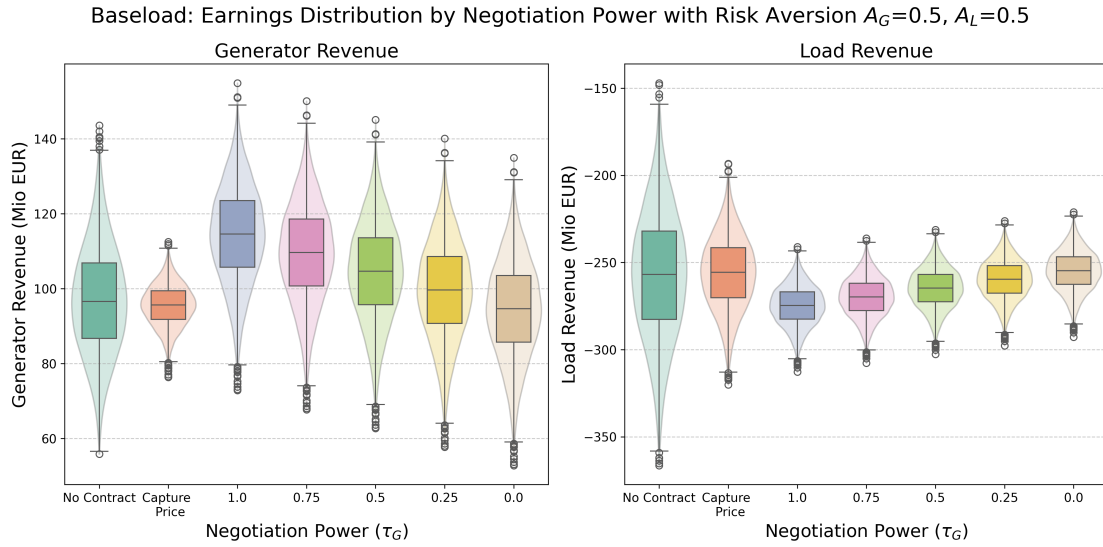


Figure 7.11: Baseload negotiation earnings given fixed risk aversion $A_G = A_L = 0.5$

million EUR with a standard deviation of 1.76 million EUR. As negotiation power shifts to the load $\tau_G = 0$, generator revenues decrease to 92.34 million EUR while maintaining the same standard deviation of 1.63 million EUR. This 17% revenue difference demonstrates that PAP structures create smaller wealth transfer opportunities compared to baseload contracts, even if the generator could set their best feasible price. Generator-dominated negotiations $\tau_G = 1$ result in load costs of 268.91 million EUR, while load negotiation power $\tau_G = 0$ reduces costs to 253.81 million EUR. Furthermore, the load cost standard deviation remains constant at 22.9 million EUR across all negotiation scenarios, as such, even if the load could choose the lowest feasible price, their hedge is ultimately reliant on the wind production reducing its risk hedging capability.

Negotiation power and risk aversion both have a direct effect on the negotiation outcome. In real negotiations, it would not be unrealistic that these parameters play in tandem with one another. Given a party coming into a negotiation, and that they are negotiating from a high risk aversion perspective, a natural interpretation would be that they have overall less negotiation power. Conversely, the same would be the case; however, given a situation where both have a similar risk profile, outside market conditions help in attributing the value of negotiation power. Finally, graphically, the negotiation power can be viewed as an alteration of allocation within a given barter set, whereas risk aversion alters the size of the barter set.

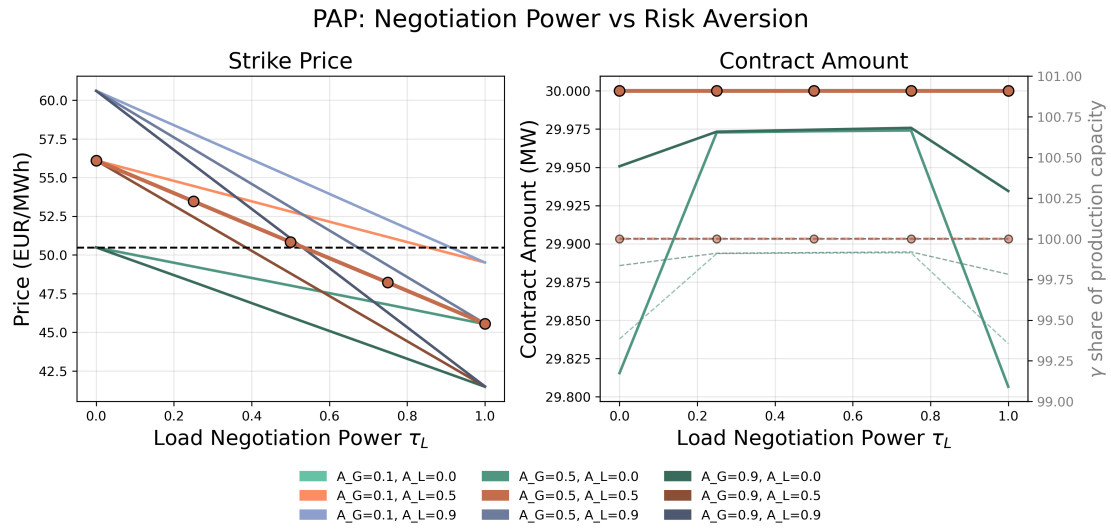


Figure 7.12: Pay as produced negotiation power τ varied across risk aversions. Left plot is the strike price in EUR/MWh. The right plot shows the contract amount γ , dashed lines represent γ while full colors are $\gamma \cdot \bar{P}^G$. The dotted line represents the base case.

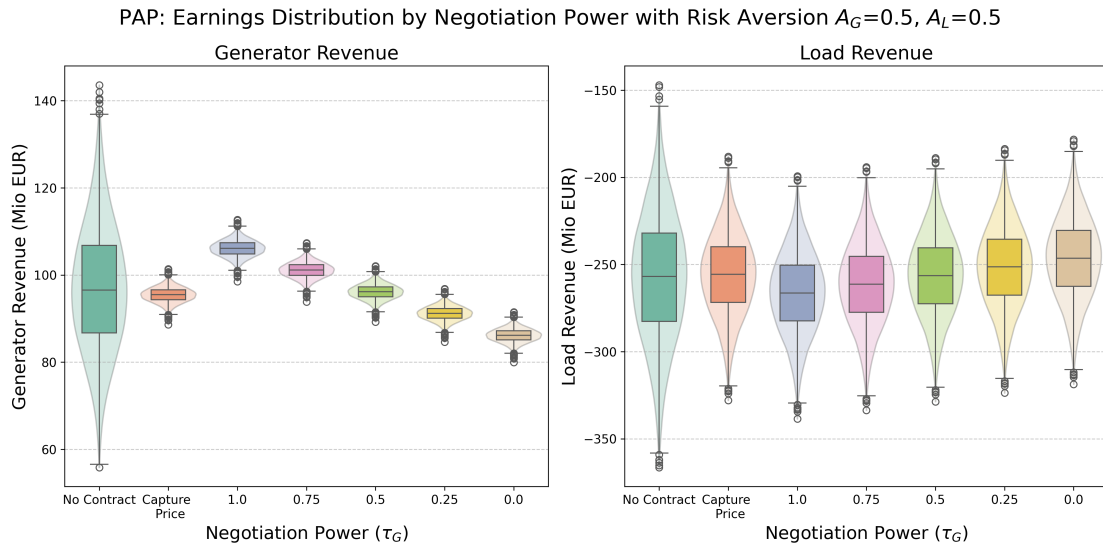


Figure 7.13: Baseload negotiation earnings given fixed risk aversion $A_G = A_L = 0.5$

7.4 Asymmetrical Information

Section 7.4 addresses Research Question 3 by examining how asymmetric information between parties affects negotiated outcomes and contract feasibility. This analysis transitions from the complete information Nash bargaining framework to incomplete information scenarios, where parties hold different expectations about stochastic parameters.

7.4.1 Price Asymmetry

Price expectation asymmetry directly tests how divergent price forecasts affect contract viability and terms. The analysis varies in prices biases K_λ^G (or K_λ^L) between -10% and +10% of the expected price $\mathbb{E}(\lambda_{t,\omega})$. It is implied that a higher value of K_λ^G (or K_λ^L) bias results in an expectation of higher prices and a negative value results in smaller prices. The results are kept at base risk-aversions of $A_G = A_L = 0.5$, but price bias results for the other risk-aversions are qualitatively similar to the base case in their behavior.

Baseload Outcomes

Figure 7.14 shows the Nash bargaining outcomes for the strike price S and contract amounts M as the price bias factor K_λ^G and K_λ^L are each systematically varied between $-10\%\mathbb{E}(\lambda_{t,\omega})$ to $10\%\mathbb{E}(\lambda_{t,\omega})$. The central no-bias case reflects the base-case results and serves as the reference point of analysis. Moving diagonally in the strike price heatmap strike prices increase from 85.48 to 97.61 EUR/MWh, however the contract amount remains around 12.15-12.12 MWh. Relative to the reference point, if the generator expects higher prices (going right), then the strike price S (generally) increases, and the contract amount M decreases; if they instead expect lower prices (going left), then S decreases and M increases. From the load's perspective, going up and down, higher expectation leads to higher S and higher M , whereas if it goes down, it results in lower S and M . If the generator and load have significantly differing views on future prices, no contract is signed. At this point, the generator sees no purpose in a contract, given the expectation of the mercantile market, as they enter the negotiation with a higher disagreement point. Likewise, the load follows a similar reasoning, as low prices reduce the necessity for a hedging strategy

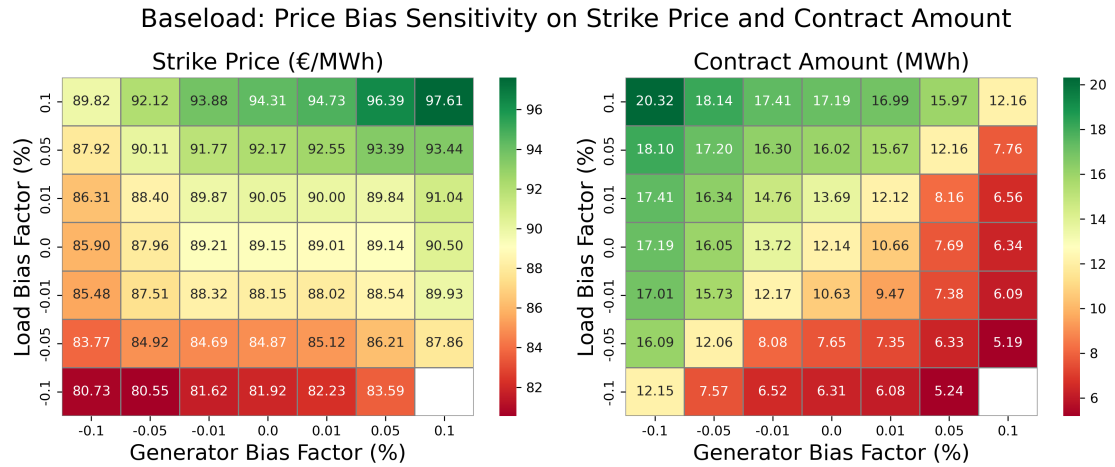


Figure 7.14: Price biased K_{λ}^G applied between $[-10\%, 10\%]$ for baseload at $A_G = A_L = 0.5$

Pay as Produced Outcomes

Figure 7.15 shows that PAP contracts exhibit a similar relationship with the baseload structures. The strike price responses follow the same directional trends in that an increase in either view of an increase in price leads to an overall higher S , and a lower expectation decreases S . Moving diagonally from lower to higher price expectations increases strike prices from 45.77 to 55.89 EUR/MWh. The contract amount γ remains the same due to the optimal Nash solution not being reached in most cases. However, at sufficient variation in observed prices, the Nash bargaining solution is found, and the same trend as with baseload is observed: a highly differing expectation of future prices leads to a decrease in contract amount.

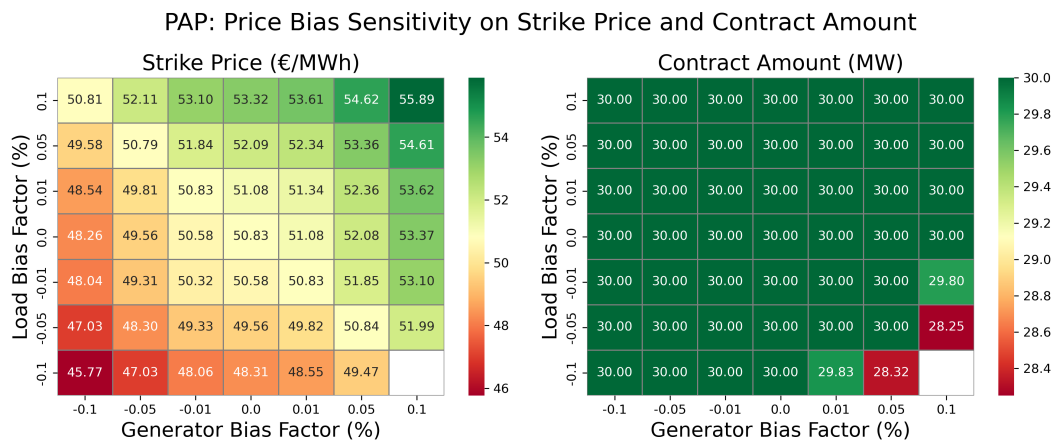


Figure 7.15: Price biased K_{λ}^G applied between $[-10\%, 10\%]$ for pay as produced at $A_G = A_L = 0.5$

7.4.2 Production Asymmetry

Production expectation asymmetry examines how divergent production forecasts affect contract outcomes, particularly relevant for renewable energy projects with inherent generation uncertainty. The results are again kept at base risk-aversions of $A_G = A_L = 0.5$, but production bias results for the other risk-aversions are qualitatively similar to the base case.

Baseload Outcomes

Figure 7.16 shows the Nash bargaining outcomes for the strike price S and contract amounts M as the price bias factor K_{PG}^G and K_{PG}^L are each systematically varied between $-10\%E(P_{t,\omega}^G)$ to $10\%E(P_{t,\omega}^G)$. The plot shows that from the load's perspective (bottom to top), the expected production has no impact on the outcome in either S or M . This is consistent as the $P_{t,\omega}^G$ is not directly part of their utility function Equation 4.11, instead they get a fixed amount M . On the other hand, the bias in production from the generator's view changes how much energy they will be able to supply the contract. At a positive bias, they can supply more 12.61 MWh, and less at a negative bias 11.61 MWh. An inverse relationship between the strike price and contract amount can be seen, as a lower expectation leads to a higher S 89.44 EUR/MWh and a lower M ; conversely, a higher expectation leads to a lower S 88.89 EUR/MWh and higher M .

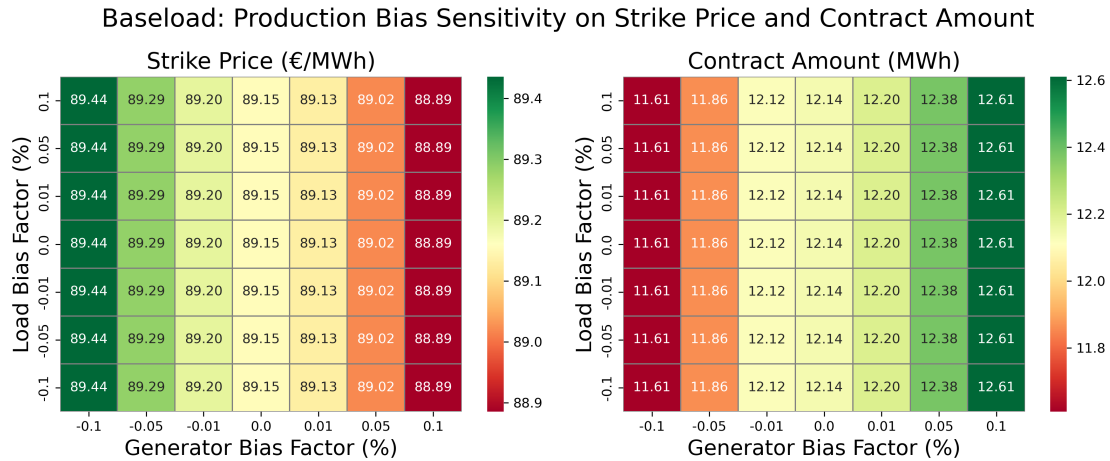


Figure 7.16: Production biased K_{PG}^G applied between $[-10\%, 10\%]$ for baseload at $A_G = A_L = 0.5$

Pay as Produced Outcomes

Looking over the production biases for the PAP in Figure 7.17 the contract share remains stable at $\gamma = 1$ as the production only determines the absolute amount that will be supplied. In the strike price S , an overall trend is visible in the diagonal, ranging from 50.86 MWh to 50.79 EUR/MWh. This trend indicates that a mutual expectation of lower prices decreases the strike price, while a mutual expectation of higher prices leads to a higher S . However, the relationship becomes non-regular at different relative risk aversions. This is likely due to relationships and breakpoints within the CVaR expressions

of both, given that the production is stochastic, and that this solution is not an optimal Nash bargaining solution.

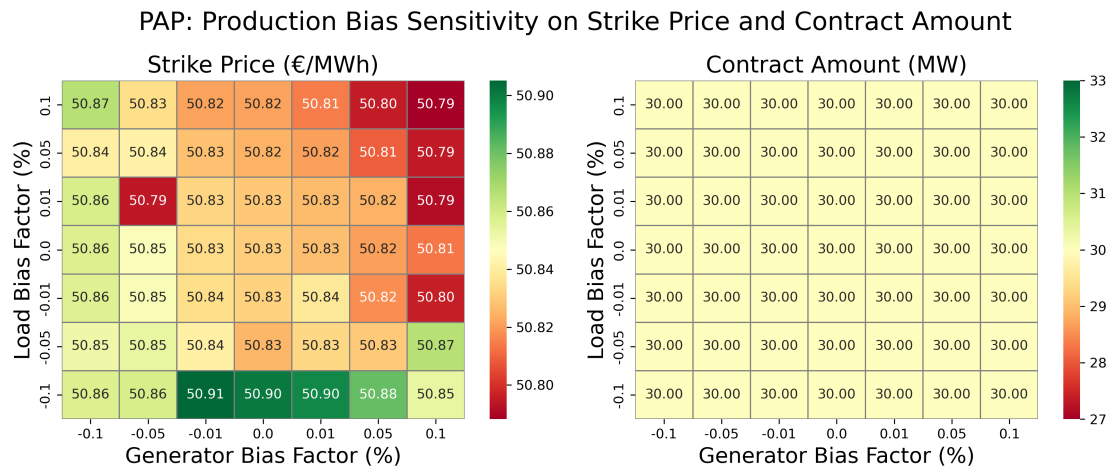


Figure 7.17: Production biased K_{PG}^G, K_{PG}^L applied between $[-10\%, 10\%]$ for pay as produced at $A_G = A_L = 0.5$

7.4.3 Bias Induced Contract Boundaries

Provided by the previous analysis of biases, there must exist an interplay between risk aversion and the bias of each party. The bias, much like the risk aversion, affects the overall shape of the barter set. The following analysis aims to identify combinations of prices and production biases that lead to an infeasible contract or a contract that has a contract amount of 0. For this analysis, we increase the range of biases to $[-30\%, 30\%]$, and plot the no-contract region. The analysis was done for five combinations of risk aversion: three symmetric and two asymmetric.

Price Boundaries of BL and PAP

As seen, for each risk-aversion case, the boundary of the contract region in the $(K_{\lambda}^L/\mathbb{E}(\lambda_{t,\omega}), K_{\lambda}^G/\mathbb{E}(\lambda_{t,\omega}))$ plane is a line, and the contract region is the shaded half-plane bounded below by the contract line. The points that exist under and on the line are biases that can lead to a feasible contract. However, if the bias is above the region with that given risk aversion, a feasible contract does not exist. For both cases of BL Figure 7.18a and Figure 7.18b, an important observation is that the contract region increases in size as the parties become more risk-averse and hence more willing to agree on a contract successfully. Another observation is that it is the lowest risk-aversion that sets the boundary, even if the counterparty is more risk-averse. This relationship is kept until a certain limit is reached in both cases, reflecting two different situations. In 7.18a, the limit is reached when the generator at a high price bias requires a strike price that is above S^U for it to be fair; anything lower would not be profitable. As such, even if the load expects their price to be higher, but has set a limit to what they are willing to offer, then the generator believes it would be unfair and rejects. In the case of the PAP in 7.18b, a similar situation occurs when the bias in prices is sufficiently low at $K_{\lambda}^L = -25\%$. In this situation, the load believes prices are sufficiently low that the lower boundary price S^R put by the generator is high, as the optimal solution lies under S^R . Insofar as the upper and lower strike prices S^R and S^U have been kept constant to emulate that in negotiation, some absolute limits have been set. Dynamically changing the upper and lower strike prices on the bias would have limited physical meaning. In the case of the lower price S^R , realistically, the generator would probably never accept a price that is below its LCOE. Having a high strike price S^U can potentially lead to credit risk, as it requires sufficient funds to purchase the buy-back energy.

7.4.4 Production Contract Boundary: BL and PAP

A similar experiment was conducted to determine if variations in bias in production, with different risk aversions, would lead to any tangible contract boundary. In both cases, a feasible contract existed in all variations of biases and combinations of risk aversions.

7.5 Parameter Sensitivity Analysis - Symmetrical & Asymmetrical

Section 7.5 completes the analysis of Research Question 3 by quantifying how variations in the parameters: prices, production, consumption, and capture rates influence the negotiated contract outcome under both symmetric and asymmetric information. Rather

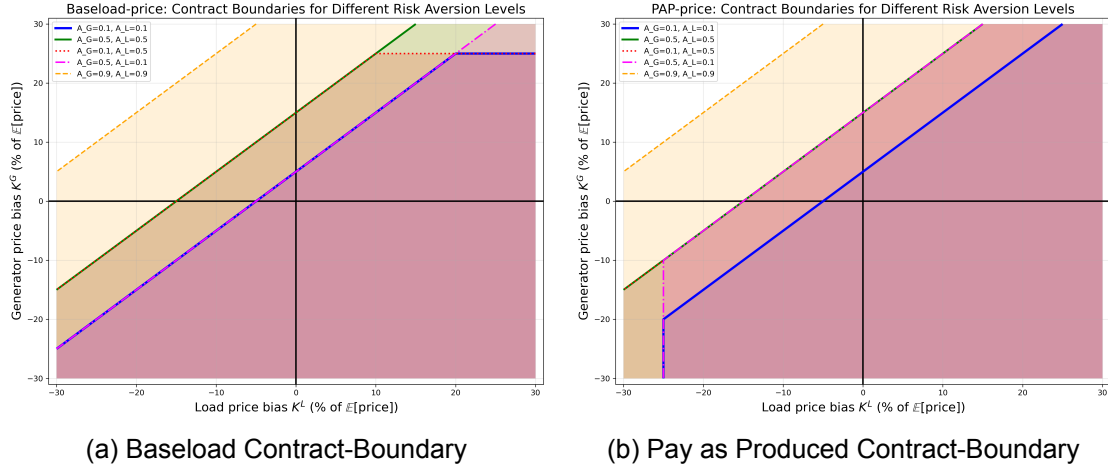


Figure 7.18: No Contract Boundaries with a fixed lower strike price $S^R = 40$ EUR/MWh and upper strike price $S^U = 1.2\mathbb{E}(\lambda_{t,\omega})$

than examining the direction of change, this analysis focuses on elasticity measures to identify which parameters most strongly drive contract outcomes and hence represent the greatest sources of negotiation influence. The evaluation follows the approach in Table 7.2, where x denotes the input parameters and y the outputs. To capture the influence of risk aversion on elasticity, the base case values $A_G = A_L = 0.5$ are compared to scenarios where each party's risk aversion is increased or decreased relative to the counterparty.

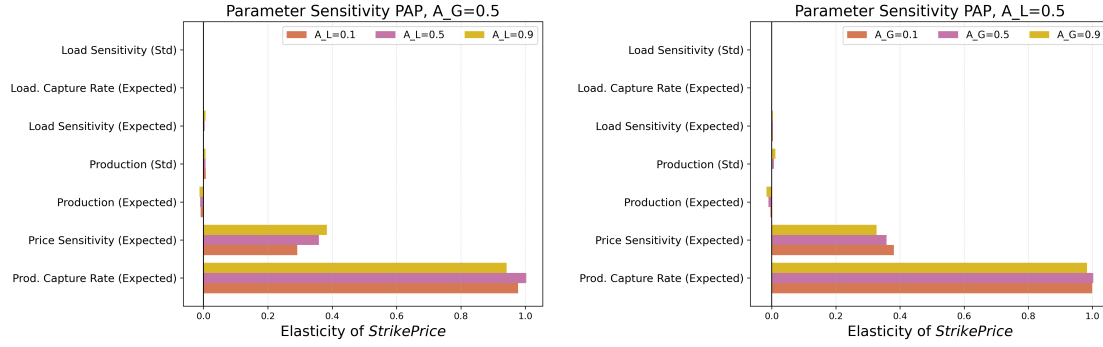
Table 7.2: Definition of Elasticity

$ \epsilon \geq 1$	elastic	y responds more than x changes
$ \epsilon = 1$	unit elastic	y responds proportionally to x
$ \epsilon \leq 1$	inelastic	y responds less than x changes

7.5.1 Symmetrical Shift in Parameters

Pay as Produced Results

In this analysis, we start with the PAP because its results are the simplest to interpret. In Figure 7.19, the only parameters that effectively change the contract outcomes are the expectation of prices and the production capture rate. The strike price has a more inelastic relationship with the price as the PAP is more directly influenced by the capture rate decreasing the value. As such, even with price changes, the PAP contract is unit elastic to the capture rate, signifying the issues that affect this contract type. In the elasticity of the price, small changes in elasticity are observed when the risk aversion is changed. There is generally less elasticity when the load has a 'winning' position $A_L < A_G$ with regard to the generator. The analysis of the contract amount γ is skipped as it shows no changes in the contract amount in this sensitivity analysis.



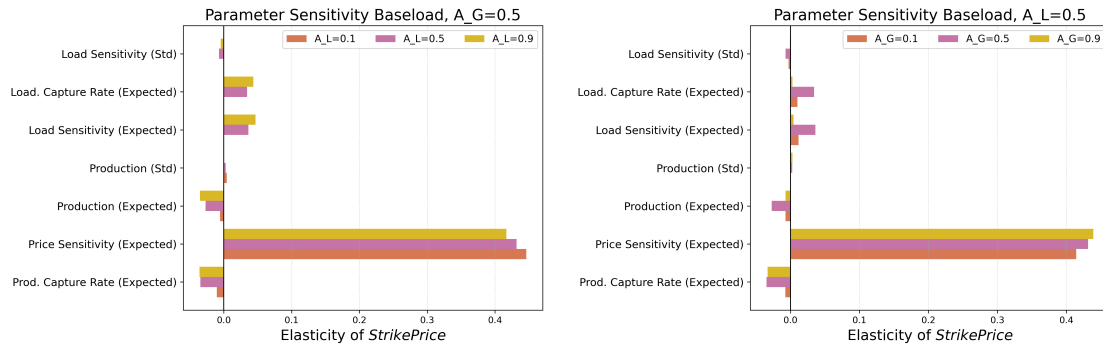
(a) Strike Price S elasticity $A_G = 0.5$ fixed

(b) Strike Price S elasticity $A_L = 0.5$ fixed

Figure 7.19: Symmetric parameter sensitivity of strike price S analysis for pay as produced contracts

Baseload - Symmetrical Results

In Figure 7.20, the primary driver that changes the strike price is the price; however, while it is the largest, it remains quite inelastic to changes in the expectation at around 0.45. The other parameters exhibit minor inelasticities with regard to the strike price. An interesting observation is the direction the bars go in. The load exhibits positive elasticity, indicating that the strike price increases with the parameter. For the parameters related to the generator, the elasticities are negative, indicating that an increase in the capture rate decreases the strike price. The effect of risk aversion is visible, however, given the absolute sizes of the elasticities, the effect is negligible for most of them.



(a) Strike Price S elasticity $A_G = 0.5$ fixed

(b) Strike Price S elasticity $A_L = 0.5$ fixed

Figure 7.20: Symmetric parameter sensitivity of strike price S analysis for baseload contracts

Looking at the contract amount changes in Figure 7.21 invites useful discoveries. One is the dynamic shown in the expected parameters of the load and the generator. If the load has a greater risk aversion than the generator $A_L > A_G$, implying that the generator is in

a better position, then the contract amount is elastic with the parameters associated with the load but inelastic with the generator. On the other hand if $A_G > A_L$, the generator parameters are elastic with the contract amount, but the load parameters are inelastic. If they have equal risk aversion, then the elasticity is spread between the parameters with the load parameters being more elastic. The behavior can be attributed to the party that has a 'winning position'. If the other party is more risk-averse, then the parameters determining their disagreement point directly correlate to the overall shape of the barter set, as they are the party more willing to barter.

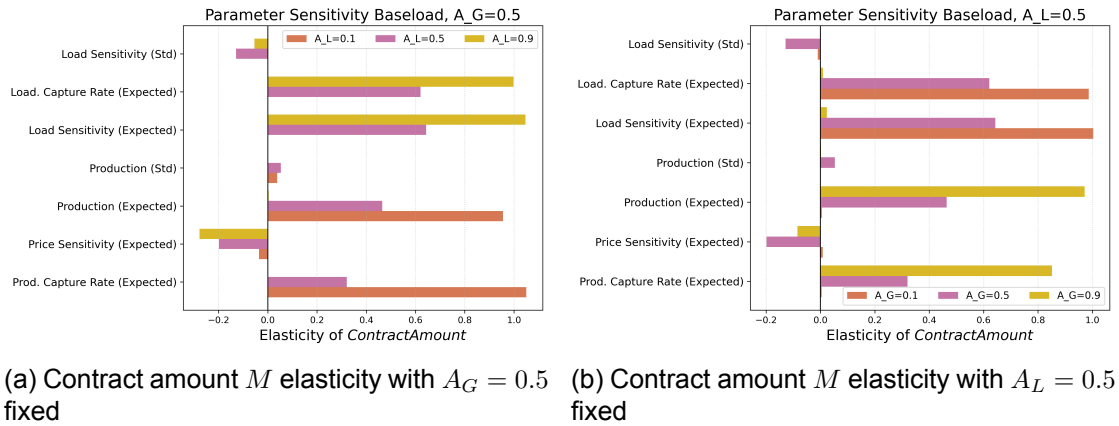


Figure 7.21: Symmetric parameter sensitivity of contract amount M analysis for baseload

7.5.2 Asymmetric Elasticity

So far, the elasticities have been calculated assuming that the change is symmetrical between the two parties. However, an indication from section 7.4 shows that perhaps much volatility in a negotiation derives from differing expectations between parties. This analysis is limited to analyzing the asymmetric biases previously used in section 7.3 and with the same risk aversions used in subsection 7.5.1. The analysis is calculated using a fixed value of $K^G = 0$ (or K^L) and calculates the elasticities of the other biases (K^L or K^G) within the range $[-10\%, 10\%]$.

Pay as Produced - Asymmetrical Results

Much like the symmetric case for the PAP, the contract amount γ has little elasticity to the parameters Figure 7.22a - 7.22b, except when $A_L = 0.1$. At this risk aversion, if the load expects a higher price, the contract's value increases; conversely, if the generator believes it will be higher than the load, the contract amount decreases, signified by the negative elasticity. The elasticity of price Figure 7.22c - 7.22d has increased from around 0.37 in the symmetric to 0.49 in the asymmetric, considering risk aversion $A_G = A_L = 0.5$, similarly, at the other risk aversion combinations, the asymmetric has a larger value. From the previous analysis in subsection 7.5.1, it would be an expected result that an even greater elasticity would be observed if the capture rate of the generator

was included in the analysis. The capture rate is after all the defining parameter of the PAP.

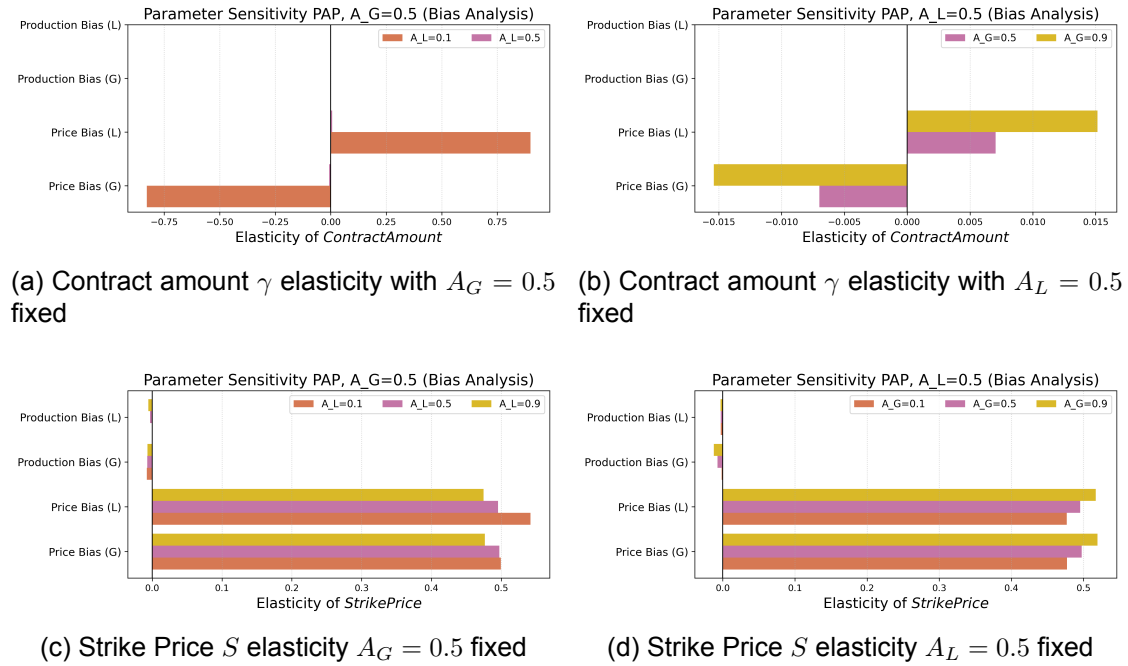


Figure 7.22: Bias - risk elasticity for PAP under varying strike price and contract amount.

Baseload - Asymmetrical Results

As exemplified by the PAP contract, the elasticities of the caseload follow the trends of the symmetrical results in subsection 7.5.1, and inflate the relationship. For the strike prices Figure 7.23a - 7.23b, two main trends are observed. Between the viewpoints of the generator and load, it is clear that what the load believes prices will be is the primary mover in elasticity in determining the strike price with its lowest elasticity of around 0.48, just slightly below the maximum elasticity generator bias in Figure 7.23d. At $A_G = 0.9$ and $A_L = 0.5$, the generator is willing to barter; then, if the load expects a higher price, the overall strike price will increase. The second trend observable is that the relative risk aversion determines the size of elasticity, given the perspective the bias is added. An inverse relationship happens between price biases when $A_G > A_L$ and $A_L > A_G$, a greater A_G emphasizes the loads' perspective and discounts the generators; conversely, a greater A_L emphasizes the generator price and discounts the loads. Lastly, the production bias for the generator shows very inelastic behavior.

Introducing biases in price expectations transforms the contract amount's elasticity from being highly inelastic in the symmetrical case (Figure 7.14) to strongly elastic in Figure 7.23c–Figure 7.23d. In the minimum case, the price elasticity is 2.2, while in the maximum case it rises to 7.3 at $A_G = 0.5$, $A_L = 0.1$. This implies that a 1% change in

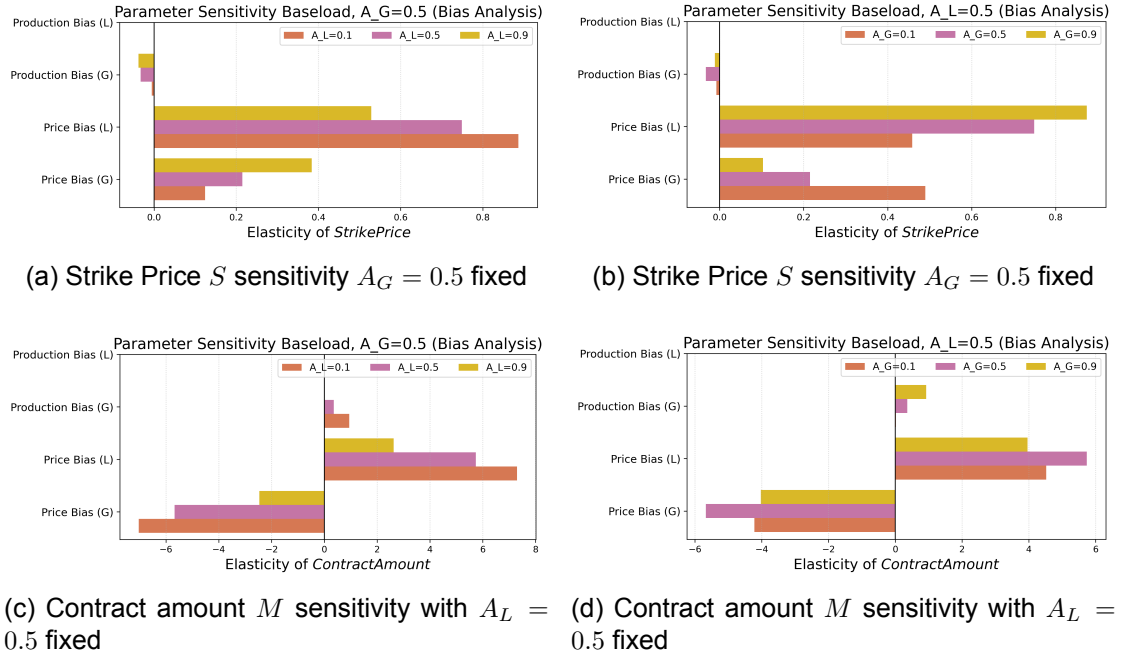


Figure 7.23: Bias - risk elasticity for Baseload under varying strike price and contract amount.

price bias leads to a 7.3% change in the contract amount for this risk aversion combination.

An inverse relationship emerges between biases in price expectations: if the load expects higher prices, the contract amount increases; if the generator expects higher prices, it decreases. With A_G fixed at 0.5, greater load risk aversion reduces price elasticity, though it remains high; lower load risk aversion increases it. At low A_L , price bias becomes a primary driver for the load's decision on contract viability.

When A_L is fixed at 0.5, the relationship between relative risk aversion and price bias is irregular: the highest elasticity occurs at $A_G = A_L = 0.5$, while in cases where one party is more risk-averse, the elasticity values are similar. The production bias exhibits the same behavior as in Figure 7.21, as it reflects the same underlying mechanism.

7.5.3 Impact of Symmetrical and Asymmetrical Bias

The sensitivity analysis yields several notable findings. For the pay as produced (PAP) contract, the generator's capture rate CR^G emerges as the most influential parameter, showing the highest absolute elasticity under symmetric variation. Under asymmetric variation, its influence would likely be even more pronounced, underscoring PAP's pronounced susceptibility to capture price fluctuations.

In contrast, the Baseload case exhibits a different dynamic: while capture prices pri-

marily drive PAP outcomes, Baseload outcomes are susceptible to differences in price risk expectations between the parties. When these expectations diverge, the resulting price and contract size can vary substantially. As such, aligning expectations leads to the most stable contract changes. Yet, even when both parties agree to use the same forecast, residual subjective bias may persist, influencing the decision-making and ultimately the negotiation outcome.

Relative risk aversion further shapes which parameters carry the most weight in negotiations. A party with higher risk aversion experiences a lower disagreement point value (Figure 7.3), making them more willing to concede. Consequently, changes to their disagreement point directly alter both their willingness and their capacity to bargain, amplifying the influence of the parameters associated with their utility function.

8 Conclusion

This thesis set out to answer the overarching question: How can an equitable negotiation outcome for pricing and contract amount in Power Purchase Agreements be predicted? Using Nash bargaining theory as a cooperative game-theoretic framework combined with stochastic optimization, the study developed and applied a model capable of capturing the strategic interplay between a renewable energy generator and a load entity under both baseload (BL) and pay as produced (PAP) PPA structures in a synthetic case study.

The first research question asked how a ‘fair’ strike price and contract amount can be determined for BL and PAP PPAs. The results show that when risk preferences are incorporated into the Nash bargaining solution, negotiated outcomes deviate systematically from naive expectations based solely on expected prices. The formalization of the barter set for both contract types provided clear boundaries for feasible negotiations with the mathematical conditions establishing when mutually beneficial agreements can exist.

The second research question examined the influence of differences in risk aversion and bargaining power. For BL contracts, higher load risk aversion leads to higher strike prices and larger contract volumes, whereas higher generator risk aversion has the opposite effect. In PAP contracts, strike prices respond predictably to relative risk aversion, but contract amounts remain largely constant unless the load is risk-neutral. Bargaining power shifts the allocation of surplus within the feasible barter set without altering its size; in BL contracts, this primarily affects strike price, while in PAP contracts, it can also marginally adjust the contracted share.

The third research question addressed the role of stochastic parameters and asymmetric information. The sensitivity analysis revealed that for PAP contracts, generator capture rate is the dominant driver of negotiated price with proportional elasticity in symmetric expectations. For BL contracts, differences in price expectations between parties can substantially alter both strike price and contract size, with asymmetric biases amplifying the volatility of the outcome. The feasible contract region shrinks when parties hold strongly divergent forecasts, underscoring the importance of aligning expectations or explicitly valuing subjective forecasts in negotiations.

Overall, the framework developed here formalizes the concept of the barter set in PPA negotiations, extends it to stochastic production and capture rate effects, and demonstrates how equilibrium outcomes emerge from the interplay of risk attitudes, bargaining power, and parameter uncertainty. The model provides a structured, transparent method of quantifying “fair” outcomes in long-term renewable energy contracting, offering both academic insight and practical guidance for PPA design.

8.1 Future Work

While the framework developed in this thesis provides a structured approach to predicting equitable outcomes in bilateral PPA negotiations, several extensions could broaden its applicability and realism:

- **Multi-Party Negotiations:** Extend the model to include multiple buyers and sellers, enabling smaller offtakers to acquire a PPA, and increase the pool of potential offtakers for sellers.
- Further convexification of the model by incorporating decomposition techniques to increase the tractability of the model.
- **Hybrid structures:** Integrate battery storage, demand flexibility, and mixed-technology renewable portfolios into the bargaining framework to assess how flexibility resources shift bargaining power and risk allocation.
- **Variable tenor lengths:** Compare short- and long-term PPAs to examine how contract duration influences negotiation outcomes, risk distribution, and the stability of agreements.
- **Alternative utility formulations:** Explore different utility functions to represent diverse stakeholder priorities, including varying definitions of risk aversion besides CVaR, alternative disagreement points besides full market exposure such as considering outside options, and the intrinsic value buyers may place on renewable energy procurement.
- **Ex-post fairness assessment:** Apply the model retrospectively to historical PPA data to evaluate how “fair” past contracts have been given the information and expectations at the time of signing, and how it compares to the realized future.
- **Change in bargaining rule:** The framework could also be extended to incorporate alternative bargaining rules, such as the Kalai-Smorodinsky, to compare how different axioms influence negotiated outcomes under identical market conditions.

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A Necessary Equations

A.1 Baseload Revenue

$$\pi_G(M, S) = \sum_{t=1}^T P_t^G \lambda_t C R_t^G + \sum_{t=1}^T [(S - \lambda_t)M] \quad (\text{A.1})$$

$$\pi_L(M, S) = \sum_{t=1}^T -P_t^L \lambda_t C R_t^L + \sum_{t=1}^T [(\lambda_t - S)M] \quad (\text{A.2})$$

A.2 Pay as Produced Revenue

$$\pi_G(\gamma, S) = \sum_{t=1}^T [(1 - \gamma) \cdot (P_t^G \lambda_t C R_t^G + \gamma(P_t^G S))] \quad (\text{A.3})$$

$$\pi_L(\gamma, S) = \sum_{t=1}^T [-P_t^L \lambda_t C R_t^L] + \sum_{t=1}^T [\gamma P_t^G (\lambda_t C R_t^G - S)] \quad (\text{A.4})$$

A.3 Utility Functions

A.3.1 Old Utility Function

$$u_G = \mathbb{E}(\pi_G(M/\gamma, S) - A_G \text{CVaR}_\alpha(-\pi_G(M/\gamma, S))) \quad (\text{A.5})$$

$$u_L = \mathbb{E}(\pi_L(M/\gamma, S) - A_L \text{CVaR}_\alpha(-\pi_L(M/\gamma, S))) \quad (\text{A.6})$$

A.3.2 Modified Utility Function

$$u_G = (1 - A_G) \mathbb{E}(\pi_G(M/\gamma, S) + A_G \text{CVaR}_\alpha(\pi_G(M/\gamma, S))) \quad (\text{A.7})$$

$$u_L = (1 - A_L) \mathbb{E}(\pi_L(M/\gamma, S) + A_L \text{CVaR}_\alpha(\pi_L(M/\gamma, S))) \quad (\text{A.8})$$

B Utility function - Risk Aversion

The definition of the barter set described so far is based on the fundamentals of [45] and the previous work of [21], which involved deterministic baseLoad PPA. In [21], their *Return-Risk* utility function differs as it is written as the following, using the right tail of the loss function:

$$\hat{u}(M, S) = E(\pi(M, S)) - A \cdot CVaR_{\alpha}(-\pi(M, S)) \quad (B.1)$$

This section seeks to verify and confirm that the results derived there, are applicable with the new utility functions A.7 and A.8, and the PAP version of the barter set.

The current utility function $u(M, S)$ with the optimums of volume (M), strike price(S), and utility value σ is derived from the following equations.

$$u(M, S) = (1 - A) \cdot E(\pi(M, S)) + A \cdot CVaR_{\alpha}(\pi(M, S)), \quad A \in [0, 1] \quad (B.2)$$

$$M^*, S^* = \arg \max(u(M, S)) \quad \& \quad \sigma^{u^*} = u(M^*, S^*) \quad (B.3)$$

For the utility function B.1 [21] it will be denoted as $\hat{u}(M, S)$ with the following optimums.

$$\hat{u}(M, S) = E(\pi(M, S)) - A \cdot CVaR_{\alpha}(-\pi(M, S)), \quad A \geq 0 \quad (B.4)$$

$$\hat{M}^*, \hat{S}^* = \arg \max(\hat{u}(M, S)) \quad \& \quad \hat{\sigma}^{\hat{u}^*} = \hat{u}(\hat{M}^*, \hat{S}^*) \quad (B.5)$$

$\hat{u}(M, S)$ uses the right-tail of the loss distribution in B.1, this can be rewritten to use the left-tail of earnings

$$\hat{u}(M, S) = E(\pi(M, S)) + A \cdot CVaR_{\alpha}(\pi(M, S)), \quad A \geq 0 \quad (B.6)$$

The risk aversion A can be rewritten as $\frac{A}{A+1} \in [0, 1[$ in $u(M, S)$. This allows us to derive the following relationship.

$$u(M, S)^{\frac{A}{A+1}} = \left(1 - \frac{A}{A+1}\right) E(\pi(M, S)) + \frac{A}{A+1} CVaR_{\alpha}(\pi(M, S)) \quad (B.7)$$

$$= \left(\frac{1}{A+1}\right) \cdot (E(\pi(M, S)) + A \cdot CVaR_{\alpha}(\pi(M, S))) \quad (B.8)$$

$$= \frac{\hat{u}(M, S)}{A+1} \quad (B.9)$$

The solution from $u(M, S)^{\frac{A}{A+1}}$ is then the following:

$$M^{\frac{A}{A+1}*}, S^{\frac{A}{A+1}*} = \widehat{M}^*, \widehat{S}^* \quad \forall A \geq 0 \quad (\text{B.10})$$

$$\sigma^{\frac{A}{A+1}*} = \frac{\widehat{\sigma}^{\widehat{u}^*}}{A+1} \quad \forall A \geq 0 \quad (\text{B.11})$$

It has then been shown that the two different *Revenue-Risk* functions arrive at the same solution for the contract volume and strike price with varying values of utility.

C Lemma 1: Derivative of Baseload Utility with Respect to Strike Price

Originally with the old utility functions A.5, A.6 and respecting revenue functions A.1 and A.2 for the baseload. Then the following lemma was presented in [21] *Given any contract amount $M \in [M^R, M^U]$, varying the strike price S from S^R to S^U maps into a straight line in U with slope $-[1 + A_L] / [1 + A_G]$.*

With the new utility functions A.7 and A.8, this changes now to be: **Lemma 2:** *Given any contract amount $M \in [M^R, M^U]$, varying the strike price S from S^R to S^U maps into a straight line in U with slope -1 for the baseload contract.*

Firstly, we derive the change in the revenue function as the strike price is varied. For simplification then the sum $\sum_{t=1}^T M$ will be denoted as $T \cdot M$ since M is a fixed value and $\sum_{t=1}^T M \equiv T \cdot M$

$$\frac{d\pi^G(M, S)}{dS} = \sum_{t=1}^T M = T \cdot M \quad (\text{C.1})$$

$$\frac{d\pi^L(M, S)}{dS} = \sum_{t=1}^T -M = -T \cdot M \quad (\text{C.2})$$

Taking the expectation of C.1 and C.2, we get

$$\frac{dE(\pi^G(M, S))}{dS} = TM \quad (\text{C.3})$$

$$\frac{dE(\pi^L(M, S))}{dS} = -TM \quad (\text{C.4})$$

For the expression of CVaR, we can from the definition of CVaR that CVaR is translation-equivariant, i.e., $CVaR(Y + c) = CVaR(Y) + c$ [62]

From the original definition [21] we get the following

$$\begin{aligned} CVaR^G(-\pi_G(M, S + \Delta S)) &= CVaR^G(-\pi_G(M, S) - TM\Delta S) \\ &= CVaR^G(-\pi_G(M, S)) - TM\Delta S \end{aligned} \quad (\text{C.5})$$

Above is comparable to using $dCVaR(\pi^G)/dS$ and observing the change based on S .

$$\frac{dCVaR_\alpha(\pi^G(M, S))}{dS} = TM \quad (\text{C.6})$$

$$\frac{dCVaR_\alpha(\pi^L(M, S))}{dS} = -TM \quad (\text{C.7})$$

Now taking the full derivative of the utility functions of A.7 and A.8

$$\frac{du_G}{dS} = (1 - A_G) \cdot (TM) + A_G \cdot TM = TM \quad (\text{C.8})$$

$$\frac{du_L}{dS} = (1 - A_G) \cdot (-TM) + A_L \cdot (-TM) = -TM \quad (\text{C.9})$$

Re-integrating Equation C.8 and Equation C.9 with respect to S the following expressions Equation C.10 and Equation C.10 is produced.

$$\int \frac{du_G}{dS} dS = \int TM dS \implies u_G + C_1 = TMS + C_2 \quad (\text{C.10})$$

$$\int \frac{du_L}{dS} dS = \int -TM dS \implies u_L + C_3 = -TMS + C_4 \quad (\text{C.11})$$

$$u_G + u_L + C_1 + C_3 = TMS + C_2 + C_4 - TMS \quad (\text{C.12})$$

$$u_G + u_L = -C_1 + C_2 - C_3 + C_4 \quad (\text{C.13})$$

Adding these expression together into Equation C.13 and re-arranging the terms , differentiating the two expressions, we get the following relationship C.14 that within the Utility Set U varying the strike price under any contract amount M will result in straight line with a slope of -1. This differs significantly from he original expression, where it depends on the risk aversions of A_G and A_L .

$$\frac{dU_L}{dU_G} = -1 \quad (\text{C.14})$$

D Derivative of Pay-as-Produced Utility with Respect to Strike Price

In this chapter, we extend the analytical framework of bilateral contract utility derivatives to the case of a Pay-as-Produced (PAP) contract. Unlike the baseload contract, where the contract volume, M , is fixed and deterministic, the PAP contract involves a stochastic production quantity, P_t^G . This complicates the analysis, particularly when differentiating Conditional Value-at-Risk (CVaR) with respect to the strike price S . The **Lemma 1**: *Given a contract amount (percentage) $\gamma \in [\gamma^R, \gamma^U]$, varying the strike price S from S^R to S^U maps under the PAP contract into a curved line in utility space U with slope given by*

$$\frac{du_L}{du_G} = \frac{(1-A_L)\mathbb{E}(\sum_{t=1}^T -P_t^G) + \frac{d}{dS}\text{CVaR}\alpha(\pi_L(\gamma, S))}{(1-A_G)\mathbb{E}(\sum_{t=1}^T P_t^G) + \frac{d}{dS}\text{CVaR}\alpha(\pi_G(\gamma, S))}.$$

The generator and load profit functions under a PAP contract are given by A.3 and A.4, and taking the derivative of those functions results in the following.

$$\frac{d\pi^G(\gamma, S)}{dS} = \frac{\sum_{t=1}^T [(1-\gamma)(P_t^G C R_t^G \lambda_t) + \gamma P_t^G S]}{dS} = \gamma \sum_{t=1}^T P_t^G \quad (\text{D.1})$$

$$\frac{d\pi^L(\gamma, S)}{dS} = \frac{\sum_{t=1}^T [-P_t^L C R_t^L \lambda_t + \gamma P_t^G (C R_t^G \lambda_t - S)]}{dS} = \gamma \sum_{t=1}^T -P_t^G \quad (\text{D.2})$$

Taking the expectation of C.1 and C.2, we get

$$\frac{d\mathbb{E}(\pi^G(M, S))}{dS} = \gamma \mathbb{E} \left(\sum_{t=1}^T P_t^G \right) \quad (\text{D.3})$$

$$\frac{d\mathbb{E}(\pi^L(M, S))}{dS} = \gamma \mathbb{E} \left(\sum_{t=1}^T -P_t^G \right) \quad (\text{D.4})$$

However, unlike in the baseload case, the PAP structure prevents us from applying the translation-equivariant property of CVaR. The stochastic nature of P_t^G makes the profit functions nonlinear in S , and thus any change in S affects both the value and the distribution of $\pi_G(\gamma, S)$ and $\pi_L(\gamma, S)$.

However, under standard assumptions of convexity and continuity of the loss distribution, [62]) show that the CVaR is convex and continuously differentiable with respect to parameters like S .

This leads to the CVaR derivatives:

$$\frac{d}{dS} \text{CVaR}_\alpha(\pi_G(\gamma, S)) = \mathbb{E} \left[\sum_{t=1}^T \gamma P_t^G \mid \pi_G \leq \text{VaR}_\alpha(\pi_G(\gamma, S)) \right], \quad (\text{D.5})$$

$$\frac{d}{dS} \text{CVaR}_\alpha(\pi_L(\gamma, S)) = -\mathbb{E} \left[\sum_{t=1}^T \gamma P_t^G \mid \pi_L \leq \text{VaR}_\alpha(\pi_L(\gamma, S)) \right]. \quad (\text{D.6})$$

Combining the derivative of the expectation and cvar into the utility function, we get the following utility gradient.

$$\frac{du_G}{dS} = (1 - A_G)\gamma \mathbb{E} \left(\sum_{t=1}^T P_t^G \right) + A_G \frac{d}{dS} \text{CVaR}_\alpha(\pi_G(\gamma, S)) \quad (\text{D.7})$$

$$\frac{du_L}{dS} = (1 - A_L)\gamma \mathbb{E} \left(\sum_{t=1}^T -P_t^G \right) + A_L \frac{d}{dS} \text{CVaR}_\alpha(\pi_L(\gamma, S)) \quad (\text{D.8})$$

The change in utility with respect to S does not produce a linear path in the utility space as in the baseload case. Instead, the PAP contract induces a curved trajectory due to the nonlinearity of the CVaR operator with respect to a stochastic quantity. This slope characterizes how sensitive the risk-adjusted utilities of both agents are to marginal changes in the strike price, forming the basis for determining optimal contract pricing under risk aversion.

Re-integrating Equation D.7 and Equation D.8 with respect to S the following expressions Equation C.10 and Equation C.10 is produced, and we can express the utilities u_G and u_L in the same expression D.11.

$$\begin{aligned} \int \frac{du_G}{dS} dS &= \int \left[(1 - A_G)\gamma \left(\sum_{t=1}^T P_t^G \right) + A_G \frac{d}{dS} \text{CVaR}_\alpha(\pi_G(\gamma, S)) \right] dS \\ \implies u_G + C_5 &= \left((1 - A_G)\gamma \mathbb{E} \left(\sum_{t=1}^T P_t^G \right) + A_G \text{CVaR}_\alpha(\pi_G(\gamma, S)) \right) S + C_6 \end{aligned} \quad (\text{D.9})$$

$$\begin{aligned} \int \frac{du_L}{dS} dS &= \int \left[(1 - A_L)\gamma \mathbb{E} \left(\sum_{t=1}^T -P_t^G \right) + A_L \frac{d}{dS} \text{CVaR}_\alpha(\pi_L(\gamma, S)) \right] dS \\ \implies u_L + C_7 &= \left((1 - A_L)\gamma \mathbb{E} \left(\sum_{t=1}^T -P_t^G \right) + A_L \text{CVaR}_\alpha(\pi_L(\gamma, S)) \right) S + C_8 \end{aligned} \quad (\text{D.10})$$

$$u_G + u_L = -C_5 + C_6 - C_7 + C_8 \quad (\text{D.11})$$

Applying the chain rule to $\frac{du_G}{dS}$ and $\frac{du_L}{dS}$ in D.11, we can compute the marginal change with regards to the strike price.

$$\frac{du_L}{du_G} = \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} = \frac{-(1 - A_L)\gamma\mathbb{E}\left(\sum_{t=1}^T -P_t^G\right) + A_L \cdot \frac{d}{dS}\text{CVaR}_\alpha(\pi_L(\gamma, S))}{(1 - A_G)\gamma\mathbb{E}\left(\sum_{t=1}^T P_t^G\right) + A_G \cdot \frac{d}{dS}\text{CVaR}_\alpha(\pi_G(\gamma, S))} \quad (\text{D.12})$$

$$\frac{du_L}{du_G} = - \frac{(1 - A_L)\gamma\mathbb{E}\left(\sum_{t=1}^T -P_t^G\right) + \frac{d}{dS}\text{CVaR}_\alpha(\pi_L(\gamma, S))}{(1 - A_G)\gamma\mathbb{E}\left(\sum_{t=1}^T P_t^G\right) + \frac{d}{dS}\text{CVaR}_\alpha(\pi_G(\gamma, S))} \quad (\text{D.13})$$

In summary, within the utility Set U, varying the strike price under any contract amount M will result in curved line with the slope shown in D.13.

E Derivative of Baseload Utility with Respect to Contract Volume

E.1 Lemma 2: Concavity - Baseload

The original paper [21] states that: If the strike price is fixed at its lowest possible level S^R , then the locus of points (u_G, u_L) traced out in the utility possibility set U under A.5 and A.6 as the contract amount M varies from M^R to M^U determines a concave curve V_1 in U . Conversely, if the strike price is fixed at its highest possible level S^U , then the locus of points (u_G, u_L) traced out in U under A.5 and A.6 as M varies from M^R to M^U determines a concave curve V_2 in U .

With the new utility functions A.7 and A.8, this lemma still holds. However, the updated objective function does change the rate of change when varying the contract amount du/dM .

E.1.1 Utility Function Definitions

The utility functions for the generator and load are defined as:

$$u_G = (1 - A_G)\mathbb{E}(\pi_G(M, S)) + A_G \cdot \text{CVaR}_\alpha(\pi_G(M, S)) \quad (\text{E.1})$$

$$u_L = (1 - A_L)\mathbb{E}(\pi_L(M, S)) + A_L \cdot \text{CVaR}_\alpha(\pi_L(M, S)) \quad (\text{E.2})$$

where $\pi_G(M, S)$ and $\pi_L(M, S)$ are the net earnings functions for the generator and load respectively.

E.1.2 Concave Curve V1

Setting the strike price S fixed at its lowest possible level S^R , the derivative with respect to M of the generator's revenue function A.1 is calculated. For simplification, $\sum_{t=1}^T S$ will be denoted as $S \cdot T$, since S is a fixed value.

$$\pi_G(M, S) = \sum_{t=1}^T \left(P_t^G \cdot CR_t^G \lambda_t + (S - \lambda_t) M \right), \quad \frac{d\pi_G(M, S)}{dM} = \sum_{t=1}^T (S - \lambda_t) = TS - \sum_{t=1}^T \lambda_t \quad (\text{E.3})$$

The expectation of the generator is then taken with regard to the derivative:

$$\frac{d\mathbb{E}(\pi_G(M, S))}{dM} = TS - \mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) \quad (\text{E.4})$$

Similarly, for the load:

$$\frac{d\mathbb{E}(\pi_L(M, S))}{dM} = \mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) - TS \quad (\text{E.5})$$

For the derivative of CVaR, much like as has been established in previous lemmas, the random combination of $P_t^G \cdot CR_t^G$ makes it so that worst-case outcomes λ_t might be offset by a value of $P_t^G \cdot CR_t^G$. As such, it is the λ_t that lie within the left tail of earnings that is accounted for in the derivative.

This leads to the CVaR derivatives:

$$\frac{d}{dM} \text{CVaR}_\alpha(\pi_G(M, S)) = \mathbb{E} \left[\sum_{t=1}^T (S - \lambda_t) \mid \pi_G \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] \quad (\text{E.6})$$

$$\frac{d}{dM} \text{CVaR}_\alpha(\pi_L(M, S)) = \mathbb{E} \left[\sum_{t=1}^T (\lambda_t - S) \mid \pi_L \leq \text{VaR}_\alpha(\pi_L(M, S)) \right] \quad (\text{E.7})$$

Two regimes (covering all realizations): (i) $M > \max_{t,\omega} P_t^G \cdot CR_t^G$ (uniformly short), (ii) $M \leq \min_{t,\omega} P_t^G \cdot CR_t^G$ (uniformly long).

Condition 1: $M > \max_{t,\omega} P_t^G \cdot CR_t^G$ (**Generator SHORT position**)

Using the *translation equivariance* of CVaR, the following equation can be rewritten. In [21] they also use the *positive homogeneity* property as they assume P_t^G is a fixed deterministic parameter with constant baseload production. This is no longer possible as production is stochastic.

$$\begin{aligned} \text{CVaR}_\alpha(\pi_G(M, S)) &= \text{CVaR}_\alpha \left(\sum_{t=1}^T [P_t^G \cdot CR_t^G \lambda_t + (S - \lambda_t)M] \right) \\ &= \text{CVaR}_\alpha \left(\sum_{t=1}^T [P_t^G \cdot CR_t^G \lambda_t - \lambda_t M] \right) + TSM \\ &= \text{CVaR}_\alpha \left(\sum_{t=1}^T [(M - P_t^G \cdot CR_t^G)(-\lambda_t)] \right) + TSM \end{aligned} \quad (\text{E.8})$$

The CVaR contains the HIGH prices of λ_t as given $M > P_t^G \cdot CR_t^G$, the generator has to buy back what is missing and is exposed to the market. It is worth remembering that the CVaR is using the left tail of earnings. The CVaR is dependent on the realizations of $M - P_t^G \cdot CR_t^G$.

Taking the derivative of this function:

$$\begin{aligned} \frac{d \text{CVaR}_\alpha(\pi_G(M, S))}{dM} &= \frac{d \left[\text{CVaR}_\alpha \left(\sum_{t=1}^T [(M - P_t^G \cdot CR_t^G)(-\lambda_t)] \right) + TSM \right]}{dM} \\ &= \mathbb{E} \left[\sum_{t=1}^T (-\lambda_t) \mid \pi_G(M, S) \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] + TS \end{aligned} \quad (\text{E.9})$$

The above derivative (E.9) is equivalent to the previously derived derivative (E.6) with the adjustment about emphasizing higher prices in the CVaR.

The full derivative of the utility function of u_G can then be formulated as the combination of (E.4) and (E.9):

$$\begin{aligned} \frac{du_G}{dM} &= (1 - A_G) \cdot (TS - \mathbb{E}(\sum_{t=1}^T \lambda_t)) \\ &\quad + A_G \cdot \left(\mathbb{E} \left[\sum_{t=1}^T (-\lambda_t) \mid \pi_G(M, S) \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] + TS \right) \\ &= TS - \left[(1 - A_G) \mathbb{E} \left(\sum_{t=1}^T \lambda_t \right) + A_G \mathbb{E} \left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S) \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] \right] \end{aligned} \quad (\text{E.10})$$

Important note regarding integration: Because the tail set $\{\pi_G \leq \text{VaR}_\alpha(\pi_G)\}$ depends on M , the derivative in (E.10) is not constant in M . Therefore one cannot conclude $u_G = C \cdot M + C'$ with constants independent of M . The rigorous statement is the absolutely continuous representation:

$$u_G(M, S) = u_G(M_0, S) + \int_{M_0}^M \frac{du_G}{dm}(m, S) dm \quad (\text{E.11})$$

where $\frac{du_G}{dm}(m, S)$ is given by (E.10) with M replaced by m .

Using the chain rule, the slope of the Pareto frontier is:

$$\frac{du_L}{du_G} = \frac{du_L/dM}{du_G/dM} = \frac{(1 - A_L)(\mathbb{E}(\sum_{t=1}^T \lambda_t) - TS) + A_L \mathbb{E}[\sum_{t=1}^T (\lambda_t - S) \mid \pi_L \leq \text{VaR}_\alpha(\pi_L)]}{TS - \left[(1 - A_G) \mathbb{E}(\sum_{t=1}^T \lambda_t) + A_G \mathbb{E}(\sum_{t=1}^T \lambda_t \mid \pi_G \leq \text{VaR}_\alpha(\pi_G)) \right]} \quad (\text{E.12})$$

Taking the second derivative with respect to u_G :

$$\frac{d^2 u_L}{du_G^2} = \frac{\frac{d^2 u_L}{dM^2} \frac{du_G}{dM} - \frac{du_L}{dM} \frac{d^2 u_G}{dM^2}}{\left(\frac{du_G}{dM} \right)^3} \quad (\text{E.13})$$

Given (E.5), $\frac{d^2 \mathbb{E}(\pi_L)}{dM^2} = 0$. Moreover, $\frac{d}{dM} \text{CVaR}_\alpha(\pi_L) = \mathbb{E}[\sum_t (\lambda_t - S) \mid \pi_L \leq \text{VaR}_\alpha(\pi_L)]$ is non-increasing in M under the left-tail selection, so $\frac{d^2 \text{CVaR}_\alpha(\pi_L)}{dM^2} \leq 0$. This establishes concavity on the region of Condition 1.

Condition 2: $M \leq \min_{t, \omega} P_t^G \cdot CR_t^G$ (Generator LONG position) **Economic Interpretation:** When $M \leq \min_{t, \omega} P_t^G \cdot CR_t^G$, the generator produces more than contracted in all scenarios and can sell excess to the spot market. The generator is exposed to LOW spot prices in the left tail of the profit distribution.

The procedure is identical to Condition 1, as such, some explanation is skipped:

$$\begin{aligned}
\text{CVaR}_\alpha(\pi_G(M, S)) &= \text{CVaR}_\alpha \left(\sum_{t=1}^T [P_t^G \cdot CR_t^G \lambda_t + (S - \lambda_t)M] \right) \\
&= \text{CVaR}_\alpha \left(\sum_{t=1}^T [P_t^G \cdot CR_t^G \lambda_t - \lambda_t M] \right) + TSM \quad (\text{E.14}) \\
&= \text{CVaR}_\alpha \left(\sum_{t=1}^T [(P_t^G \cdot CR_t^G - M)(\lambda_t)] \right) + TSM
\end{aligned}$$

$$\begin{aligned}
\frac{d\text{CVaR}_\alpha(\pi_G(M, S))}{dM} &= \frac{d \left[\text{CVaR}_\alpha \left(\sum_{t=1}^T [(P_t^G \cdot CR_t^G - M)(\lambda_t)] \right) + TSM \right]}{dM} \quad (\text{E.15}) \\
&= \mathbb{E} \left[\sum_{t=1}^T (-\lambda_t) \mid \pi_G(M, S) \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] + TS
\end{aligned}$$

The full derivative of the utility function of u_G can then be formulated as the combination of (E.4) and (E.15):

$$\begin{aligned}
\frac{du_G}{dM} &= (1 - A_G) \cdot (TS - \mathbb{E}(\sum_{t=1}^T \lambda_t)) \\
&\quad + A_G \cdot \left(\mathbb{E} \left[\sum_{t=1}^T (-\lambda_t) \mid \pi_G(M, S) \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] + TS \right) \\
&= TS - \left[(1 - A_G) \mathbb{E} \left(\sum_{t=1}^T \lambda_t \right) + A_G \mathbb{E} \left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S) \leq \text{VaR}_\alpha(\pi_G(M, S)) \right] \right] \quad (\text{E.16})
\end{aligned}$$

As in Condition 1, $u_G(M, S)$ satisfies the integral representation:

$$u_G(M, S) = u_G(M_0, S) + \int_{M_0}^M \frac{du_G}{dm}(m, S) dm \quad (\text{E.17})$$

The concavity conclusion follows analogously using (E.13).

In summary, when the strike price S is fixed at the lowest possible level S^R , it has been shown that the contract amount interval from M^R to $\min_{t,\omega} P_t^G \cdot CR_t^G$ and the contract amount interval between $\max_{t,\omega} P_t^G \cdot CR_t^G$ to M^U each map under A.7 and A.8 into a concave curve in the utility set U traced out by $u_L(u_G)$ as M varies in the M intervals. The function is continuous at the boundary points under mild regularity, so the entire V_1 curve is concave for $M \in [M^R, M^U]$.

The method for the concave curve V_2 is a replica of above, and as such will be skipped.

E.2 Lemma 3: Monotonicity - Baseload

This part seeks to affirm the Pareto frontier made by $u_L(u_G)$ and find the strike price thresholds where the frontier is upheld.

If S^R is less than S^{R*} , then with the strike price fixed at S^R , as the contract amount M increases, u_G decreases and u_L increases. On the other hand, if S^U is greater than S^{U*} , then with the strike price fixed at S^U as the contract amount M is increased, u_G increases and u_L decreases.

Case 1: Finding S^{R*} when strike price is fixed at S^R

We establish the threshold strike prices where the utility derivatives with respect to M equal zero, based on the methodology presented in the previous sections.

For the Load at S^R : Setting $\frac{du_L}{dM} = 0$ and using the general form, the zero-derivative threshold is:

$$S_L^{R*} = \frac{(1 - A_L)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_L\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_L(M, S^R) \leq \text{VaR}_\alpha(\pi_L(M, S^R))\right]}{T} \quad (\text{E.18})$$

For the Generator at S^R - Condition 1: $M > \max_{t,\omega} P_t^G \cdot CR_t^G$ From equation (E.10), setting $\frac{du_G}{dM} = 0$:

$$S_{G1}^{R*} = \frac{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_G\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S^R) \leq \text{VaR}_\alpha(\pi_G(M, S^R))\right]}{T} \quad (\text{E.19})$$

For the Generator at S^R - Condition 2: $M \leq \min_{t,\omega} P_t^G \cdot CR_t^G$ From equation (E.16), setting $\frac{du_G}{dM} = 0$:

$$S_{G2}^{R*} = \frac{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_G\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S^R) \leq \text{VaR}_\alpha(\pi_G(M, S^R))\right]}{T} \quad (\text{E.20})$$

Important Note: While the algebraic forms of S_{G1}^{R*} and S_{G2}^{R*} are identical, the conditional expectations are evaluated over different tail sets. In Condition 1, the tail set includes scenarios where the generator must buy back shortfall (over-contracting), while in Condition 2, the tail set includes scenarios where the generator has excess capacity (under-contracting).

Monotonicity when $S^R < S^{R*}$: Define $S^{R*} = \min\{S_{G1}^{R*}, S_{G2}^{R*}, S_L^{R*}\}$. Since $S^R < S^{R*}$:

For the Generator: Since $S^R < S_{Gi}^{R*}$ (for the relevant condition):

$$\left. \frac{du_G}{dM} \right|_{S=S^R} < 0 \quad (\text{E.21})$$

Therefore, as M increases, u_G decreases.

For the Load: Since $S^R < S_L^{R*}$:

$$\left. \frac{du_L}{dM} \right|_{S=S^R} > 0 \quad (\text{E.22})$$

Therefore, as M increases, u_L increases.

Case 2: Finding S^{U*} when strike price is fixed at S^U

For the Load at S^U : Setting $\frac{du_L}{dM} = 0$:

$$S_L^{U*} = \frac{(1 - A_L)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_L\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_L(M, S^U) \leq \text{VaR}_\alpha(\pi_L(M, S^U))\right]}{T} \quad (\text{E.23})$$

For the Generator at S^U - Condition 1: $M > \max_{t,\omega} P_t^G \cdot CR_t^G$ From equation (E.10), setting $\frac{du_G}{dM} = 0$:

$$S_{G1}^{U*} = \frac{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_G\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S^U) \leq \text{VaR}_\alpha(\pi_G(M, S^U))\right]}{T} \quad (\text{E.24})$$

For the Generator at S^U - Condition 2: $M \leq \min_{t,\omega} P_t^G \cdot CR_t^G$ From equation (E.16), setting $\frac{du_G}{dM} = 0$:

$$S_{G2}^{U*} = \frac{(1 - A_G)\mathbb{E}\left(\sum_{t=1}^T \lambda_t\right) + A_G\mathbb{E}\left[\sum_{t=1}^T \lambda_t \mid \pi_G(M, S^U) \leq \text{VaR}_\alpha(\pi_G(M, S^U))\right]}{T} \quad (\text{E.25})$$

Monotonicity when $S^U > S^{U*}$: Define $S^{U*} = \max\{S_{G1}^{U*}, S_{G2}^{U*}, S_L^{U*}\}$. Since $S^U > S^{U*}$:

For the Generator: Since $S^U > S_{Gi}^{U*}$ (for the relevant condition):

$$\left. \frac{du_G}{dM} \right|_{S=S^U} > 0 \quad (\text{E.26})$$

Therefore, as M increases, u_G increases.

For the Load: Since $S^U > S_L^{U*}$:

$$\left. \frac{du_L}{dM} \right|_{S=S^U} < 0 \quad (\text{E.27})$$

Therefore, as M increases, u_L decreases.

E.2.1 Boundary Definitions

Finally, we define S^{R*} and S^{U*} that define the boundaries where a barter can be constructed:

$$S^{R*} = \min \{S_{G1}^{R*}, S_{G2}^{R*}, S_L^{R*}\} \quad (\text{E.28})$$

$$S^{U*} = \max \{S_{G1}^{U*}, S_{G2}^{U*}, S_L^{U*}\} \quad (\text{E.29})$$

where each threshold is evaluated at the strike price level being considered (S^R or S^U), with:

- S_{G1}^{R*}, S_{G1}^{U*} : Evaluated for $M > \max_{t,\omega} P_t^G \cdot CR_t^G$
- S_{G2}^{R*}, S_{G2}^{U*} : Evaluated for $M \leq \min_{t,\omega} P_t^G \cdot CR_t^G$
- S_L^{R*}, S_L^{U*} : Evaluated consistently across all cases

The feasible contract region requires $S^{R*} \leq S^R < S^U \leq S^{U*}$ for both monotonicity patterns to be achievable

E.3 Lemma 4: Barter Set Existence - Baseload

In the original paper, the following lemma is proposed:

$$\left. \frac{du_L}{du_G} \right|_{M^R, S^R} < -\frac{1 + A_L}{1 + A_G} \quad (\text{E.30})$$

$$\left. \frac{du_L}{du_G} \right|_{M^U, S^U} > -\frac{1 + A_L}{1 + A_G} \quad (\text{E.31})$$

The new lemma that is proposed in this paper is:

$$\left. \frac{du_L}{du_G} \right|_{M^R, S^R} < -1 \quad (\text{E.32})$$

$$\left. \frac{du_L}{du_G} \right|_{M^U, S^R} > -1 \quad (\text{E.33})$$

The purpose of this lemma is to show that when you consider a contract starting at the threat point M^R , what is the slope of the concave curves V_1, V_2 . Conversely, what is the slope at the maximum contract amount M^U . The reason for the inequalities compared to -1 is from Lemma 2, wherein it was defined that given any contract amount M , varying the strike price maps under the utility functions A.7, A.8 into a straight line with slope -1. If inequality (E.32) doesn't hold, then there is no contract with a beneficial outcome. Conversely, if (E.33) doesn't hold, it means that the curve never descends, such that the load (buyer) is willing to buy electricity at the highest feasible price.

Starting the proof, we first consider (E.32) and observe that this is equivalent to Condition 2 of $M \leq \min_{t,\omega} P_t^G \cdot CR_t^G$, as such we can use (E.16) and (E.12) to write

$$\frac{du_L}{du_G} \Big|_{(M^R, S^R)} = \frac{(1 - A_L)(\mathbb{E} \sum_{t=1}^T \lambda_t - TS^R) + A_L \mathbb{E} \left[\sum_{t=1}^T (\lambda_t - S^R) \mid \pi_L(M^R, S^R) \leq \text{VaR}_\alpha(\pi_L(M^R, S^R)) \right]}{TS^R - \left[(1 - A_G) \mathbb{E} \sum_{t=1}^T \lambda_t + A_G \mathbb{E} \left(\sum_{t=1}^T \lambda_t \mid \pi_G(M^R, S^R) \leq \text{VaR}_\alpha(\pi_G(M^R, S^R)) \right) \right]} \quad (\text{E.34})$$

Therefore the condition becomes

$$\frac{(1 - A_L)(\mathbb{E} \sum_{t=1}^T \lambda_t - TS^R) + A_L \mathbb{E} \left[\sum_{t=1}^T (\lambda_t - S^R) \mid \pi_L(M^R, S^R) \leq \text{VaR}_\alpha(\pi_L(M^R, S^R)) \right]}{TS^R - \left[(1 - A_G) \mathbb{E} \sum_{t=1}^T \lambda_t + A_G \mathbb{E} \left(\sum_{t=1}^T \lambda_t \mid \pi_G(M^R, S^R) \leq \text{VaR}_\alpha(\pi_G(M^R, S^R)) \right) \right]} < -1. \quad (\text{E.35})$$

For (E.33) (equivalent to Condition 1 with $M > \max_{t,\omega} P_t^G \cdot CR_t^G$), evaluate at (M^U, S^R) :

$$\frac{du_L}{du_G} \Big|_{(M^U, S^R)} = \frac{(1 - A_L)(\mathbb{E} \sum_{t=1}^T \lambda_t - TS^R) + A_L \mathbb{E} \left[\sum_{t=1}^T (\lambda_t - S^R) \mid \pi_L(M^U, S^R) \leq \text{VaR}_\alpha(\pi_L(M^U, S^R)) \right]}{TS^R - \left[(1 - A_G) \mathbb{E} \sum_{t=1}^T \lambda_t + A_G \mathbb{E} \left(\sum_{t=1}^T \lambda_t \mid \pi_G(M^U, S^R) \leq \text{VaR}_\alpha(\pi_G(M^U, S^R)) \right) \right]} \quad (\text{E.36})$$

and the condition is

$$\frac{(1 - A_L)(\mathbb{E} \sum_{t=1}^T \lambda_t - TS^R) + A_L \mathbb{E} \left[\sum_{t=1}^T (\lambda_t - S^R) \mid \pi_L(M^U, S^R) \leq \text{VaR}_\alpha(\pi_L(M^U, S^R)) \right]}{TS^R - \left[(1 - A_G) \mathbb{E} \sum_{t=1}^T \lambda_t + A_G \mathbb{E} \left(\sum_{t=1}^T \lambda_t \mid \pi_G(M^U, S^R) \leq \text{VaR}_\alpha(\pi_G(M^U, S^R)) \right) \right]} > -1. \quad (\text{E.37})$$

These inequalities (E.35) and (E.37) establish the necessary conditions for the existence of a non-empty barter set. The first inequality ensures that at the minimum contract level M^R , the trade-off between utilities favors further contracting. The second inequality implies that at a maximum contract level, a contract level has been found before M^U , and further increasing has diminishing returns for one party.

F Derivative of Pay-as-Produced Utility with Respect to Contract Percentage

F.1 Lemma 2: Concavity - Pay-as-Produced

As established with the previous baseload example, the lemmas are extended to include the pay-as-produced framework as well. These lemmas are less robust, as they do not have previously established proofs to rely on from [21]. However, the methodology of approach is the same and similar results are proven, and with reasoning from [21] and the general concept of cooperative game theory [45], we extend to include Pay-as-Produced.

If the strike price is fixed at its lowest possible level S^R , then the locus of points (u_G, u_L) traced out in the utility possibility set U under A.7 and A.8 as the contract percentage γ varies from γ^R to γ^U determines a concave curve V_1 in U . Conversely, if the strike price is fixed at its highest possible level S^U , then the locus of points (u_G, u_L) traced out in U under A.7 and A.8 as γ varies from γ^R to γ^U determines a concave curve V_2 in U .

F.1.1 Utility Function Definitions

The utility functions for the generator and load under the pay-as-produced framework are:

$$u_G = (1 - A_G)\mathbb{E}(\pi_G(\gamma, S)) + A_G \cdot \text{CVaR}_\alpha(\pi_G(\gamma, S)) \quad (\text{F.1})$$

$$u_L = (1 - A_L)\mathbb{E}(\pi_L(\gamma, S)) + A_L \cdot \text{CVaR}_\alpha(\pi_L(\gamma, S)) \quad (\text{F.2})$$

where the revenue functions are:

$$\pi_G(\gamma, S) = \sum_{t=1}^T [(1 - \gamma)P_t^G \cdot CR_t^G \lambda_t + \gamma P_t^G \cdot CR_t^G S] \quad (\text{F.3})$$

$$\pi_L(\gamma, S) = \sum_{t=1}^T [\gamma P_t^G \cdot CR_t^G (\lambda_t - S)] \quad (\text{F.4})$$

F.1.2 Concave Curve V_1

Setting the strike price S fixed at its lowest possible level S^R , the derivative with respect to γ of the generator's revenue function (A.3) is calculated:

$$\frac{d\pi_G(S^R, \gamma)}{d\gamma} = \frac{d}{d\gamma} \left[\sum_{t=1}^T (1 - \gamma)P_t^G \cdot CR_t^G \lambda_t + \gamma P_t^G \cdot CR_t^G S \right] = \sum_{t=1}^T P_t^G \cdot CR_t^G (S - \lambda_t) \quad (\text{F.5})$$

The expectation of the generator's derivative is:

$$\frac{d\mathbb{E}(\pi_G(\gamma, S))}{d\gamma} = \mathbb{E} \left(\sum_{t=1}^T P_t^G \cdot CR_t^G (S - \lambda_t) \right) \quad (\text{F.6})$$

Similarly, for the load:

$$\frac{d\mathbb{E}(\pi_L(\gamma, S))}{d\gamma} = \mathbb{E} \left(\sum_{t=1}^T P_t^G \cdot CR_t^G (\lambda_t - S) \right) \quad (\text{F.7})$$

For the derivative of CVaR, much like as has been established in previous lemmas, the random combination of $P_t^G \cdot CR_t^G \lambda_t$ makes it so that worst-case outcomes λ_t might be offset by values of $P_t^G \cdot CR_t^G$. As such, it is the λ_t that lie within the left tail that are accounted for in the derivative.

This leads to the CVaR derivatives:

$$\frac{d}{d\gamma} \text{CVaR}_\alpha(\pi_G(\gamma, S)) = \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G (S - \lambda_t) \mid \pi_G \leq \text{VaR}_\alpha(\pi_G(\gamma, S)) \right] \quad (\text{F.8})$$

$$\frac{d}{d\gamma} \text{CVaR}_\alpha(\pi_L(\gamma, S)) = \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G (\lambda_t - S) \mid \pi_L \leq \text{VaR}_\alpha(\pi_L(\gamma, S)) \right] \quad (\text{F.9})$$

Important Note: Unlike the baseload contract, we cannot utilize the *translation equivariance* or *positive homogeneity* properties for any of the parameters as they are all combined with random variables. The contract percentage γ lies in the interval $[0, 1]$, where $\gamma = 0$ represents no contract and $\gamma = 1$ represents full pay-as-produced commitment.

The full derivative of the utility function u_G can then be formulated as the combination of (F.6) and (F.8):

$$\begin{aligned} \frac{du_G}{d\gamma} = & (1 - A_G) \cdot \mathbb{E} \left(\sum_{t=1}^T P_t^G \cdot CR_t^G (S - \lambda_t) \right) \\ & + A_G \cdot \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G (S - \lambda_t) \mid \pi_G(\gamma, S) \leq \text{VaR}_\alpha(\pi_G(\gamma, S)) \right] \end{aligned} \quad (\text{F.10})$$

Important note regarding integration: As with the baseload case, because the tail set $\{\pi_G \leq \text{VaR}_\alpha(\pi_G)\}$ depends on γ , the derivative in (F.10) is not constant in γ . The rigorous representation is:

$$u_G(\gamma, S) = u_G(\gamma^R, S) + \int_{\gamma^R}^{\gamma^U} \frac{du_G}{d\gamma}(\gamma, S) d\gamma \quad (\text{F.11})$$

where $\frac{du_G}{d\xi}(\xi, S)$ is given by (F.10) with γ replaced by ξ .

Using the chain rule to find the slope of the Pareto frontier:

$$\frac{du_L}{du_G} = \frac{du_L/d\gamma}{du_G/d\gamma} = \frac{(1 - A_L) \frac{d\mathbb{E}(\pi_L(\gamma, S))}{d\gamma} + A_L \frac{d\text{CVaR}(\pi_L(\gamma, S))}{d\gamma}}{(1 - A_G) \frac{d\mathbb{E}(\pi_G(\gamma, S))}{d\gamma} + A_G \frac{d\text{CVaR}(\pi_G(\gamma, S))}{d\gamma}} \quad (\text{F.12})$$

Taking the second derivative with respect to u_G :

$$\frac{d^2 u_L}{du_G^2} = \frac{1}{\left(\frac{du_G}{d\gamma}\right)^2} \left[(1 - A_L) \frac{d^2 \mathbb{E}(\pi_L)}{d\gamma^2} + A_L \frac{d^2 \text{CVaR}_\alpha(\pi_L)}{d\gamma^2} \right] \quad (\text{F.13})$$

Given the earlier definition of (F.7), it is evident that:

$$\frac{d^2 \mathbb{E}(\pi_L(\gamma, S))}{d\gamma^2} = 0 \quad (\text{F.14})$$

As was established in Lemma 1 in [21], the CVaR of the load's revenue function $\text{CVaR}_\alpha(-\pi_L(\gamma, S))$ was derived to be a convex function. Their interpretation is that the $-\pi_L(\gamma, S)$ is using a right-tail of the loss-function, whereas in our case it is an earnings left-tail perspective. Multiplying by (-1) to $\text{CVaR}_\alpha(-\pi_L(\gamma, S))$, it follows that $\text{CVaR}_\alpha(\pi_L(\gamma, S))$ is a concave function:

$$\frac{d^2 \text{CVaR}(\pi_L(\gamma, S))}{d\gamma^2} \leq 0 \quad (\text{F.15})$$

Therefore, the curve of points (u_G, u_L) that is traced out in the utility space U as γ varies from 0 to $\gamma^U = 1$ has been shown to follow a concave curve.

In summary, when the strike price S is fixed at the lowest possible level S^R , it has been shown that the contract percentage interval from 0 to $\gamma^U = 1$ maps under A.7 and A.8 into a concave curve in the utility set U traced out by $u_L(u_G)$ in U as γ varies in the γ interval.

The method for the concave curve V_2 is a replica of above, and as such will be skipped.

F.2 Lemma 3: Monotonicity - Pay-as-Produced

This part seeks to affirm the Pareto frontier made by $u_L(u_G)$ and find the strike price thresholds where the frontier is upheld.

If S^R is less than S^{R*} , then with the strike price fixed at S^R , as the contract percentage γ increases, u_G decreases and u_L increases. On the other hand, if S^U is greater than S^{U*} , then with the strike price fixed at S^U as the contract percentage γ is increased, u_G increases and u_L decreases.

F.2.1 CVaR Derivatives at the Boundary

We establish based on the methodology for (F.7), the derivatives with respect to γ at $\gamma = 0$:

$$\left. \frac{d\text{CVaR}(\pi_L(\gamma, S))}{d\gamma} \right|_{\gamma=0} = \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(\lambda_t - S) \mid \pi_L(\gamma, S) \leq \text{VaR}_\alpha(\pi_L(\gamma, S)) \right] \quad (\text{F.16})$$

$$\left. \frac{d\text{CVaR}(\pi_G(\gamma, S))}{d\gamma} \right|_{\gamma=0} = \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(S - \lambda_t) \mid \pi_G(\gamma, S) \leq \text{VaR}_\alpha(\pi_G(\gamma, S)) \right] \quad (\text{F.17})$$

Key Difference from Baseload: In comparison to the baseload case, where S can be brought outside the CVaR expression as it is deterministic, this is not the case for PAP. The strike price S is multiplied by the stochastic production $P_t^G \cdot CR_t^G$, making analytical solutions impossible. Additionally, the value of γ impacts the tail set in the CVaR expressions in (F.16) and (F.17).

F.2.2 Boundary Strike Price Definitions

As such, to find the optimal lower and upper bounds is no longer independent of the contract volume but part of it. The upper and lower contract bounds are described when $\gamma = 0$, which gives the starting point of what the initial price boundary is. The numerical solutions for S^{R*} and S^{U*} are presented here:

$$S^{R*} = \arg \min_{S \in \mathbb{R}} \left\{ S : (1 - A_L) \cdot \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(\lambda_t - S) \right] + A_L \cdot \left. \frac{d\text{CVaR}(\pi_L(\gamma, S))}{d\gamma} \right|_{\gamma=0} = 0 \right\} \quad (\text{F.18})$$

$$S^{U*} = \arg \max_{S \in \mathbb{R}} \left\{ S : (1 - A_G) \cdot \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(S - \lambda_t) \right] + A_G \cdot \left. \frac{d\text{CVaR}(\pi_G(\gamma, S))}{d\gamma} \right|_{\gamma=0} = 0 \right\} \quad (\text{F.19})$$

Economic Interpretation:

- S^{R*} represents the threshold below which S^R must lie for monotonic increase in load utility

- S^{U*} represents the threshold above which S^U must lie for monotonic increase in generator utility
- These thresholds must be solved numerically due to the interaction between S and stochastic production

F.3 Lemma 4: Barter Set Existence - Pay-as-Produced

This lemma follows the same approach as [21] and the previously described lemma for the baseload case. At the disagreement point with the lowest possible strike price S^R , if the curve slope is below the gradient from Lemma 2 - PAP, then a barter set exists. If the gradient hasn't become greater than this threshold, the barter set is unbounded with regards to the contract volume.

The conditions for barter set existence are:

$$\left. \frac{du_L}{du_G} \right|_{\gamma^R, S^R} < \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (\text{F.20})$$

$$\left. \frac{du_L}{du_G} \right|_{\gamma^U, S^R} > \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (\text{F.21})$$

With $\frac{\frac{du_L}{dS}}{\frac{du_G}{dS}}$ defined from D.13

$$\frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} = - \frac{(1 - A_L)\gamma \mathbb{E}(\sum_t (-P_t^G)) + \frac{d}{dS} \text{CVaR}_\alpha(\pi_L(\gamma, S))}{(1 - A_G)\gamma \mathbb{E}(\sum_t P_t^G) + \frac{d}{dS} \text{CVaR}_\alpha(\pi_G(\gamma, S))} \quad (\text{F.22})$$

F.3.1 Proof at Lower Bound

Starting with (F.20), we observe that this applies when $\gamma^R = 0$. Using the chain rule:

$$\left. \frac{du_L}{du_G} \right|_{\gamma^R, S^R} = \frac{(1 - A_L) \frac{d\mathbb{E}(\pi_L(\gamma^R, S^R))}{d\gamma} + A_L \frac{d\text{CVaR}(\pi_L(\gamma^R, S^R))}{d\gamma}}{(1 - A_G) \frac{d\mathbb{E}(\pi_G(\gamma^R, S^R))}{d\gamma} + A_G \frac{d\text{CVaR}(\pi_G(\gamma^R, S^R))}{d\gamma}} \quad (\text{F.23})$$

Substituting the specific derivatives gives:

$$\frac{(1 - A_L) \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(\lambda_t - S^R) \right] + A_L \frac{d\text{CVaR}(\pi_L(\gamma^R, S^R))}{d\gamma}}{(1 - A_G) \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(S^R - \lambda_t) \right] + A_G \frac{d\text{CVaR}(\pi_G(\gamma^R, S^R))}{d\gamma}} < \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (\text{F.24})$$

F.3.2 Proof at Upper Bound

For (F.21), inserting $\gamma^U = 1$:

$$\left. \frac{du_L}{du_G} \right|_{\gamma^U, S^R} = \frac{(1 - A_L) \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(\lambda_t - S^R) \right] + A_L \frac{d\text{CVaR}(\pi_L(\gamma^U, S^R))}{d\gamma}}{(1 - A_G) \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(S^R - \lambda_t) \right] + A_G \frac{d\text{CVaR}(\pi_G(\gamma^U, S^R))}{d\gamma}} \quad (\text{F.25})$$

Resulting in the following inequality for the barter set:

$$\frac{(1 - A_L) \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(\lambda_t - S^R) \right] + A_L \frac{d\text{CVaR}(\pi_L(\gamma^U, S^R))}{d\gamma}}{(1 - A_G) \mathbb{E} \left[\sum_{t=1}^T P_t^G \cdot CR_t^G(S^R - \lambda_t) \right] + A_G \frac{d\text{CVaR}(\pi_G(\gamma^U, S^R))}{d\gamma}} > \frac{\frac{du_L}{dS}}{\frac{du_G}{dS}} \quad (\text{F.26})$$

- Inequality (F.20): At zero contract ($\gamma^R = 0$), the utility trade-off must be steep enough to justify entering into a contract
- Inequality (F.21): At full contract ($\gamma^U = 1$), the utility trade-off is too steep, and one party is benefiting more than another. A lower γ provides an optimum.
- The right-hand side represents the rate of substitution between utilities when varying strike price
- Unlike baseload, these conditions involve ratios of CVaR derivatives due to the stochastic nature of production

G Risk Earnings Comparison Across Contract Types and Risk Aversion Levels

Table G.1: Risk Scenario Analysis for Different Contract Types

Risk Scenario	Contract Type	Generator Revenue	Load Cost	Generator Std Dev	Load Std Dev
No Contract	Baseline	96.56 Mio EUR	-256 Mio EUR	14.55 Mio EUR	37.0 Mio EUR
AG=0.5, AL=0.1	Baseload	115 Mio EUR	-240 Mio EUR	8.5 Mio EUR	28.0 Mio EUR
AG=0.5, AL=0.1	Pay-as-Produced	91.4 Mio EUR	-260 Mio EUR	1.6 Mio EUR	35.0 Mio EUR
AG=0.5, AL=0.5	Baseload	110 Mio EUR	-245 Mio EUR	12.0 Mio EUR	20.0 Mio EUR
AG=0.5, AL=0.5	Pay-as-Produced	91.4 Mio EUR	-255 Mio EUR	3.5 Mio EUR	30.0 Mio EUR
AG=1.0, AL=0.5	Baseload	108 Mio EUR	-248 Mio EUR	15.0 Mio EUR	15.0 Mio EUR
AG=0.5, AL=1.0	Pay-as-Produced	91.4 Mio EUR	-260 Mio EUR	1.6 Mio EUR	23.0 Mio EUR

H Risk Earnings Comparison Across Contract Types and Negotiation Power

I Monte Carlo Simulations

I.1 Price

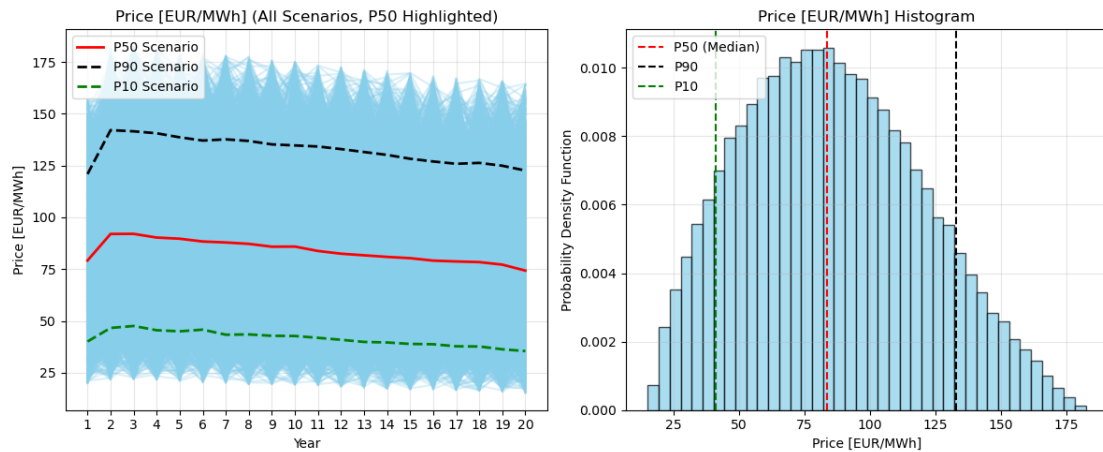


Figure I.1: Monte Carlo Prices

I.2 Production

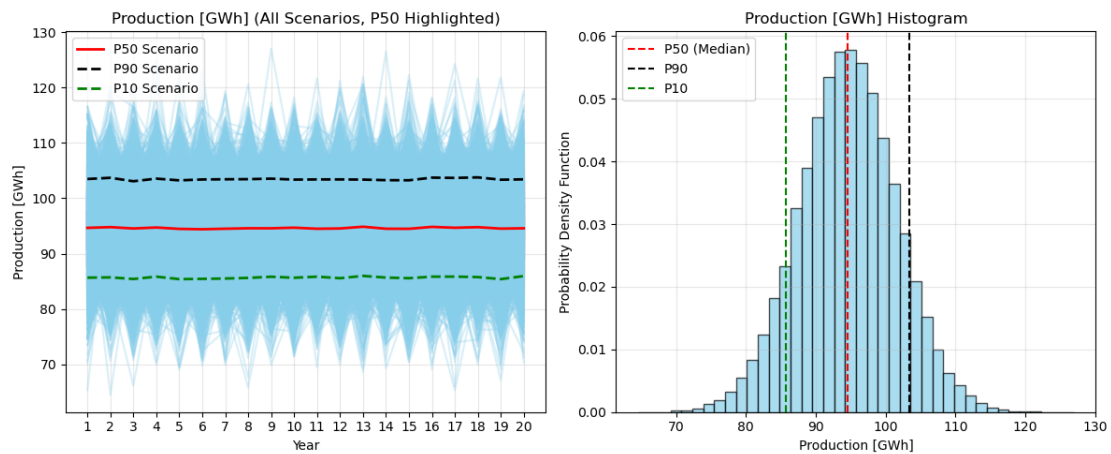


Figure I.2: Monte Carlo Production

I.3 Consumption

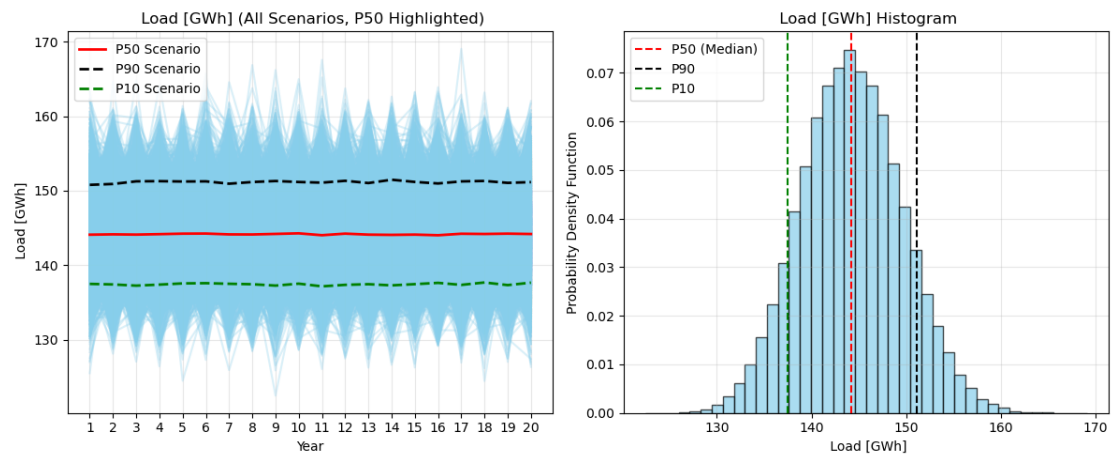


Figure I.3: Monte Carlo Load Consumption

I.4 Capture Rate

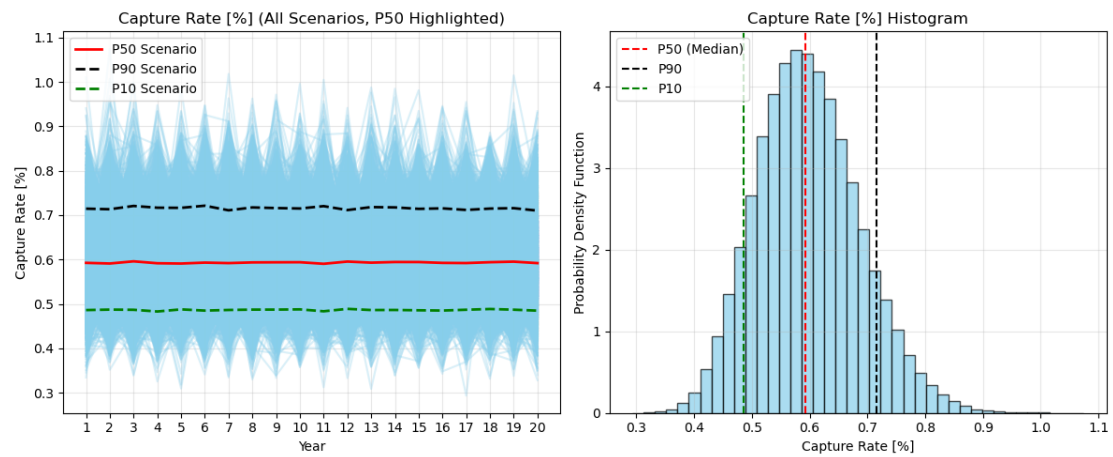


Figure I.4: Monte Carlo Capture Rates

I.5 Load Capture Rate

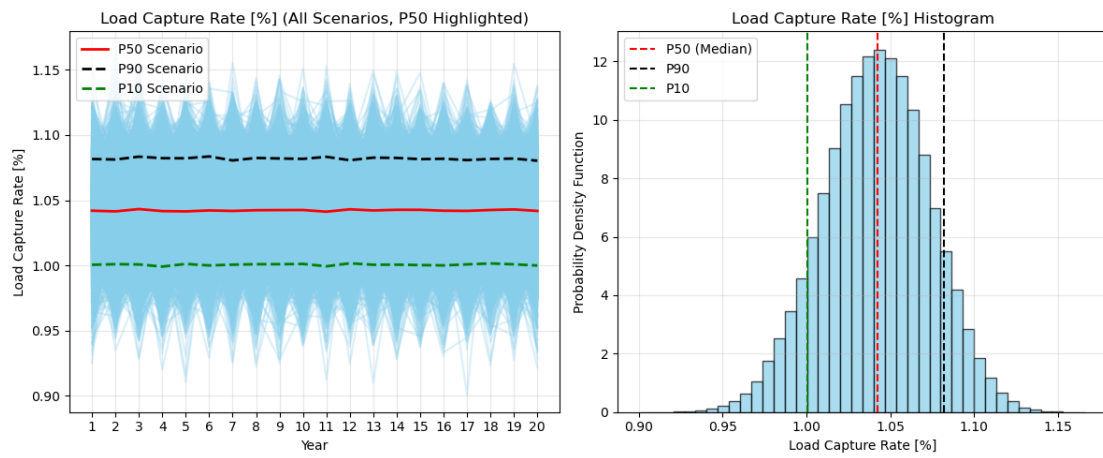


Figure I.5: Monte Carlo Load Capture Rates

I.6 Revenue G

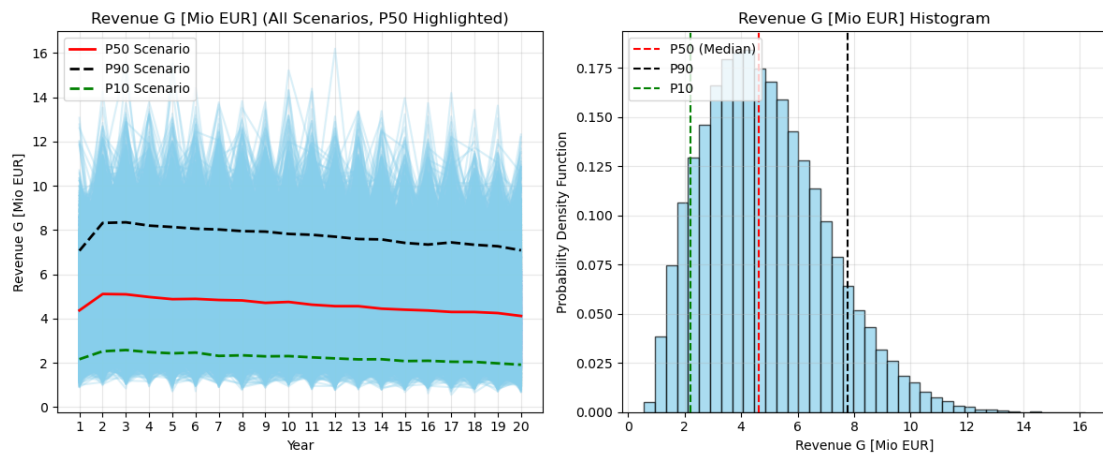


Figure I.6: Monte Carlo Revenue G

I.7 Revenue L

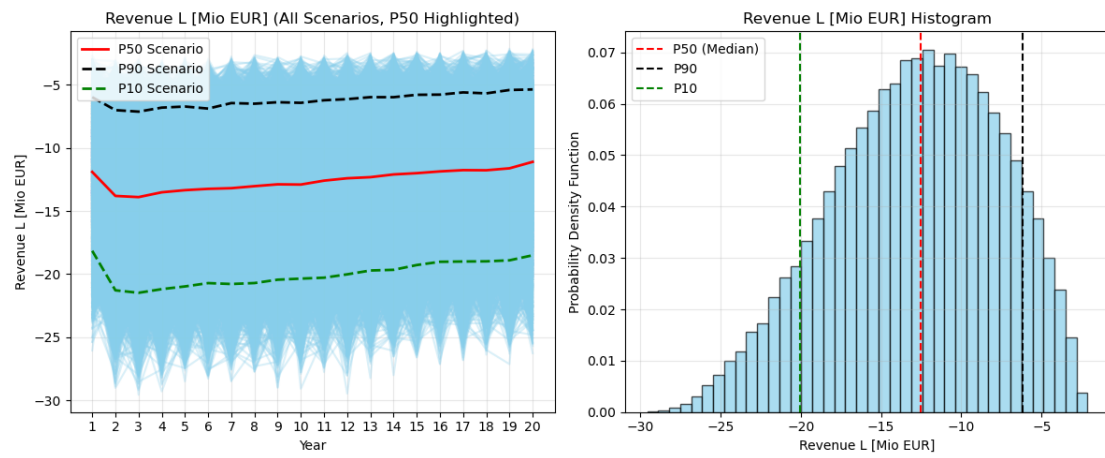


Figure I.7: Monte Carlo Revenue L

J Scenario Reduction - Kmeans 5000

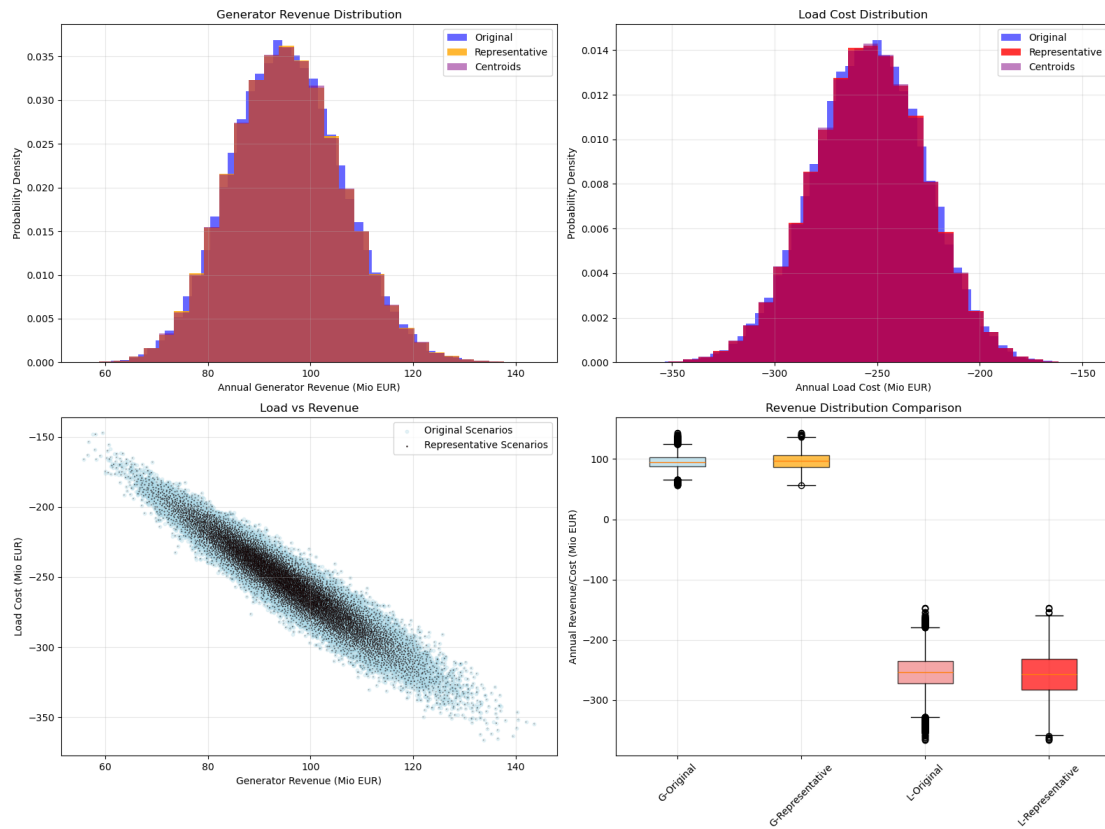


Figure J.1: Kmeans scenario reduction

K Scenario Reduction - Kmeans 500

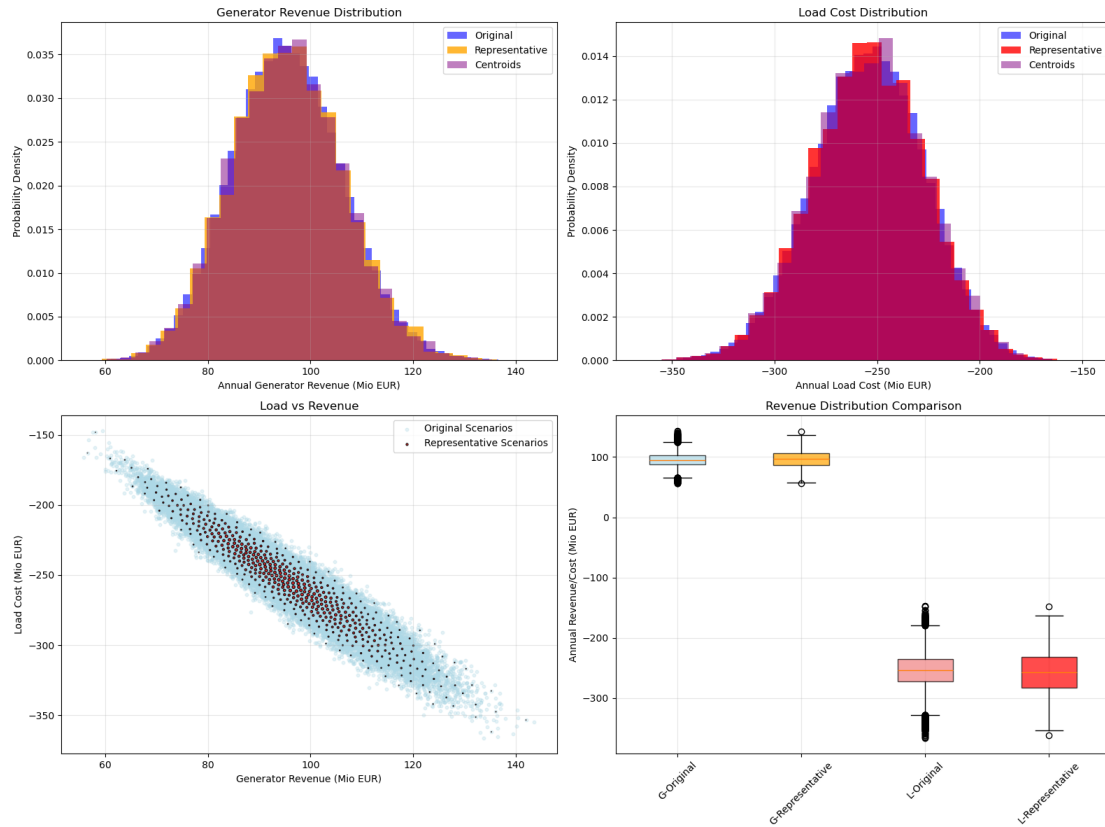


Figure K.1: Kmeans scenario reduction

L Barter Set Case 2 - No Barter Set

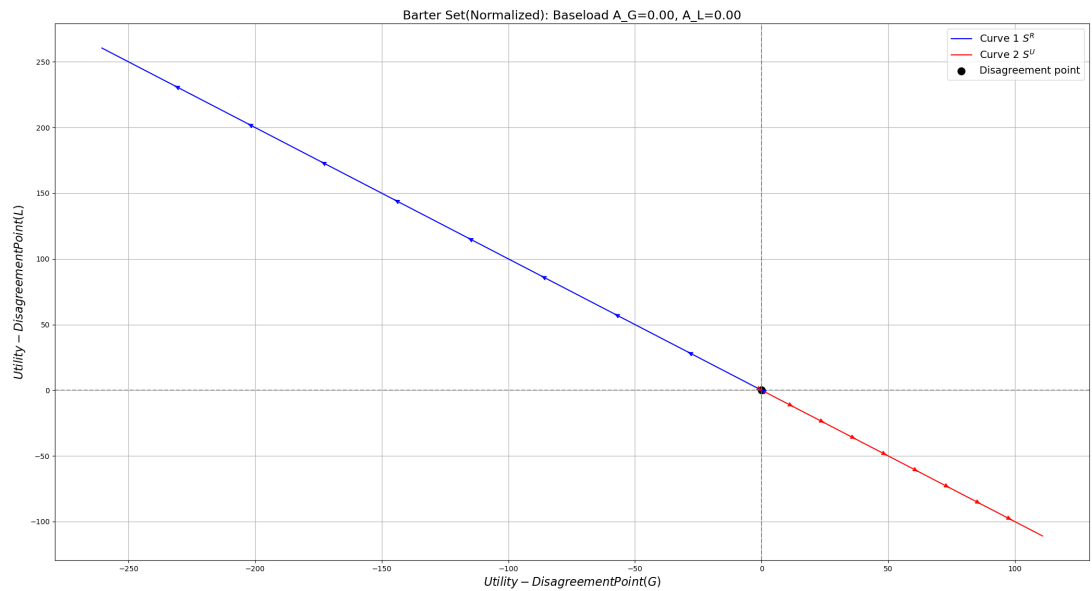


Figure L.1: Case 2: BL and PAP

M Barter Set Case 3 - Base Case in PAP

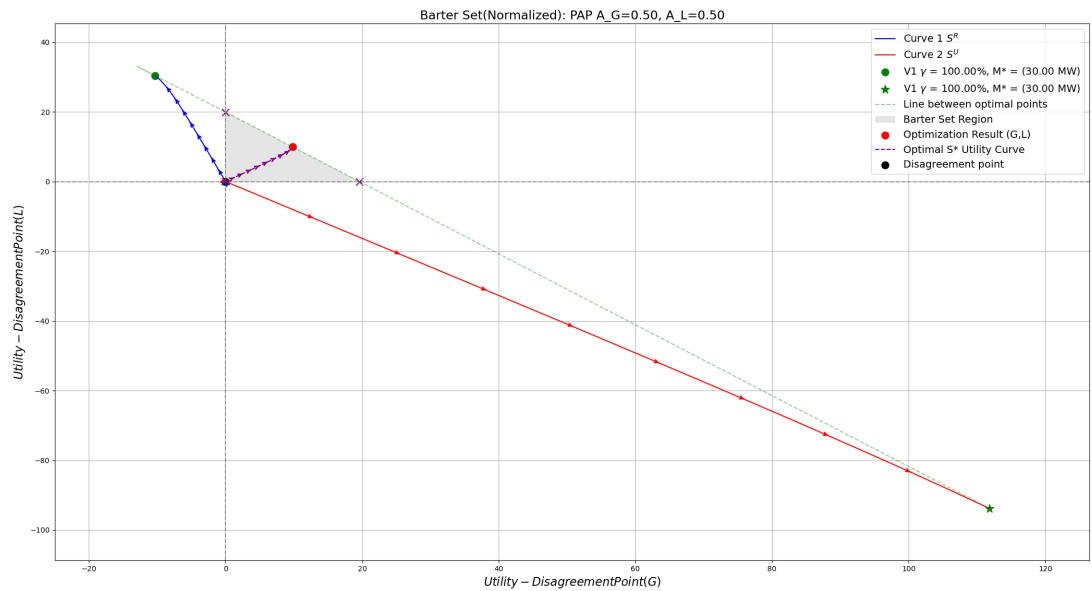


Figure M.1: Case 3: PAP Base Case

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